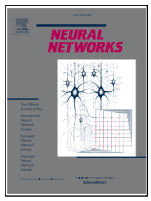
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Full Length Article



A Lagrangian-inspired force-based loss for deep metric learning

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ABSTRACT

Deep metric learning objectives typically define desired embedding distributions while leaving interaction directions and magnitudes implicit. We introduce a Lagrangian-inspired force-first framework prescribing and integrating conservative embedding-space interactions into a scalar objective. Under a fixed interaction configuration, the objective's negative gradient recovers the prescribed field. The framework identifies and quantifies three force-design principles: intrinsic hard mining, margin control through anchor-adaptive interaction ranges, and cardinality-weighted force balancing. Their force-based realization defines the potential energy metric loss, while an annulus-scaled projection preserves radial variation and bounds embedding norms. In frozen-encoder linear evaluation, our loss achieved the highest observed top-1 accuracies among compared objectives: 96.04%, 77.61%, and 85.94% on CIFAR-10, CIFAR-100, and ImageNet-100, respectively, with gains over the supervised contrastive loss reaching 2.88 percentage points. On long-tailed CIFAR-100 (imbalance ratio 200), accuracy was 55.54% versus 33.43% for the supervised contrastive loss, a 22.11-percentage-point gain. At this imbalance, the scale-independent ratio of within-class dispersion to between-centroid distance was lower overall (0.7387 versus 0.8372) and among the ten rarest classes (1.0607 versus 1.3000), indicating better-preserved class-level organization under cardinality skew. In class-disjoint fine-grained retrieval, our loss achieved the highest observed Recall@K at every evaluated depth on CUB-200-2011 and Cars196, with a minimum 4.07-percentage-point margin over the corresponding second-best result, demonstrating effective transfer to held-out categories. The annulus-scaled projection improved all five evaluated Euclidean metric-learning objectives over ℓ_2 normalization by 1.91–7.16 percentage points. These results establish force-first objective construction as a principled and empirically effective route from desired embedding interactions to metric-learning objectives.

1. Introduction

Deep metric learning aims to learn a discriminative representation in which semantically related samples are placed nearby while unrelated samples are separated (Kaya & Bilge, 2019). The strength and range of these attractive and repulsive interactions are governed largely by the training objective. Loss design is therefore a central avenue for controlling the geometry, discrimination, and transferability of learned metric representations.

Existing deep metric-learning objectives organize supervision through pair- or tuple-level constraints, proxy-based class representations, or batchwise contrastive objectives (Khosla et al., 2020; Qian et al., 2019; Schroff et al., 2015). Despite differences in the entities they compare and the manner in which those comparisons are aggregated, these approaches generally follow a loss-first formulation: a scalar objective is defined to favor a desired representation geometry. For example, triplet loss explicitly encourages intra-class compactness and inter-class separation (Schroff et al., 2015), whereas contrastive objectives can be interpreted as optimizing alignment by bringing positive pairs together (Wang & Isola, 2020). This perspective specifies the geometric relations that the learned representation should satisfy, but it does not necessarily prescribe a target vector field governing how individual embeddings should evolve during optimization. Consequently, although the canonical principle of pulling positives together and pushing negatives apart underlies most metric-learning formulations, the intended direction and magnitude of each interaction, together with their dependence on pair hardness and class cardinality, often remain implicit.

Several studies, including (Zhang et al., 2020) and (Khosla et al., 2020), analyze the gradients of metric losses with respect to their embedding vectors to clarify how the objectives act in representation space. Such analyzes reveal the update field induced by an already defined loss. However, the customary top-down workflow of defining a scalar

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objective first and examining its gradients afterward does not ensure that the induced field matches a desired interaction field in both direction and magnitude. We therefore invert this workflow: we first prescribe the desired embedding-space interaction field and then derive a compatible scalar objective, adopting a bottom-up, force-first paradigm for metric-loss construction.

The central requirement in this inverse construction is that the prescribed embedding-space field be integrable into a scalar objective. A Lagrangian-inspired force–potential framework is particularly well suited to this requirement because it provides a direct inverse relation for conservative interactions. For such interactions, a potential-energy difference is obtained from the negative work integral, whereas taking the negative gradient of the resulting potential recovers the prescribed field under a fixed interaction configuration. We therefore specify pairwise embedding interactions as conservative forces and identify their aggregated potential energy with the metric-learning loss. The distinction of this formulation lies in the direction of objective construction rather than in the energy interpretation itself: the desired interaction field is prescribed before the corresponding scalar potential is obtained. This force–potential correspondence provides a principled basis for constructing the objective and interpreting the structure of its induced embedding-space updates, as developed in Section 3.1.

With this foundation, effective force construction requires interaction rules that reflect both metric-learning objectives and the composition of the contrast set. The analysis in Section 3.2 identifies three central design principles: intrinsic hard mining, margin constraints realized through adaptive interaction ranges, and cardinality-weighted force balancing. Intrinsic hard mining requires the interaction magnitude to vary consistently with pair hardness; adaptive interaction ranges determine which positive and negative pairs remain active; and cardinality-weighted aggregation prevents the number of sampled members of a represented class from directly determining that class's aggregate contribution. The analysis further shows that the per-pair interaction magnitudes induced by normalized softmax-based objectives such as InfoNCE (van den Oord et al., 2018) and SupCon (Khosla et al., 2020) need not follow the desired monotonic hardness ordering. Guided by these principles, we construct bounded, monotone attractive and repulsive forces whose active ranges adapt to the local configuration around each anchor and whose aggregation balances the contributions of positive samples and represented negative classes. Integrating these forces yields closed-form pairwise potentials, and their cardinality-weighted aggregation defines the scalar objective that we call the *potential energy metric loss* (PEML), developed in Section 3.4.

Because PEML is formulated through Euclidean distances, retaining both angular and radial variation is important for realizing its intended interactions. Conventional ℓ_2 normalization bounds the embedding space but restricts every embedding to a single radius and thereby suppresses radial updates. We therefore introduce the annulus-scaled projection (ASP) in Section 3.3. ASP maps embeddings to a bounded spherical annulus, preserving radial variation while retaining bounded output norms. This isotropic mapping is compatible with the central-force construction of PEML and can also serve as an output parameterization for other Euclidean-distance-based metric-learning objectives.

The empirical study evaluates the formulation across balanced and long-tailed representation learning, class-disjoint retrieval, and complementary geometric and mechanistic analyzes. On CIFAR-10 and CIFAR-100 (Krizhevsky, 2009) and ImageNet-100 (Barbano et al., 2023; Deng et al., 2009), PEML records the highest observed frozen-encoder linear-evaluation accuracy among the compared objectives. Clustering and centroid-relative analyzes further indicate that these improvements are accompanied by more coherent class-level organization rather than being confined to the fitted linear classifier. On CIFAR100-LT, PEML preserves linear-evaluation accuracy markedly better as the imbalance ratio increases in the comparison with SupCon and the evaluated representation objective from ProCo (Du et al., 2024); at an imbalance ratio of 200, it exceeds SupCon by 22.11 percentage points. In class-disjoint retrieval, PEML records the highest observed Recall@K at every reported depth on CUB-200-2011 (Wah et al., 2011) and Cars196 (Krause et al., 2013), supporting effective transfer to categories held out from benchmark-specific training. Stanford Online Products (Song et al., 2016) and In-Shop Clothes Retrieval (Liu et al., 2016), in contrast, exhibit a more method- and depth-dependent ordering, thereby delineating the benchmark-dependent scope of the retrieval gains. Moreover, replacing ℓ_2 normalization with ASP improves all five Euclidean-distance-based objectives evaluated under the common CIFAR-100 projection comparison. Together, these findings indicate that the proposed construction benefits class-level organization, preservation under severe cardinality skew, and fine-grained transfer while maintaining a clearly characterized scope across different retrieval settings.

Our primary contributions are summarized as follows:

- We introduce a force-first framework for constructing deep metric-learning objectives. Rather than defining a scalar loss and retrospectively analyzing its gradients, the framework prescribes a conservative embedding-space pair-interaction field and derives the corresponding potential by integration. Under a fixed interaction configuration, the negative gradient of the constructed potential recovers the prescribed field, making the intended directions and magnitudes of embedding-space interactions explicit design quantities.
- We identify and analytically characterize three key force-design principles for deep metric learning: intrinsic hard mining, margin control through anchor-adaptive interaction ranges, and cardinality-weighted force balancing. We translate these principles into quantitative prescriptions for the direction, magnitude, active range, and aggregation of attractive and repulsive interactions, thereby providing a theoretical framework that converts deep metric-learning requirements into integrable conservative force fields and corresponding scalar objectives. PEML is a concrete realization of this framework, obtained by constructing bounded monotone forces that satisfy these prescriptions and integrating them into closed-form pairwise potentials.
- We demonstrate strong cross-regime effectiveness of PEML across balanced representation learning, severe class-frequency imbalance, and class-disjoint retrieval. Among the compared objectives, PEML achieves the highest observed frozen-encoder linear-evaluation accuracy on CIFAR-10, CIFAR-100, and ImageNet-100; maintains the highest accuracy at every evaluated CIFAR100-LT imbalance ratio in a comparison including the representation objective from ProCo, exceeding SupCon by 22.11 percentage points at an imbalance ratio of 200; and records the highest observed Recall@K at every reported depth on both CUB-200-2011 and Cars196. These results establish strong class-level discrimination, pronounced robustness to cardinality skew, and effective transfer to categories held out from benchmark-specific training.
- We introduce ASP as a bounded, isotropic relaxation of ℓ_2 normalization that preserves radial variation in Euclidean embedding spaces. Its consistent improvements for PEML and four other Euclidean-distance-based objectives demonstrate that it provides a reusable output parameterization rather than a mechanism specialized to the proposed loss.
- We establish that PEML's empirical gains are accompanied by coherent, imbalance-resilient representation geometry without substantial additional computational cost. Across standard representation analyzes and controlled imbalance settings, PEML produces more class-consistent clustering and more favorable centroid-relative organization than SupCon, with substantially better preservation of this structure under severe cardinality skew. Mechanistic ablations support the roles of monotone hardness-dependent interactions and cardinality-weighted force balancing, while the alternative-kernel analysis demonstrates that the force-based construction extends beyond the default force profile. These benefits are obtained with the same asymptotic loss-computation time and auxiliary-memory

complexity as SupCon and only marginal measured runtime and VRAM overhead.

2. Related work and preliminaries

2.1. Losses for deep metric learning

Deep metric learning objectives can be organized along several complementary axes. We consider how embedding interactions are constructed and normalized, the geometry and similarity measure through which they act, and the degree to which the training objective is aligned with downstream retrieval or verification criteria. These perspectives are not mutually exclusive; together, they provide a basis for locating the formulation developed later within the broader landscape of metric-learning losses.

2.1.1. Interaction structure and normalization

From an interaction-structure perspective, metric-learning objectives can first be classified into pairwise and batch-wide normalized constructions. Pairwise objectives are assembled from explicit terms over labeled sample pairs, following the classical formulation (Chopra et al., 2005; Hadsell et al., 2006). This local construction naturally expresses distance- or margin-based constraints and supports flexible relation weighting and hard-example mining, including tuple-based extensions such as triplet learning (Schroff et al., 2015). By contrast, batch-wide normalized objectives employ log-softmax (cross-entropy) normalization over a candidate set, typically the mini-batch, as in InfoNCE/NT-Xent and supervised batch-contrastive objectives (Chen et al., 2020; Khosla et al., 2020; van den Oord et al., 2018). This form admits a probabilistic interpretation as discriminating positives from competing candidates, closely relates to noise-contrastive estimation, and efficiently reuses many negatives with well-scaled gradients (Gutmann & Hyvärinen, 2010). Its normalization term, however, couples the candidates within a mini-batch rather than specifying their contributions independently.

Sample–proxy objectives further broaden this axis by changing both the entities that interact and the way their relations are aggregated. SoftTriple represents each class using multiple learnable centers and forms the sample-to-class similarity through a soft assignment over those centers, thereby modeling intra-class variation without explicit triplet sampling (Qian et al., 2019). Proxy-AN organizes positive relations in a proxy-centric manner and negative relations in a sample-centric manner, whereas PD-Loss constructs batch-wide genuine and impostor distributions from sample-to-proxy similarities (Peng et al., 2026; Silva et al., 2025). Within this taxonomy, the formulation developed in this work lies on the explicit pairwise-interaction side, which permits the contribution of each labeled pair to the embedding update to be stated directly.

2.1.2. Embedding geometry and similarity metrics

Contrastive objectives can also be grouped by the geometry and similarity metric imposed on the embedding space. A classical line of metric-learning work defines interactions directly in Euclidean space through objectives that act on explicit ℓ_2 distances, including pairwise contrastive losses and their margin-based variants (Chopra et al., 2005; Hadsell et al., 2006), as well as triplet and neighborhood-based objectives (Goldberger et al., 2004; Schroff et al., 2015; Weinberger & Saul, 2009). In these methods, attraction and repulsion are defined in terms of inter-sample distances, providing a direct mechanism for shaping the embedding space. By contrast, modern contrastive frameworks such as SimCLR and SupCon adopt an angular view by ℓ_2-normalizing embeddings and operating on cosine similarity with temperature scaling and softmax-based objectives (Chen et al., 2020; Khosla et al., 2020). Hyperspherical normalization yields scale-invariant comparisons and can improve discrimination, as reflected in angular-margin formulations and ℓ_2-normalized representations (Deng et al., 2019; Liu et al., 2017; Wang et al., 2017). Although cosine similarity can be expressed in terms of Euclidean distance for normalized embeddings, this construction explicitly constrains the norm and prioritizes angular separation on the hypersphere.

Recent methods broaden this geometric axis beyond Euclidean and fixed-radius hyperspherical embeddings. Deep metric learning in projected-hypersphere space uses a positive-curvature projected hypersphere, whereas mixed-curvature metric learning combines positive-, zero-, and negative-curvature components for image retrieval (Xu et al., 2025, 2026). Hierarchical ranking in hyperbolic space further couples a non-Euclidean representation of latent hierarchy with order-aware metric supervision (Zhang & Li, 2026). The present formulation uses Euclidean-distance-based interactions, which provide a direct coordinate-space basis for the force–potential construction developed later.

2.1.3. Alignment with retrieval and verification criteria

A third, orthogonal axis concerns how directly the training objective is aligned with a downstream retrieval or verification criterion. Numerous metric-learning objectives form discriminative embedding geometry without explicitly approximating a particular retrieval statistic; a complementary line couples learning more directly to retrieval order or evaluation criteria. In person re-identification, parameterized retrieval and verification loss search constructs searchable surrogates for otherwise non-differentiable evaluation operations, whereas Rank-in-Rank Loss optimizes relative-order consistency within the embedding space (Xu et al., 2022; Yuan et al., 2023b). For image retrieval, OSAP-Loss constructs an average-precision-oriented surrogate that also accounts for samples ranked after positive items, while the rank-loss framework of Ramzi et al. supports objectives aligned with average precision, Recall@K, and hierarchical ranking criteria (Ramzi et al., 2025; Yuan et al., 2023a).

The formulation developed in this work is not constructed as a surrogate for a particular retrieval or verification statistic. Instead, it starts from the pairwise interactions used to organize labeled embeddings; criterion-aligned optimization and interaction-level loss design therefore provide complementary perspectives.

2.2. Energy and potential-field formulations for deep metric learning

Energy-based learning associates a scalar energy with a configuration of variables and learns an energy function that assigns lower values to task-compatible configurations than to incompatible ones (LeCun et al., 2007). Classical contrastive metric learning fits this general perspective: inter-embedding distance serves as a compatibility energy, and the objective lowers it for similar pairs while enforcing a margin for dissimilar pairs (Chopra et al., 2005; Hadsell et al., 2006).

Potential-field formulations make the spatial structure of this energy explicit in deep metric learning. PFML superposes distance-decaying attractive and repulsive potential fields induced by embeddings and proxies and minimizes the resulting total potential energy (Bhatnagar & Ahuja, 2025). NPFML extends this line through direction-sensitive, non-isotropic potential fields and hierarchical decay (Li et al., 2026). These methods make the force–potential view of embedding organization explicit and thus constitute the closest conceptual line to the present formulation. In our formulation, the design direction is force-first: a desired conservative pairwise interaction field is specified before the corresponding scalar potential is derived by integration. This construction leads naturally to the Lagrangian-mechanics perspective reviewed next and is formalized in Section 3.1.

2.3. Lagrangian mechanics

Lagrangian mechanics has been used in machine learning mainly as a variational tool for specifying or analyzing dynamics. In learned

dynamical-system models, methods such as Lagrangian neural formulations parameterize a scalar Lagrangian and recover equations of motion through the Euler–Lagrange conditions, often with additional structure such as constraints or structure-preserving discretizations (Cranmer et al., 2020; Finzi et al., 2020; Lutter et al., 2019; Saemundsson et al., 2020). Lagrangian viewpoints have also been applied to optimization by interpreting accelerated methods through continuous-time variational principles whose stationary conditions yield Euler–Lagrange dynamics (Su et al., 2016; Wibisono et al., 2016), and to training principles that regularize learning via action-like functionals over parameter trajectories (Karkar et al., 2021). Finally, in geometric metric learning, calculus-of-variations arguments characterize geodesic distances on manifolds, which can be used as the distance primitive for downstream tasks (Hauberg et al., 2012). Taken together, prior work uses Lagrangian mechanics either as an inductive bias for learning physical dynamical systems or as a variational lens for algorithmic dynamics in optimization and related geometric constructions. Against this background, our work adopts a Lagrangian-inspired, force-first construction for deep metric learning: prescribed conservative inter-embedding force laws determine the corresponding potential-energy objective.

3. Method

3.1. Force-based loss derivation

This section develops the theoretical foundations of force-based loss derivation. Although rationales for gradient analysis vary, the following update mechanism lies at their core.

$$f(s) \leftarrow f(s) - \eta \frac{\partial \mathcal{L}(D(s))}{\partial f(s)}, \tag{1}$$

where $f : S \to \mathbb{R}^{d_{\text{emb}}}$ denotes the embedding map, $s \in S$ is an input sample, $\eta > 0$ is the step size, and $\mathcal{L}$ is a scalar loss constructed from the embedding-space distances $D(s) = \{d(f(s), f(q)) \mid q \in S \setminus \{s\}\}$. Although this abstraction accommodates differentiable distances more generally, we use the Euclidean distance, which is standard in deep metric learning (Ghojogh et al., 2022). We refer to Eq. (1) as the Deep Metric-Learning Scheme (DMLS). It specifies the negative-gradient displacement prescribed by the loss when the embeddings are treated directly as optimization variables. In network training, however, the embeddings are coupled through shared model parameters, and the induced embedding displacement is transformed by the local network Jacobian.

To characterize this relation for a pairwise objective, we make the parameter dependence explicit by writing the embedding map as $f(\cdot; w)$, where $w \in \mathbb{R}^m$. For $s, q \in S$, let $a(w) = f(s; w)$ and $t(w) = f(q; w)$ belong to $\mathbb{R}^{d_{\text{emb}}}$. Consider a differentiable scalar pair objective $\ell : \mathbb{R}^{d_{\text{emb}}} \times \mathbb{R}^{d_{\text{emb}}} \to \mathbb{R}$, and define its parameter-space composition as $\widetilde{\ell}_{s,q}(w) = \ell(a(w), t(w))$. Treating all gradients as column vectors, define

$$\mathbf{z}_{s,q}(w) = \begin{pmatrix} a(w) \\ t(w) \end{pmatrix} \in \mathbb{R}^{2d_{\text{emb}}}, \; J_{s,q}(w) = D_w \mathbf{z}_{s,q}(w) \in \mathbb{R}^{2d_{\text{emb}} \times m}, \tag{2}$$

$$\mathbf{g}_{s,q} = \nabla_{\mathbf{z}_{s,q}} \ell(a,t) \in \mathbb{R}^{2d_{\text{emb}}}. \tag{3}$$

For one direct, unpreconditioned parameter-gradient step, let

$$w^+ = w - \eta \nabla_w \widetilde{\ell}_{s,q}(w), \quad \Delta \mathbf{z}_{\text{DMLS}} = -\eta \mathbf{g}_{s,q}, \tag{4}$$

$$\Delta \mathbf{z}_{\text{param}} = \mathbf{z}_{s,q}(w^+) - \mathbf{z}_{s,q}(w). \tag{5}$$

The following proposition states the resulting first-order correspondence.

Proposition 1. *At any parameter point where $\mathbf{z}_{s,q}$ is Fréchet differentiable, the matrix $J_{s,q}(w)J_{s,q}(w)^\top$ is symmetric positive semidefinite, and the parameter-induced embedding displacement satisfies*

$$\Delta \mathbf{z}_{\text{param}} = J_{s,q}(w) J_{s,q}(w)^\top \Delta \mathbf{z}_{\text{DMLS}} + \mathbf{r}_\eta, \qquad \frac{\|\mathbf{r}_\eta\|_2}{\eta} \longrightarrow 0, \tag{6}$$

$$\mathbf{g}_{s,q}^\top \Delta \mathbf{z}_{\text{param}} = -\eta \left\| J_{s,q}(w)^\top \mathbf{g}_{s,q} \right\|_2^2 + o(\eta) \qquad \text{as } \eta \to 0^+. \tag{7}$$

Thus, parameter optimization induces a positive-semidefinite Jacobian–Gram-transformed first-order counterpart of the direct DMLS update, rather than an unconditional identity with it. If $J_{s,q}(w)^\top \mathbf{g}_{s,q} \neq \mathbf{0}$, then $\widetilde{\ell}_{s,q}(w^+) < \widetilde{\ell}_{s,q}(w)$ for every sufficiently small positive step size.[1]

For the Euclidean pair objective $\ell(a,t) = \mathcal{L}(d_{at})$, where $d_{at} = \|a - t\|_2$, suppose that $a \neq t$ and that $\mathcal{L}$ is differentiable at d_{at}. Direct differentiation then gives $\nabla_t \ell(a,t) = -\nabla_a \ell(a,t)$. With $J_s(w) = D_w a(w)$ and $J_q(w) = D_w t(w)$, the anchor block of Eq. (6) becomes

$$a(w^+) = a(w) - \eta M_{s,q}(w) \nabla_a \ell(a,t) + \mathbf{r}_{a,\eta}, \qquad \frac{\|\mathbf{r}_{a,\eta}\|_2}{\eta} \longrightarrow 0, \tag{8}$$

$$M_{s,q}(w) = J_s(w)\big(J_s(w) - J_q(w)\big)^\top \in \mathbb{R}^{d_{\text{emb}} \times d_{\text{emb}}}. \tag{9}$$

Unlike the joint Jacobian–Gram operator, $M_{s,q}(w)$ need not be symmetric or positive semidefinite and therefore need not preserve the direct DMLS direction. For $\nabla_a \ell(a,t) \neq \mathbf{0}$, its leading-order anchor displacement agrees in direction with the direct DMLS displacement precisely when

$$M_{s,q}(w) \nabla_a \ell(a,t) = \lambda \nabla_a \ell(a,t) \qquad \text{for some } \lambda > 0. \tag{10}$$

When $d_{\text{emb}} = 1$, the anchor operator reduces to the scalar

$$\zeta_{s,q}(w) = J_s(w)\big(J_s(w) - J_q(w)\big)^\top, \tag{11}$$

and the anchor update takes the compact form,

$$a(w^+) = a(w) - \eta\, \zeta_{s,q}(w) \frac{\partial \mathcal{L}(d_{at})}{\partial a} + r_{a,\eta}, \qquad r_{a,\eta} = o(\eta). \tag{12}$$

Thus, the original scalar expression is recovered as a special case. For a nonzero anchor gradient, its leading-order displacement agrees in direction with the direct DMLS displacement precisely when $\zeta_{s,q}(w) > 0$, rather than merely when $\zeta_{s,q}(w) \neq 0$.

For a complete mini-batch objective, the same relation holds after stacking all embeddings entering the objective and using the Jacobian of the resulting stacked vector. This result is local to the current parameter point and concerns a direct, unpreconditioned parameter-gradient step; in stochastic training, it is conditional on the current mini-batch. The effects of momentum and adaptive optimization on the induced trajectory are discussed separately in Section I.5.

Suppose a target gradient field over the representation space is specified. If embeddings are updated along this field at every training step, will the resulting final distribution satisfy the training objective? If so, how can one construct a loss whose gradient reproduces this field? To answer these questions, we select a well-established physical model from mechanics and demonstrate that its governing theory can be applied to gradient-based updates to establish the result.

Physical principles have been widely adopted in machine learning. For example, cross-entropy admits a statistical-mechanical interpretation through the Boltzmann–Gibbs framework (Goodfellow et al., 2016; Jaynes, 1957; LeCun et al., 2007), and diffusion models are grounded in stochastic dynamics and nonequilibrium thermodynamics (Ho et al., 2020; Sohl-Dickstein et al., 2015). In the same spirit, Lagrangian mechanics provides a compact description of conservative interactions through scalar potentials, and we adopt the Lagrangian viewpoint to analyze and model the update dynamics of high-dimensional embedding variables. Specifically, we treat the collection of embeddings as a multibody-like system evolving under loss-induced interactions, building on the classical Lagrangian formalism to derive a force–potential relation suited to embedding evolution under gradient-based updates.

Let $x_i, x_j \in \mathbb{R}^d$ ($d \geq 1$) be the positions of two pointlike charges with labels c_i, c_j. Let $r = x_i - x_j$ and $\hat{r} = r/\|r\|$. Then, we specify the force on

[1] A complete vector-valued proof, including the explicit Jacobian chain rule, the higher-order Taylor remainder, the scalar special case, and the mini-batch extension, is provided in Section A.1.

i due to j as a central field:

$$F_{ij} = \begin{cases} -g(\|r\|)\hat{r} & \text{for } c_i = c_j \\ h(\|r\|)\hat{r} & \text{for } c_i \neq c_j \end{cases}, \tag{13}$$

where $g : [0,\infty) \to \mathbb{R}_{\geq 0}$ and $h : (0,\infty) \to \mathbb{R}_{\geq 0}$ are smooth. Thus, for the same labels, F_{ij} is an attractive force on i pointing from x_i toward x_j. When the labels differ, the force is repulsive. Assuming that $g \in C^1([0,\infty))$ and $h \in C^1((0,\infty))$ have no singularities, F_{ij} is a smooth central field on the non-collision domain $\{r \neq 0\} \subset \mathbb{R}^d$. Hence, F_{ij} is a conservative field (Spivak, 2018). Accordingly, F_{ij} can be treated as a generalized force in the Lagrangian framework (Goldstein et al., 2011). This implies the existence of a scalar potential-energy function V satisfying

$$F = -\nabla_x V. \tag{14}$$

We next compute the potential energy associated with a directed pair interaction. Let $A = ((x_i)_A, (x_j)_A)$ and $B = ((x_i)_B, (x_j)_B)$ denote two configurations. For the contribution indexed by (i,j), let $F_{ij}^{pair} = -F_{ij}$ be the equal-and-opposite force associated with the j coordinate. This paired force is distinct from the separately indexed reverse interaction F_{ji}; when the interaction parameters depend on the anchor, F_{ij}^{pair} need not equal F_{ji}, and the directed potential energies V_{ij} and V_{ji} need not coincide. The change in potential energy between the two configurations is the negative work performed by the interaction:

$$V_{ij}(B) - V_{ij}(A) = -\int_A^B \left(F_{ij} \cdot dx_i + F_{ij}^{pair} \cdot dx_j \right). \tag{15}$$

Because the central field is conservative on its stated domain, the potential-energy difference in Eq. (15) depends only on the fixed endpoints A and B, not on the path connecting them (Goldstein et al., 2011). The potential-energy function is determined up to an additive constant, which is fixed by assigning a value at a selected reference configuration.

Let $d_A = \|(x_i)_A - (x_j)_A\|$ and $d_B = \|(x_i)_B - (x_j)_B\|$ denote the scalar pair separations at the two endpoints. Since $F_{ij}^{pair} = -F_{ij}$ and $dr = dx_i - dx_j$, Eq. (15) reduces to an integral of the radial force component. For a pair with the same label, Eq. (13) gives

$$\begin{aligned} V_{ij}^{pos}(B) - V_{ij}^{pos}(A) &= -\int_{d_A}^{d_B} \big(-g(u)\big)\, du \\ &= \int_{d_A}^{d_B} g(u)\, du \\ &= G(d_B) - G(d_A). \end{aligned} \tag{16}$$

Similarly, for a pair with different labels,

$$\begin{aligned} V_{ij}^{neg}(B) - V_{ij}^{neg}(A) &= -\int_{d_A}^{d_B} h(u)\, du \\ &= H(d_A) - H(d_B), \end{aligned} \tag{17}$$

where G and H are antiderivatives of g and h, respectively, i.e., $G'(u) = g(u)$ and $H'(u) = h(u)$. After fixing $V_{ij}(A)$, we write V_{ij} for the pairwise potential energy at configuration B.

Finally, we define the system's total potential energy

$$V_{\text{total}} = \sum_{i \neq j} V_{ij}. \tag{18}$$

Note that in this setting $V_{ij} \neq V_{ji}$, so the directed interaction is asymmetric and no factor of $1/2$ is introduced, unlike in symmetric pair-potential formulations in classical mechanics. The relevant descent property does not require pairwise action–reaction symmetry. Once the directed pair potentials are collected into the scalar total potential V_{total}, an update along $-\nabla_X V_{\text{total}}$, where $X = (x_1, \ldots, x_N)$, decreases V_{total} under the usual local smoothness and sufficiently small step-size condition. This remains true even when $V_{ij} \neq V_{ji}$, because V_{ij} and V_{ji} are treated as distinct directed contributions to the scalar objective. The corresponding descent argument is provided in Section A.3. Consequently, equilibrium configurations are stationary points of V_{total}, while local minima are candidate stable equilibria under additional regularity conditions. The same picture applies to metric learning: the generalized coordinate x corresponds to an embedding, the potential energy V_{total} to the loss, and the force to the gradient used during training updates. The following proposition formalizes this correspondence.

Proposition 2. *Under the stated idealized short-time approximation of uniform force, the potential-induced coordinate displacement and the DMLS embedding update share the same negative-gradient form.*[2]

The mechanics correspondence provides a principled framework for loss construction and interpretation. For a prescribed conservative field with a fixed interaction configuration, the force–potential relation in Eq. (14) is exact: taking the negative gradient of the constructed scalar potential recovers the prescribed force field. Under the idealized discretization underlying Proposition 2, the potential-induced coordinate displacement and the DMLS embedding update further share the same negative-gradient form, thereby providing a formal mechanical interpretation of their common update structure. The correspondence thus formalizes the loss construction and the shared update structure. The optimization analysis then proceeds directly in the discrete setting: descent for a fixed scalar potential is established in Section A.3, while the PEML-specific fixed-threshold descent and detached-threshold one-step surrogate descent results are provided in Sections I.3 and I.4, respectively. The optimizer-dependent scope of these descent results, including the effects of momentum and adaptive optimization on the induced trajectory, is discussed separately in Section I.5. Accordingly, rather than defining the loss directly, we first specify the target force field and derive the corresponding potential energy, which is then used as the loss.

3.2. Properties of forces

In the previous section, we examined the role of forces in deep metric learning and showed how to derive a loss from a prescribed force field. In this section, we ask what properties the force must satisfy to meet deep metric learning objectives. From a broad review of prior works, together with an analytical characterization of the resulting update dynamics, we identify three key principles: intrinsic hard mining, the margin constraint, and cardinality-weighted force balancing. Then, these principles are translated into quantitative prescriptions for the loss.

3.2.1. Intrinsic hard mining

In deep metric learning, hard mining is the practice of prioritizing hard pairs or triplets during training to obtain higher-quality embeddings (Wu et al., 2017). Specifically, a hard positive is a same-class pair that lies far apart in the current embedding space, whereas a hard negative is a different-class pair that lies close. Although we do not provide a formal analytical proof of the effectiveness of hard mining in deep metric learning, its efficacy has been demonstrated empirically in numerous studies. In particular, works by Harwood et al. (2017) and Zhai et al. (2023) present experimental evidence that prioritizing hard pairs improves performance.

Margin-based losses such as triplet loss (Schroff et al., 2015) require explicit hard mining because each anchor interacts with only a single or a small number of contrastive pairs. However, this requirement increases computational cost. Naive hard mining scales as $\mathcal{O}(N^2)$ in the number of samples N (Harwood et al., 2017). This quadratic scaling implies that mining time grows rapidly with dataset size. One approach is to reduce the cost of the mining stage using techniques such as reweighting (Wang et al., 2019; Wu et al., 2017). However, a more appealing alternative is to employ losses whose optimization naturally emphasizes hard examples, thereby avoiding explicit mining. Softmax-based metric losses, including N-pair loss (Sohn, 2016), InfoNCE, and SupCon loss, consider all positives and negatives within a mini-batch and thereby

[2] The mathematical proof of the proposition is provided in Section A.2.

aggregate their contributions during training. This in-batch aggregation often acts as an implicit form of hard example emphasis through the softmax weighting. Wang and Liu (2021) and Khosla et al. (2020) demonstrate this intrinsic hard mining ability of the loss with ℓ_2 normalization via gradient analysis. Since the gradient can be regarded as a force, an intrinsic hard mining capability is necessary for a metric loss.

Then, what properties must the force satisfy to realize intrinsic hard mining? Harder negatives, those closer to the anchor, should be prioritized. Therefore, the repulsive force should be larger at smaller distances and decrease monotonically as the distance increases. Conversely, harder positives, those farther from the anchor, should be emphasized. Hence, the attractive force should increase monotonically with distance. Equivalently, letting d and $F(d)$ denote pairwise distance and the corresponding force, the magnitudes satisfy:

$$\text{Repulsive force: } d_1 < d_2 \rightarrow |F(d_1)| \geq |F(d_2)|\,, \tag{19}$$

$$\text{Attractive force: } d_1 < d_2 \rightarrow |F(d_1)| \leq |F(d_2)|\,. \tag{20}$$

Softmax-based metric losses possess intrinsic hard mining, but their per-pair tangential contributions do not satisfy Eq. (19). For a fixed ℓ_2-normalized anchor a and current mini-batch $\mathcal{A}$, let

$$P(a) = \{t \in \mathcal{A} \setminus \{a\} : y_t = y_a\}, \tag{21}$$

$$N(a) = \{t \in \mathcal{A} \setminus \{a\} : y_t \neq y_a\}, \tag{22}$$

$$A(a) = P(a) \cup N(a), \tag{23}$$

and define

$$\pi_a(t) = \frac{\exp(a^\top t/\tau)}{\sum_{u \in A(a)} \exp(a^\top u/\tau)}, \qquad \theta_{at} = \arccos(a^\top t). \tag{24}$$

The magnitudes of the individual positive- and negative-pair contributions to the tangential component of the per-anchor gradient are

$$\mu_p(a, p; \mathcal{A}) = \left|\pi_a(p) - \frac{1}{|P(a)|}\right| \sin\theta_{ap}\,, \tag{25}$$

$$\mu_n(a, n; \mathcal{A}) = \pi_a(n) \sin\theta_{an}\,. \tag{26}$$

These quantities are deterministic once the anchor and current mini-batch are fixed; no expectation is taken over the data distribution, mini-batch sampling distribution, or embedding-direction distribution. When $a = \alpha/\|\alpha\|_2$, the corresponding terms in $\partial\ell/\partial\alpha$ additionally contain the common positive factor $1/(\tau\|\alpha\|_2)$. In the many-negative regime considered here, under the condition $\pi_a(p) \leq 1/|P(a)|$, the positive-pair contribution is nondecreasing with angular separation on $[0, \pi/2]$. The negative-pair contribution is non-monotonic and diminishes as the pair angle approaches zero. Fig. 1 illustrates these two behaviors. A rigorous derivation is provided in Section B. Softmax-based metric losses cast representation learning as an N-way identification problem: given an anchor a and a candidate set $\mathcal{I} = \{l_1, \ldots, l_N\}$, model $p_\theta(j|a, \mathcal{I}) = \exp(s_\theta(a, l_j)/\tau)/\sum_{k=1}^{N} exp(s_\theta(a, l_k)/\tau)$ and train by maximum likelihood (cross-entropy). Here, $j \in \{1, \ldots, N\}$ indexes the candidate containing the true positive, $s_\theta(\cdot,\cdot)$ denotes a similarity score such as the dot product on ℓ_2-normalized embeddings, and τ is a positive temperature (Gutmann & Hyvärinen, 2012; van den Oord et al., 2018). Consequently, although softmax-based metric losses exhibit some intrinsic hard mining, they are grounded in a probabilistic modeling perspective rather than in a formulation that applies similarity-induced forces in the embedding space. As a result, they retain the hard mining limitations discussed above. Therefore, to achieve fully intrinsic hard mining, we introduce a loss that is not based on softmax.

3.2.2. Implementation of the margin constraint

We investigate the effective interaction range of the force and how it is induced by the margin constraint. Margin-based losses impose a margin constraint for an anchor a:

$$d_{an} \geq d_{ap} + \epsilon, \quad \forall p \in P(a), \forall n \in N(a)\,, \tag{27}$$

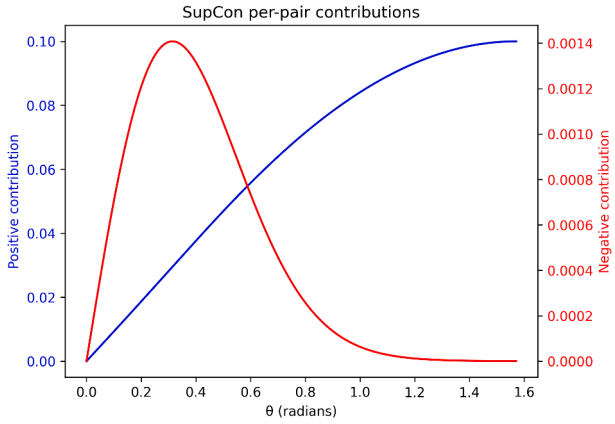


Fig. 1. Per-pair tangential contribution magnitudes induced by the SupCon loss as functions of angular separation θ. Under the stated condition $\pi_a(\theta) \leq 1/|P(a)|$, the positive-pair contribution is nondecreasing, whereas the negative-pair contribution is non-monotonic (note the different y-axis scales).

where $\epsilon > 0$ is the margin and $N(a) = \{t \in \mathcal{A} : y_t \neq y_a\}$. The triplet loss realizes this constraint by penalizing violations (Schroff et al., 2015). The margin perspective also connects to softmax-based metric losses. For example, Barbano et al. (2023) derive InfoNCE from the margin constraint.

Mutual information (MI) between random variables Z and Y is $I(Z, Y)$:

$$I(Z, Y) = \mathbb{E}_{p(z,y)}\left[log\,\frac{p(z, y)}{p(z)p(y)}\right] = H(Z) - H(Z|Y) \geq 0\,, \tag{28}$$

where H denotes entropy, $z \in Z$, and $y \in Y$ (Cover & Thomas, 2006). In our setting, the relevant quantity is $I(X, X^+)$ where X denotes an anchor and X^+ is its positive. Numerous studies have linked this margin viewpoint to MI. Boudiaf et al. (2020) show that margin-based losses can be interpreted as MI-aligned surrogates. In the case of softmax-based metric losses, Poole et al. (2019) derive InfoNCE as a lower bound and a leave-one-out (InfoL1O) upper bound on MI, yielding a double inequality (sandwich). Moreover, Barbano et al. (2023) show that an ϵ-margin formulation of supervised contrastive loss interpolates between InfoNCE($\epsilon = 0$) and InfoL1O($\epsilon \rightarrow \infty$), so tuning ϵ can tighten the approximation to MI.

Achieving a tighter MI approximation can improve optimization. First, a model becomes more robust to dataset bias. Barbano et al. (2023) report performance gains on both generic and biased datasets. Second, the tighter MI approximation can enhance optimization stability. As shown in Poole et al. (2019), different MI limits trade bias for gradient variance, and adopting ϵ allows selecting a point along this spectrum (Barbano et al., 2023), which produces a more consistent optimization. Consequently, introducing well-tuned margins into both margin-based and softmax-based losses is theoretically linked to MI and, in practice, can improve representation quality and downstream performance.

We now examine how to incorporate the margin constraint into a force-based loss. A canonical mechanical model corresponds to the case $\epsilon = \infty$, in which forces act at any distance. Transposed to the embedding space in deep metric learning, and assuming optimization can drive the loss to its global minimum, each class cluster collapses to a point, after which the class centroids repel one another until they become maximally separated. Although in practice hard mining, as in InfoL1O, prevents class collapse during optimization, the $\epsilon = \infty$ case remains undesirable from the point of view of the training objective. The margin can therefore be realized by imposing a finite interaction range on the force. Eq. (27) is equivalent to the following pair of constraints:

$$d_p^{max}(a) + \epsilon \leq d_{an}, \text{ where } d_p^{max}(a) = \max_p d_{ap}\,, \tag{29}$$

$$d_{ap} \le d_n^{min}(a) - \epsilon, \text{ where } d_n^{min}(a) = \min_n d_{an} . \tag{30}$$

For notational convenience, let

$$k_n(a) = d_p^{max}(a) + \epsilon , \tag{31}$$

$$k_p(a) = d_n^{min}(a) - \epsilon . \tag{32}$$

By Eq. (29), the repulsive force on a negative should vanish once the distance exceeds $k_n(a)$. Then, by Eq. (19) together with Eq. (29), the magnitude of the repulsive force attains its maximum at $d = 0$ and decreases monotonically to zero on $d \in [0, k_n(a)]$, and is zero for $d \ge k_n(a)$. Similarly, by Eq. (30) together with Eq. (20), the attractive force on a positive should be zero for $d \in [0, k_p(a)]$, and then increase monotonically for $d \ge k_p(a)$, pulling only those positives that lie beyond the threshold back toward the anchor. Note that $d_p^{\max}(a)$ and $d_n^{\min}(a)$ depend on the anchor a and on the current embedding distribution. Consequently, unlike the global margin ϵ, the thresholds $k_p(a)$ and $k_n(a)$ are dynamic and vary across anchors and over training. By expressing the forces as functions with dynamic thresholds $k_p(a)$ and $k_n(a)$, the margin constraint can be implemented within the force-based metric loss, ensuring short-range repulsion for negatives and long-range attraction for positives in a manner consistent with the margin principle.

3.2.3. Cardinality-weighted force balancing

To examine the force balancing, we first consider SupCon loss, which includes multiple positives and negatives per anchor. If one sums the per-positive log terms, anchors that have more positives produce gradients whose magnitude scales roughly with $|P(a)|$. Averaging by $1/|P(a)|$ equalizes per-anchor contributions and reduces variance. SupCon loss makes this explicit by contrasting outside-log averaging with inside-log normalization. Placing $1/|P(a)|$ outside the log changes the gradient, whereas moving it inside the log only adds a constant and therefore does not affect gradients, so it cannot debias cardinality. However, Park et al. (2022) show that per-anchor normalization by $1/|P(a)|$ does not correct group-level cardinality imbalance and propose additional group-wise normalizers. Hence, effective force balancing under cardinality requires a group-wise balancing component in addition to per-anchor averaging.

Let $F_P^{tot}(a)$ denote the total force exerted on anchor a by all of its positives in the current batch, and $F_N^{tot}(a)$ denote the total force from all negatives. From a force-based loss perspective, these per-anchor averaging and group-wise balancing operations are represented by

$$F_P^{tot}(a) = \frac{1}{|P(a)|} \sum_{p \in P(a)} F_P(d_{ap}) , \tag{33}$$

$$F_N^{tot}(a) = \frac{1}{|C(a)|} \sum_{c \in C(a)} \left(\frac{1}{|N_c(a)|} \sum_{n \in N_c(a)} F_N(d_{an}) \right) , \tag{34}$$

where $F_P(d_{ap})$ denotes the force exerted on a by its positive pair p that is at distance d_{ap} from a, and $F_N(d_{an})$ denotes the force applied to a due to a negative pair n separated by d_{an}. Also, let $C(a) = \{y_t : t \in \mathcal{A}, y_t \ne y_a\}$ be the set of distinct negative classes in the batch. Then, for each $c \in C(a)$, let $N_c(a) = \{t \in \mathcal{A} : y_t = c\}$ define the class-wise negatives. Because forces are additive, both expressions can be written as summations. Per-anchor averaging is applied via the factors $1/|P(a)|$ and $1/|N_c(a)|$. Group-wise balancing is encoded by the outer factor $1/|C(a)|$. Since there is only one positive class group, no additional group-wise divisor is applied in $F_P^{tot}(a)$.

These cardinality effects are particularly relevant in long-tailed contrastive learning, where class-frequency skew changes the composition and multiplicity of the positive and negative sets associated with an anchor. Existing long-tailed contrastive objectives address these effects through distinct mechanisms involving positive selection, target weighting, parametric class representations, or class-conditional distribution estimation. For example, KCL (Kang et al., 2021) controls positive cardinality by selecting at most k additional same-class positives for each anchor, while retaining a sample-wise negative denominator. DSCL (Xuan & Zhang, 2024) assigns fixed aggregate weights to the paired augmented view and the remaining same-class positives, but likewise retains sample-wise negative aggregation. PaCo (Cui et al., 2021) combines non-parametric sample contrast with learnable class centers and incorporates class-frequency information into the center terms in its long-tailed formulation; its sample-negative component is not explicitly normalized by class. ProCo (Du et al., 2024) replaces finite contrast sets with prior-weighted expectations under online class-conditional feature distributions, addressing finite-pair scarcity through distribution estimation rather than equal-weight aggregation of finite sample-wise negative contributions across classes. Relative to them, the novel aspect of our force balancing is its simultaneous cardinality normalization of finite sample-wise attractive and repulsive forces: attractive contributions are averaged over positives, whereas repulsive contributions are averaged first within each negative class and then across the negative classes represented in the batch. Consequently, the number of observed samples from a negative class does not directly scale that class's total repulsive contribution. A detailed objective-level comparison is provided in Section E.

3.3. Annulus-scaled projection

In this section, we analyze the role of the projection head with a focus on its output mapping, discuss its limitations, and introduce a projection tailored to our loss formulation as its replacement. The projection head, though discarded for downstream tasks, confers several benefits during training. It improves the quality of learned representations (Chen et al., 2020), enhances generalization (Gupta et al., 2022), and increases robustness to distribution shift (Xue et al., 2024). Applying ℓ_2 normalization to the projection head's outputs constrains embeddings to lie on the unit hypersphere, thereby bounding the logits and improving numerical stability (Deng et al., 2019; Wang et al., 2017). In particular, ℓ_2 normalization is typically required with cosine-similarity-based metric losses to impart intrinsic hard-mining behavior to the objective, as shown in Eq. (25) and Eq. (26). However, we do not adopt ℓ_2 normalization in our loss formulation. Our approach relies on Euclidean distance, while ℓ_2 normalization eliminates radial variation. Once embeddings are normalized to the unit hypersphere, updates are confined to the tangent space, so the induced forces are purely rotational. To illustrate, let α be the pre-normalized embedding of the normalized anchor a, where $a = \alpha/\|\alpha\|$. Then,

$$\frac{\partial a}{\partial \alpha} = \frac{I - aa^\top}{\|\alpha\|} , \tag{35}$$

For the anchor a, its paired embedding t, and any loss L,

$$\nabla_a L(\|a - t\|) = L'(\|a - t\|) \frac{a - t}{\|a - t\|} . \tag{36}$$

By the chain rule, $\nabla_\alpha L(\|a - t\|) = (\partial a/\partial \alpha) \nabla_a L(\|a - t\|)$. Taking the dot product with α and using $a^\top (I - aa^\top) = 0$ yields

$$\begin{aligned} \alpha^\top \nabla_\alpha L(\|a - t\|) &= \frac{L'(\|a - t\|)}{\|a - t\|} a^\top (I - aa^\top)(a - t) \\ &= 0 . \end{aligned} \tag{37}$$

Hence, $\alpha \perp \nabla_\alpha L$; the update therefore has no radial component. This fails to fully realize the fundamental objective of Euclidean-distance–based deep metric-learning losses, namely to decrease the distances between positive pairs while increasing those between negative pairs. To address these issues, we propose an annulus-scaled projection as a replacement for ℓ_2 normalization. For an embedding x, define

$$\psi(x) = \left(R_A + (R_B - R_A) \sigma\left(\frac{z}{\tau_r} \right) \right) \frac{x}{\|x\|}, \quad z = \frac{\|x\| - \mu}{s} , \tag{38}$$

where $0 < R_A < R_B$ are the inner and outer radii of the target annulus, $\tau_r > 0$ denotes the radius temperature, μ and s are the mini-batch mean and standard deviation of $\|x\|$, and $\sigma(u) = 1/(1 + e^{-u})$ is the logistic sigmoid. Thus, ψ preserves direction and maps x into the open

annulus $\{ y \in \mathbb{R}^d : R_A < \|y\| < R_B \}$. This mapping retains the numerical stability benefits of ℓ_2 normalization by bounding the output norm while preserving radial variation. In Eq. (38), we use a sigmoid-based radial mapping rather than linear rescaling to $[0, 1]$. The sigmoid soft-clips extreme values, limiting the influence of outliers. Because the sigmoid saturates in the tails, the effective scaling is less sensitive to batch-to-batch variance than a linear min–max map, which reduces scale jitter and gradient variance across iterations.

Although annulus-scaled projection imposes a bounded geometric constraint, it should be understood as a relaxation of the single-radius hyperspherical constraint rather than as an unconstrained mapping. This design is particularly well suited to the bounded Euclidean metric-learning setting considered in this work. In applications where embedding magnitudes are intentionally left unbounded or are themselves used to encode semantic information, a different radial parameterization may be more appropriate. Within its intended scope, however, the projection can be applied to other Euclidean-distance-based metric losses without altering their underlying objectives, as demonstrated in the experiments.

The spherical annulus is adopted as the simplest bounded isotropic extension of the single-radius hypersphere. Because the mapping depends only on $\|x\|$ and preserves the direction $x/\|x\|$, it maintains the rotational symmetry of Euclidean distance and remains consistent with the central-force formulation developed in Section 3.1. By contrast, non-isotropic geometric constraints, such as ellipsoidal constraints, introduce direction-dependent scaling and cause the projected geometry and pairwise interactions to depend on orientation as well as distance. Therefore, the spherical annulus provides a simple, interpretable, and parameter-efficient means of retaining bounded radial variation. In this paper, we use the empirically determined setting $(R_A, R_B, \tau_r) = (1, 2, 0.5)$. A detailed analysis of the roles of R_A, R_B, and τ_r, together with their sensitivity, is provided in Section K.

3.4. Loss formulation

In this section, we formalize the force field and derive the corresponding loss function. As a first step, we adapt the parameters $k_p(a)$ and $k_n(a)$ introduced in Section 3.2.2 to the force formulation. Under the annulus-scaled projection, the projected embeddings satisfy $R_A < \|\psi(x)\|_2 < R_B$. In what follows, a, p, and n denote post-projection embeddings, with $d_{ap} = \|a - p\|_2$ and $d_{an} = \|a - n\|_2$. Hence, these pairwise distances obey

$$d_{ap}, d_{an} \in [0, d_{max}] , \tag{39}$$

where $d_{\max} = 2R_B$ in this case. Additive margins in Eqs. (31) and (32) may violate the feasible range $[0, d_{\max}]$ for $k_p(a)$ and $k_n(a)$ when ϵ is chosen without regard to $d_p^{\max}(a)$ and $d_n^{\min}(a)$. To obtain range-preserving, anchor-adaptive thresholds, we replace the additive margin with multiplicative, dimensionless factors $\rho_p \in [0, 1]$ and $\rho_n \in [0, 1)$:

$$k_p(a) = \rho_p \, d_n^{min}(a) , \tag{40}$$

$$k_n(a) = d_{max} - \rho_n (d_{max} - d_p^{max}(a)) . \tag{41}$$

Note that, at each optimization step, $k_p(a)$ and $k_n(a)$ are recomputed from the current mini-batch configuration and then held fixed during differentiation. By construction, $k_p(a) \in [0, d_n^{\min}(a)]$ and $k_n(a) \in (d_p^{\max}(a), d_{\max}]$. Geometrically, ρ_p sets $k_p(a)$ to a fixed fraction of the gap from the anchor up to the nearest negative $d_n^{\min}(a)$, while ρ_n places $k_n(a)$ a fixed fraction inward from the outer bound $d_{\max}$ toward the farthest positive $d_p^{\max}(a)$. This parameterization yields bounded, scale-aware margins that adapt to the local configuration around each anchor. In this paper, we use empirically determined settings of $(\rho_p, \rho_n) = (0.5, 0.4)$ for datasets with a large number of classes, such as CIFAR-100, and $(\rho_p, \rho_n) = (0.1, 0.2)$ for CIFAR-10.

For numerical regularity, each pairwise force magnitude must remain finite. To place the per-pair contributions entering the cardinality-weighted aggregation in Eqs. (33) and (34) on a common normalized scale, we use the same lower bound, upper bound, and endpoint normalization across anchors. As discussed in Section 3.2.1, the force magnitude should be monotonic with pair hardness over its active interval. These requirements constrain, but do not uniquely determine, the functional form of the force. We use the squared trigonometric profile as the default. It is bounded, admits a closed-form primitive, and introduces no additional shape parameter. Among the candidates evaluated in this work, it is also the only profile whose scalar force magnitude has zero slope at both endpoints of the normalized active interval. Consequently, its magnitude turns on with zero first-order variation at the inactive boundary and saturates with zero first-order variation at the hardest endpoint. This is a local endpoint property that motivates the design choice rather than a standalone guarantee of optimization stability.

A unified construction for other admissible monotone profiles and their corresponding potentials is provided in Section C, and their controlled empirical comparison is reported in Section 4.6. For an anchor a, we define the force exerted by a paired example as follows:

$$F_P(d_{ap}) = \sin^2 \left(\frac{\pi (d_{ap} - k_p(a))}{2(d_{max} - k_p(a))} \right) \mathbf{1}_{[k_p(a), d_{max}]}(d_{ap}) \hat{r}_{ap} , \tag{42}$$

$$F_N(d_{an}) = \cos^2 \left(\frac{\pi d_{an}}{2k_n(a)} \right) \mathbf{1}_{[0, k_n(a)]}(d_{an}) \hat{r}_{an} , \tag{43}$$

where $\mathbf{1}_{[u_1, u_2]}(u)$ denotes the indicator function of $[u_1, u_2]$: it equals 1 when $u \in [u_1, u_2]$ and 0 otherwise. Following the direction convention for $\hat{r}$ in Eq. (13), we define the pairwise unit directions as $\hat{r}_{ap} = (p - a)/d_{ap}$ and $\hat{r}_{an} = (a - n)/d_{an}$. The indicator terms, parameterized by $k_p(a)$ and $k_n(a)$, enforce the margin constraints by restricting the forces to their respective active intervals. Moreover, squared sine and cosine kernels ensure monotonic behavior of the force over these intervals. To obtain the pairwise potential energies, we choose reference separations at the respective force-inactive ends of the feasible distance range. For the attractive positive branch, Eq. (42) vanishes for $0 \le d_{ap} \le k_p(a)$, and Eq. (16) with $d_A = 0$ and $d_B = d_{ap}$ gives

$$V_P(d_{ap}) = \int_0^{d_{ap}} \|F_P(u)\| \, du. \tag{44}$$

For the repulsive negative branch, Eq. (43) vanishes for $k_n(a) \le d_{an} \le d_{\max}$, and Eq. (17) with $d_A = d_{\max}$ and $d_B = d_{an}$ gives

$$V_N(d_{an}) = \int_{d_{an}}^{d_{max}} \|F_N(u)\| \, du. \tag{45}$$

The opposite reference endpoints reflect attraction at small distances and repulsion at large distances; because the force magnitudes vanish throughout the respective inactive intervals, the resulting potential energies are zero throughout those intervals. Evaluating these work integrals yields the following closed forms:

$$V_P(d_{ap}) = \left(\frac{d_{ap} - k_p(a)}{2} - \frac{d_{max} - k_p(a)}{2\pi} \sin \left(\frac{\pi (d_{ap} - k_p(a))}{d_{max} - k_p(a)} \right) \right) \mathbf{1}_{[k_p(a), d_{max}]}(d_{ap}) , \tag{46}$$

$$V_N(d_{an}) = \left(\frac{k_n(a) - d_{an}}{2} - \frac{k_n(a)}{2\pi} \sin \left(\frac{\pi d_{an}}{k_n(a)} \right) \right) \mathbf{1}_{[0, k_n(a)]}(d_{an}) . \tag{47}$$

Finally, by identifying the constructed potential energy with the loss and using Eq. (18), we obtain the total objective, which we call the potential energy metric loss (PEML):

$$\mathcal{L} = \sum_{a \in \mathcal{A}} \left(\frac{1}{|P(a)|} \sum_{p \in P(a)} V_P(d_{ap}) + \frac{1}{|C(a)|} \sum_{c \in C(a)} \left(\frac{1}{|N_c(a)|} \sum_{n \in N_c(a)} V_N(d_{an}) \right) \right) . \tag{48}$$

A formal convergence and asymptotic-stationarity analysis of the PEML embedding dynamics is provided in Section I.

Table 1
Top-1 classification accuracy (%) on CIFAR-10/100 and ImageNet-100. Columns marked with † are reported from Barbano et al. (2023). Columns marked with ∗ are our reproductions under matched loss-independent training conditions and a common frozen-encoder linear-evaluation protocol; objective-specific output mappings and hyperparameters are retained. All models use a ResNet-50 encoder. The highest observed accuracy in each row is shown in bold.

Dataset	SimCLR†	CE†	SoftTriple∗	SupCon∗	PEML∗
CIFAR-10	91.74	94.73	95.64	95.98	**96.04**
CIFAR-100	68.94	73.43	74.57	76.49	**77.61**
ImageNet-100	66.14	82.10	82.43	83.06	**85.94**

4. Experiments

In this section, we first benchmark our framework on standard vision benchmarks, including CIFAR-10 and CIFAR-100 (Krizhevsky, 2009), as well as ImageNet-100, constructed from ImageNet (Deng et al., 2009) using the 100-class synset list of Barbano et al. (2023). These experiments use frozen-encoder linear evaluation. We next evaluate transfer to classes held out from benchmark-specific training on four standard class-disjoint deep metric-learning retrieval benchmarks under a shared training and evaluation protocol. We then characterize the learned representations through unsupervised clustering, nonparametric k-nearest-neighbor (k-NN) classification, qualitative visualization, and centroid-relative geometry in the original encoder space. Additional controlled studies evaluate the cardinality-weighted force-balancing mechanism and the principal design choices of PEML.

4.1. Image classification

We compare the representation quality of models trained with our proposed loss to baselines by using image classification as a standard downstream task. We follow the original experimental setup of Khosla et al. (2020) and Barbano et al. (2023). Training follows a two-stage protocol. In Stage 1, a ResNet-50 encoder (He et al., 2016) and a two-layer fully connected projection head with a 128-dimensional output are trained from scratch for each reproduced loss–dataset combination. For Stage 1 of these image-classification experiments, we use a batch size of 1024 and generate two augmented views per image. All reproduced methods share the same loss-independent training conditions while retaining their objective-specific output mappings and hyperparameters. For SoftTriple (Qian et al., 2019), we use its standard unit-normalized formulation with $(K, \lambda, \gamma, \delta, \tau) = (10, 20, 0.1, 0.01, 0.2)$. In Stage 2, we discard the projection head and any objective-specific auxiliary parameters, freeze the encoder, and train a newly initialized linear classifier with cross-entropy loss on the raw encoder representation. The resulting model is used for evaluation. The experiments in this subsection are conducted on eight NVIDIA RTX A5000 GPUs.

As shown in Table 1, PEML records the highest observed accuracy on all three datasets among the compared methods. Relative to SupCon, PEML improves top-1 accuracy by 0.06 percentage points (pp) on CIFAR-10, 1.12 pp on CIFAR-100, and 2.88 pp on ImageNet-100. Among SoftTriple, SupCon, and PEML—the three reproduced metric-learning objectives evaluated across all three datasets in Table 1—the observed ranking is consistently PEML, SupCon, and SoftTriple, from highest to lowest, although the magnitude of the differences is dataset dependent. A complementary comparison of Circle Loss, MS, ProxyNCA++-Obj., SupCon, and PEML on CIFAR-10 and CIFAR-100 is reported along with their class-disjoint retrieval results in Section 4.2.

Under the two-view setting, PEML and SupCon have the same asymptotic time complexity for loss forward and backward computation, $\mathcal{O}(B^2 d_{\text{emb}})$, and the same auxiliary memory complexity, $\mathcal{O}(B^2)$, where B denotes the global batch size in original images and d_{emb} denotes the embedding dimension. Because PEML constructs $P(a)$, $C(a)$, and $N_c(a)$ only from the current mini-batch and introduces no loss-specific parameters or persistent buffers indexed by the dataset-wide class set, its Stage 1 loss-specific per-step computation and auxiliary-memory requirements have no direct dependence on the total number of classes or images in the dataset when B and d_{emb} are fixed under this two-view setting. A larger training set nevertheless increases the number of mini-batches per epoch and hence the total training time under a fixed-epoch schedule. A controlled profiling experiment using the CIFAR-100 training protocol described in this section shows that PEML incurs only marginal runtime and VRAM overhead relative to SupCon. A detailed analysis of computational complexity and empirical overhead is provided in Section H.

4.2. Standard class-disjoint retrieval benchmarks and cross-regime objective comparison

We complement the frozen-encoder linear evaluation in Section 4.1 with four class-disjoint benchmarks widely used in deep metric-learning retrieval: CUB-200-2011 (CUB) (Wah et al., 2011), Cars196 (Cars) (Krause et al., 2013), Stanford Online Products (SOP) (Song et al., 2016), and In-Shop Clothes Retrieval (In-Shop) (Liu et al., 2016). In each benchmark, the training and evaluation classes are disjoint, so performance measures transfer to categories unseen during benchmark-specific training. We use the standard class-disjoint training and test partitions for CUB, Cars, and SOP, and the prescribed training, query, and gallery partitions for In-Shop.

We compare Circle Loss (Sun et al., 2020), Multi-Similarity (MS) Loss (Wang et al., 2019), the ProxyNCA++ objective (Teh et al., 2020), SupCon (Khosla et al., 2020), and PEML. Circle Loss assigns self-paced weights to pair similarities and jointly optimizes positive and negative similarities within a unified objective, whereas MS combines relative-similarity mining with similarity-dependent weighting of informative pairs. The complete ProxyNCA++ method combines a low-temperature proxy-based neighborhood component analysis objective with method-specific architectural and optimization refinements. To distinguish the controlled objective-level instantiation used here from the complete method-specific recipe, we refer to it as ProxyNCA++-Obj. Accordingly, the retrieval comparison evaluates these objectives under shared loss-independent conditions rather than reproducing every method-specific architecture and optimization recipe.

For the retrieval experiments, the ResNet-50 encoder is initialized from ImageNet-1K V1 pretrained weights, whereas the two-layer projection head, whose loss-space output dimension is 128, is initialized anew. The learnable class proxies used by ProxyNCA++-Obj. are also initialized anew. Each global mini-batch is constructed using a class-balanced $P \times K_{\text{img}}$ sampler with $P = 64$ classes and $K_{\text{img}} = 4$ original images per class, yielding 256 original images. Two independently augmented views are generated for each image, so each loss is evaluated over 512 embeddings per global mini-batch. We train for 1000 epochs using SGD with Nesterov momentum of 0.9, weight decay 3×10^{-4}, a 20-epoch warmup to a peak learning rate of 0.05, and cosine learning-rate decay thereafter. No memory queue is used. All loss-independent training conditions, including the data partitions, sampler, augmentations, image exposures, encoder and projection-head architectures, and optimization budget, are shared across the compared objectives.

For Circle Loss, we use $\gamma = 80$ and $m = 0.4$ and stop the gradient through the self-paced weighting coefficients, as prescribed by the original objective. For MS, we retain its relative-similarity pair mining and weighting and use the authors' released-code settings $(\alpha, \beta, \lambda, \epsilon) = (2, 40, 0.5, 0.1)$. ProxyNCA++-Obj. uses one learnable proxy per training class, ℓ_2-normalized embeddings and proxies, squared Euclidean distance, a softmax denominator over all class proxies, and temperature $T = 1/9$. Its proxies use a learning-rate multiplier of 1.0 and no weight decay. During retrieval training, PEML uses $(\rho_p, \rho_n) = (0.5, 0.4)$ and the annulus-scaled projection with $(R_A, R_B, \tau_r) = (1, 2, 0.5)$ introduced in Section 3.3, whereas the remaining objectives use ℓ_2-normalized loss embeddings.

Table 2
Controlled comparison across two distinct training and evaluation regimes. CUB, Cars, SOP, and In-Shop report Recall@K (%) after benchmark-specific fine-tuning of a ResNet-50 encoder initialized with ImageNet-1K V1 pretrained weights, using class-disjoint training and evaluation partitions. CIFAR-10 and CIFAR-100 report top-1 frozen-encoder linear-evaluation accuracy (%) after the ResNet-50 encoder is trained from scratch with each objective and then frozen for linear evaluation. Within each regime, all loss-independent conditions are shared across the five objectives. The highest observed value in each row is shown in bold.

Dataset	Metric	Circle	MS	ProxyNCA++-Obj.	SupCon	PEML
ImageNet-1K-pretrained encoder fine-tuning: class-disjoint retrieval						
CUB	R@1	21.08	22.21	22.55	12.31	**26.62**
	R@2	30.37	32.22	32.68	18.79	**37.58**
	R@4	41.29	44.21	45.09	27.99	**49.70**
	R@8	53.81	57.31	58.34	39.74	**63.12**
Cars	R@1	36.98	43.16	41.96	24.82	**48.53**
	R@2	48.83	55.57	54.54	35.51	**62.11**
	R@4	59.91	67.13	66.29	46.62	**73.00**
	R@8	70.31	77.36	76.78	58.80	**82.62**
SOP	R@1	53.69	**66.68**	66.57	52.03	56.59
	R@10	73.01	**83.78**	83.22	74.37	78.34
	R@100	86.98	**93.02**	92.96	89.41	91.27
	R@1000	95.60	**97.76**	97.60	97.03	97.30
In-Shop	R@1	74.57	**77.89**	77.21	71.59	71.72
	R@10	90.96	**93.93**	93.71	91.26	90.90
	R@20	93.29	**96.25**	95.65	94.01	94.31
	R@30	94.43	96.65	96.60	95.39	**96.66**
	R@40	95.07	**97.77**	97.15	96.08	97.75
	R@50	95.55	97.97	97.46	96.55	**98.28**
From-scratch encoder training: frozen-encoder linear evaluation						
CIFAR-10	Top-1 Acc.	92.79	95.34	94.36	95.98	**96.04**
CIFAR-100	Top-1 Acc.	74.66	73.18	70.77	76.49	**77.61**

For all methods, retrieval is performed using the raw 2,048-dimensional global-average-pooled encoder representation before the projection head. We apply neither additional normalization nor post-processing and rank samples by Euclidean distance. For CUB, Cars, and SOP, each test image is used as a query against all remaining test images, with the query itself excluded. For In-Shop, each query is evaluated against the prescribed gallery partition. Recall@K is the percentage of queries for which at least one gallery image belonging to the same class occurs among the K nearest retrieved samples.

To complement the class-disjoint retrieval evaluation, we additionally evaluate the same five objectives on CIFAR-10 and CIFAR-100 under the two-stage image-classification protocol described in Section 4.1. We use the prespecified common peak learning rate adopted for Circle Loss, MS, and ProxyNCA++-Obj. in the controlled retrieval comparison. All other loss-independent conditions are held fixed across the five objectives within each dataset.

As shown in Table 2, PEML records the highest observed Recall@K at every reported depth on CUB and Cars, exceeding the next-best objective by 4.07–4.90 pp and 5.26–6.54 pp, respectively. On the two product-retrieval benchmarks, the ordering is more benchmark- and depth-dependent. On SOP, MS and ProxyNCA++-Obj. obtain higher values, while PEML exceeds Circle Loss and SupCon at every reported depth and comes within 0.46 pp of the highest value at Recall@1000. On In-Shop, PEML records the highest observed values at Recall@30 and Recall@50 and differs from MS by only 0.02 pp at Recall@40, while MS leads at the shallower reported depths. Under the complementary CIFAR linear-evaluation regime, however, Circle Loss, MS, and ProxyNCA++-Obj. all remain below SupCon and PEML on both datasets. This regime-dependent ordering highlights PEML's cross-regime profile: among the compared objectives, PEML alone records the highest observed values in both evaluation regimes, leading at every reported depth on CUB and Cars and attaining the highest top-1 frozen-encoder linear-evaluation accuracy on both CIFAR datasets, while remaining competitive at deeper retrieval depths on SOP and In-Shop.

Because the compared objectives interact differently with mini-batch size, composition, and sampling, both absolute performance and the relative ordering of the objectives may vary under other training configurations. The results therefore characterize the objectives under the two reported regime-specific protocols rather than their individually optimized performance ceilings. Taken together, PEML combines the highest observed frozen-encoder linear-evaluation accuracy on CIFAR-10 and CIFAR-100 with the highest observed Recall@K at every reported depth on CUB and Cars. This cross-regime pattern indicates that PEML supports strong class-level linear separability under the CIFAR protocol and effective transfer to held-out fine-grained categories, while the product-retrieval benchmarks retain a method-dependent ordering.

4.3. Representation quality evaluations

We assess the representation quality of PEML relative to SupCon from three complementary perspectives. First, we evaluate k-means clustering using normalized mutual information (NMI), adjusted mutual information (AMI), and adjusted Rand index (ARI). Second, we use k-NN classification to probe local neighborhood structure without training an additional classifier. Third, we examine the balanced CIFAR-100 endpoint through a full-dataset t-SNE visualization (van der Maaten & Hinton, 2008) and scale-invariant centroid-relative measurements in the original high-dimensional encoder space.

4.3.1. Clustering and local-neighborhood quality

The table in Fig. 2 reports the unsupervised clustering quality obtained by applying k-means to embeddings from ResNet-50 encoders trained with PEML and SupCon. Across both CIFAR-100 and ImageNet-100 and all three metrics, PEML consistently outperforms SupCon. On CIFAR-100, PEML improves over SupCon by 5.48 pp in NMI, 6.21 pp in AMI, and 14.78 pp in ARI. On ImageNet-100, the corresponding gains are 8.96 pp in NMI, 11.98 pp in AMI, and 16.59 pp in ARI. The particularly large ARI gains indicate more accurate recovery of class-consistent cluster assignments, in agreement with the linear-evaluation results in Section 4.1.

The graph in Fig. 2 presents top-1 and top-5 k-NN accuracies on CIFAR-100 for neighborhood sizes from 5 to 100. For both methods, top-1 accuracy varies little across k, with PEML matching or slightly outperforming SupCon by approximately 0.1 pp at most values. For top-5 accuracy, PEML improves by nearly 5 pp at $k = 5$; this advantage gradually decreases as k increases, and the two methods attain nearly identical performance at $k = 100$. Thus, PEML's stronger global clustering quality is obtained without sacrificing top-1 local-neighborhood performance, while its top-5 advantage at small k indicates that the local neighborhoods provide more useful evidence among the leading class predictions. This pattern is consistent with the intrinsic hard-mining behavior of the PEML-induced forces discussed in Section 3.2.1.

4.3.2. Embedding visualization and centroid-relative organization

We next examine the balanced CIFAR-100 endpoint using the raw 2,048-dimensional pre-projection encoder representations of all 10,000 test images. For each method, we first reduce the representation to 50 dimensions by principal component analysis and then fit t-SNE independently using Euclidean distance, perplexity 30, 2000 iterations, and seed 1. Before PCA, we apply neither ℓ_2 feature normalization nor coordinate-wise variance standardization; PCA otherwise uses its standard mean centering.

The qualitative difference in Fig. 3 is evaluated in the original encoder space using training-set class centroids. For method $m \in \{\text{SupCon}, \text{PEML}\}$, let $\boldsymbol{h}_i^{(m)}$ be the raw test representation and let $\boldsymbol{\mu}_c^{(m),\text{tr}}$ be the mean raw representation of class c over the 50,000-image training split. For a test sample with label y_i, define the scale-invariant nearest-

Metric	SupCon		PEML	
	CIFAR-100	ImageNet-100	CIFAR-100	ImageNet-100
NMI	72.21	74.34	**77.69**	**83.30**
AMI	68.38	67.53	**74.59**	**79.51**
ARI	41.68	44.93	**56.46**	**61.52**

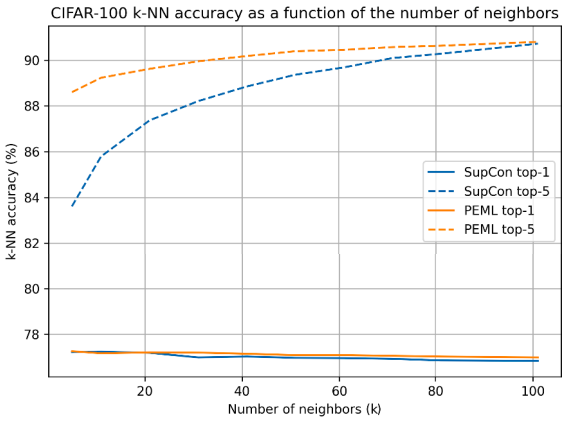


Fig. 2. Evaluation of the representation quality of PEML and SupCon. **Left**: Unsupervised clustering quality (NMI/AMI/ARI, %) of k-means on frozen features from ResNet-50 encoders trained with SupCon and PEML on CIFAR-100 and ImageNet-100. **Right**: k-NN top-1 and top-5 accuracy on CIFAR-100 as a function of k for ResNet-50 encoders.

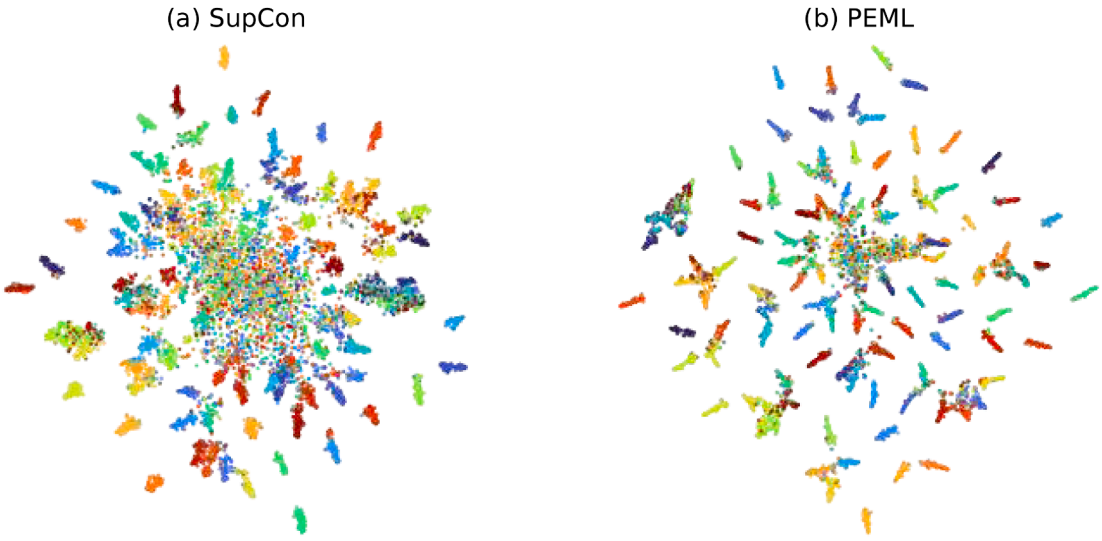


Fig. 3. t-SNE visualization of the raw pre-projection encoder representations for all 10,000 balanced CIFAR-100 test images. **(a)** SupCon. **(b)** PEML. PCA to 50 dimensions and t-SNE are fitted independently for each method using perplexity 30, 2000 iterations, Euclidean distance, and seed 1; neither ℓ_2 normalization nor coordinate-wise variance standardization is applied before PCA. Colors denote the 100 fine classes under a shared class–color mapping. PEML exhibits less cross-class color mixing and more visually distinct local class groupings. Because the projections are fitted independently, their coordinates, orientations, inter-cluster distances, areas, and shapes are not directly comparable across panels; the original high-dimensional geometry is evaluated separately in Table 3.

wrong-centroid ratio

$$r^{(m)}_{\text{wrong}}(i) = \frac{\min_{c \neq y_i} \left\| \boldsymbol{h}^{(m)}_i - \boldsymbol{\mu}^{(m),\text{tr}}_c \right\|_2}{\left\| \boldsymbol{h}^{(m)}_i - \boldsymbol{\mu}^{(m),\text{tr}}_{y_i} \right\|_2}. \tag{49}$$

A value greater than one means that the true-class centroid is closer than every competing centroid. Nearest-centroid accuracy is therefore the proportion of test samples satisfying $r^{(m)}_{\text{wrong}}(i) > 1$.

As shown in Table 3, PEML improves nearest-centroid accuracy from 68.72% to 73.30%, a gain of 4.58 pp, and increases the median nearest-wrong-centroid ratio from 1.213 to 1.872. These original-space measurements support the visual pattern in Fig. 3 without relying on the two-dimensional projection. They also complement the temporal analysis in Section J, which tracks the scale-invariant global ratio $Q = D_{\text{intra}}/D_{\text{inter}}$. From epoch 50 onward, PEML attains a lower Q than SupCon at every recorded checkpoint, and the final difference arises primarily from reduced within-class dispersion relative to the between-centroid scale, as shown in Fig. J.1(e) and Table J.1. A detailed endpoint analysis of shared and method-specific linear-classification errors, including category-balanced severe centroid-inversion examples, is provided in Section J.4. Overall, the qualitative, high-dimensional, and temporal results consistently connect PEML's improved clustering and linear separability to a more distinctly organized class-level representation.

Table 3
Centroid-relative organization of the raw 2,048-dimensional CIFAR-100 encoder representations. Class centroids are fitted on the training split, whereas both metrics are evaluated over all 10,000 test images. A larger r_{wrong} indicates stronger separation from the nearest competing class centroid relative to the true-class centroid.

Metric	SupCon	PEML
Nearest-centroid accuracy (%)	68.72	**73.30**
Median r_{wrong} over all test samples	1.213	**1.872**

4.4. Evaluation of cardinality-weighted force balancing

To examine the effectiveness of the cardinality-weighted force balancing discussed in Section 3.2.3, we conduct experiments under a class-imbalanced setting. Our evaluation comprises two complementary data constructions: (i) a controlled synthetic embedding-evolution setup based on class-conditional Gaussians that isolates the force-balancing mechanism; and (ii) an end-to-end long-tailed resampling of CIFAR-100 that mirrors common long-tail benchmarks.

4.4.1. Synthetic embedding evolution

To directly examine the cardinality-weighted force balancing in Section 3.2.3, we use a controlled embedding-evolution experiment in which the sample coordinates themselves are optimized without an encoder. For each seed, class-conditional Gaussian samples are generated in $\mathbb{R}^3$, and nested long-tailed subsets are constructed for IR $\in \{1, 10, 50, 100, 200, 500\}$. The class-conditional geometry is held fixed as the class-frequency profile changes. The complete data construction, including the dimension-aware overlap calibration and pseudocode, is provided in Section D.1. SupCon and PEML are initialized from identical coordinates and use the same class labels, mini-batch order, stochastic views, and number of optimizer updates at each IR. SupCon is applied to ℓ_2-normalized views, whereas PEML uses the annulus-scaled projection. All runs use a batch size of 1,024, 500 epochs, SGD with zero momentum and zero weight decay, and a per-step cosine schedule without warmup. Because the two objectives induce different numerical gradient scales, the method-specific optimization settings were calibrated using direct gradient- and raw-coordinate-update diagnostics; the reported pair was selected as the closest empirically balanced setting among the evaluated controls. Results are averaged over five seeds.

We evaluate k-means clustering on the raw, unaugmented endpoint coordinates. This is the common pre-projection space and serves as the synthetic analog of the encoder output used in the downstream evaluations. Let $\bar{m}_X(\mathrm{IR})$ denote the five-seed mean of a clustering metric $m \in \{\mathrm{NMI}, \mathrm{AMI}, \mathrm{ARI}\}$ for method X. Its relative degradation from the balanced setting is

$$d_{X,m}(\mathrm{IR}) = 100\left(1 - \frac{\bar{m}_X(\mathrm{IR})}{\bar{m}_X(1)}\right). \tag{50}$$

A smaller $d_{X,m}$ indicates better preservation of the balanced performance. We additionally report $Q = D_{\mathrm{intra}}/D_{\mathrm{inter}}$, where D_{intra} is the class-macro mean distance to the corresponding empirical centroid and D_{inter} is the mean distance over unordered centroid pairs. We define $G_Q = Q_{\mathrm{final}}/Q_{\mathrm{initial}}$. Lower values of Q_{final} and G_Q indicate stronger within-class compactness relative to the between-centroid scale. For $z \in \{Q_{\mathrm{final}}, G_Q\}$, let $\bar{z}_X(\mathrm{IR})$ denote the five-seed mean of z for method X at the indicated imbalance ratio. We define its relative decrease from the balanced setting as

$$r_{X,z}(\mathrm{IR}) = 100\left(1 - \frac{\bar{z}_X(\mathrm{IR})}{\bar{z}_X(1)}\right), \tag{51}$$

where a positive value denotes an improvement relative to the balanced setting.

As shown in Table 4, PEML achieves higher mean NMI, AMI, and ARI than SupCon at every evaluated IR. The improvements range from 0.34 to 4.66 pp for NMI, from 0.34 to 4.70 pp for AMI, and from 2.94 to 7.03 pp for ARI. At the seed level, PEML obtains the higher score in 28 of the 30 paired $(\mathrm{IR}, \mathrm{seed})$ comparisons for each of the three metrics. The relative-degradation results provide a more specific view of robustness. At IR = 10, PEML has slightly larger NMI and AMI degradation despite retaining higher absolute scores, while its ARI degradation is already smaller. For every evaluated IR ≥ 50, however, PEML exhibits smaller relative degradation in all three clustering metrics. From IR = 1 to IR = 500, the relative declines in NMI, AMI, and ARI are 31.20%, 31.78%, and 61.14% for PEML, compared with 33.51%, 34.16%, and 63.11% for SupCon. The geometric diagnostics show the same overall advantage. PEML attains lower Q_{final} and G_Q at every IR and in all 30 paired seed-level comparisons. At IR = 500, PEML improves Q_{final} and G_Q relative to their balanced-setting values by 10.37% and 4.67%, respectively, whereas the corresponding SupCon values worsen by 2.61% and 9.17%. Thus, PEML preserves higher clustering quality while developing more favorable centroid-relative geometry under severe cardinality skew. This result is consistent with the intended role of cardinality-weighted force balancing, which limits the extent to which head-class multiplicity dominates the aggregate pairwise interactions.

4.4.2. Long-tailed resampling of CIFAR-100 (CIFAR100-LT)

We construct class-imbalanced variants of CIFAR-100 by subsampling the training set according to a nominal imbalance ratio and evaluate IR $\in \{10, 50, 100, 200\}$, with the balanced setting IR = 1 retained as a reference. We impose a common lower bound $n_j \geq n_{\min} = 2$ on every class. Because integer rounding and this minimum-count constraint can make the realized head-to-tail ratio differ from its nominal value, the exact class-count construction and realized sample counts are reported in Section D.2. We use ProCo-Rep to denote the representation objective $\mathcal{L}_{\mathrm{ProCo}}$ of Probabilistic Contrastive Learning (ProCo), evaluated without its jointly trained logit-adjusted classification branch (Du et al., 2024), thereby yielding an objective-level comparison rather than a reproduction of the complete ProCo framework. For every setting, SupCon, ProCo-Rep, and PEML follow the same encoder-training and linear-evaluation protocol used for the balanced CIFAR-100 experiments in Section 4.1.

To compare robustness after normalizing by each method's balanced accuracy, let $m_X(r)$ denote the top-1 accuracy of method X at nominal imbalance ratio r. We define its relative degradation from the balanced setting as

$$d_X(r) = 100\left(1 - \frac{m_X(r)}{m_X(1)}\right). \tag{52}$$

A smaller $d_X(r)$ indicates better preservation of balanced performance.

As shown in Table 5, PEML achieves the highest accuracy at every evaluated IR. The maximum difference among the three methods in the balanced setting is 1.49 pp. Across the long-tailed settings, SupCon exceeds ProCo-Rep at every evaluated IR by 1.45–1.90 pp, whereas PEML's advantage over SupCon increases from 8.34 pp at IR = 10 to 22.11 pp at IR = 200. This separation persists after normalization by each method's balanced accuracy in Fig. 4. At IR = 200, the relative degradation is 28.44% for PEML, compared with 56.29% for SupCon and 58.22% for ProCo-Rep. Thus, at the most severe evaluated setting, PEML incurs approximately one-half of the relative degradation of either baseline. This robustness pattern is consistent with the intended role of cardinality-weighted force balancing: by preventing the number of samples observed from a negative class from directly scaling that class's aggregate repulsive contribution for an anchor, PEML limits interaction-level cardinality bias. A broader comparison of related long-tailed contrastive objectives is provided in Section E.

To characterize the representation underlying the PEML–SupCon accuracy difference beyond linear evaluation, we further examine the frozen-encoder representation space at IR = 200 in Section F. Whereas the balanced-data experiments in Section 4.3 show concurrent improvements in class-level clustering and local k-NN quality, the complementary analysis combines a tail-class visualization, centroid-relative geometry, Recall@K, and mAP@R to show that class-level organization and local retrieval quality become partially decoupled under severe class imbalance. PEML exhibits a lower centroid-relative intra/inter ratio and a slightly higher mAP@R, indicating more favorable class-level organization and broader positive-set ranking, whereas SupCon achieves higher Recall@K, indicating more frequent retrieval of a positive sample among the nearest neighbors. Taken together, these findings show that PEML better preserves the class-level structure relevant to downstream linear discrimination under severe cardinality skew.

Table 4
Raw embedding endpoints under increasing class imbalance. Each clustering entry reports the five-seed mean score followed by its relative degradation d from IR = 1. Each geometry entry reports the five-seed mean followed by its relative decrease r from IR = 1. Boldface indicates the better value within each IR; lower degradation, lower Q_{final} and G_Q, and higher relative decrease r are preferred.

IR	Method	Score (%) / degradation d (%)			Value / relative decrease r (%)	
		NMI ↑	AMI ↑	ARI ↑	Q_{final} ↓	G_Q ↓
1	SupCon	79.28 / 0.00	79.25 / 0.00	78.24 / 0.00	0.354 / 0.00	0.806 / 0.00
	PEML	**82.29** / 0.00	**82.26** / 0.00	**81.83** / 0.00	**0.343** / 0.00	**0.781** / 0.00
10	SupCon	68.60 / **13.47**	68.45 / **13.62**	60.03 / 23.28	0.365 / −2.96	0.832 / −3.11
	PEML	**68.94** / 16.22	**68.80** / 16.37	**65.62** / **19.81**	**0.347** / **−1.11**	**0.791** / **−1.26**
50	SupCon	61.71 / 22.16	61.42 / 22.49	42.57 / 45.60	0.367 / −3.52	0.832 / −3.11
	PEML	**64.78** / **21.28**	**64.51** / **21.58**	**49.59** / **39.40**	**0.335** / **2.44**	**0.759** / **2.84**
100	SupCon	61.23 / 22.77	60.88 / 23.17	39.41 / 49.63	0.356 / −0.51	0.816 / −1.18
	PEML	**64.98** / **21.04**	**64.66** / **21.40**	**43.60** / **46.72**	**0.314** / **8.43**	**0.720** / **7.84**
200	SupCon	61.29 / 22.69	60.89 / 23.16	37.25 / 52.40	0.351 / 0.79	0.802 / 0.53
	PEML	**65.95** / **19.87**	**65.59** / **20.26**	**41.57** / **49.20**	**0.305** / **11.12**	**0.695** / **11.02**
500	SupCon	52.72 / 33.51	52.17 / 34.16	28.86 / 63.11	0.363 / −2.61	0.880 / −9.17
	PEML	**56.62** / **31.20**	**56.12** / **31.78**	**31.80** / **61.14**	**0.307** / **10.37**	**0.745** / **4.67**

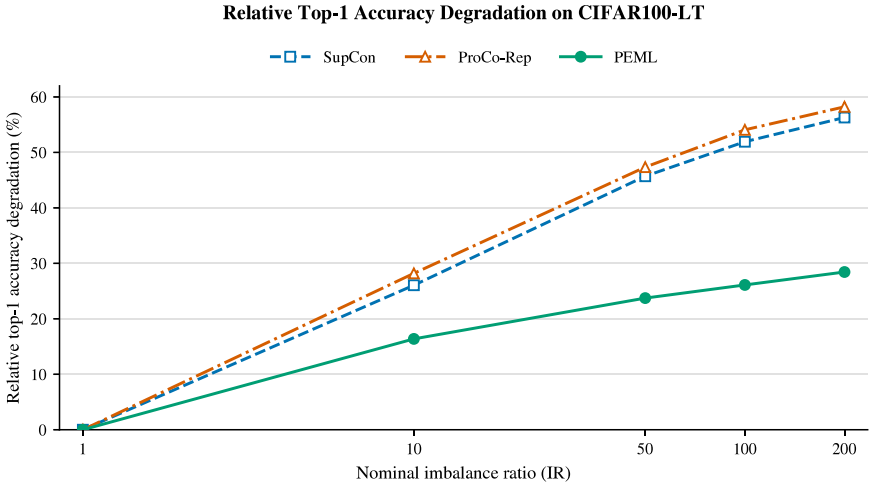


Fig. 4. Relative degradation in frozen-encoder linear-evaluation accuracy on CIFAR100-LT as the nominal imbalance ratio increases. For method X, the degradation is computed using Eq. (52); lower values indicate greater robustness to class imbalance. Markers denote the evaluated nominal IRs on a logarithmic horizontal scale, and connecting lines are included only as visual guides.

Table 5
Frozen-encoder linear-evaluation top-1 accuracy (%) on CIFAR100-LT under the common two-stage protocol. ProCo-Rep denotes the objective-only ProCo variant. The highest accuracy at each evaluated IR is shown in bold.

	Nominal imbalance ratio (IR)				
Method	1	10	50	100	200
SupCon	76.49	56.56	41.53	36.77	33.43
ProCo-Rep	76.12	54.66	40.08	34.96	31.80
PEML	**77.61**	**64.90**	**59.19**	**57.35**	**55.54**

4.5. Effect of force monotonicity on intrinsic hard mining

We evaluate whether the intrinsic hard-mining behavior of the loss, induced by the monotonicity properties of the forces (Eq. (19) and Eq. (20)), improves metric-learning performance. To this end, we consider variants of PEML in which the original monotonic forces in Eq. (42) and Eq. (43) are modified to become non-monotonic, analogous to the repulsive force of SupCon analyzed in Section 3.2.1. Specifically, in Eq. (42) and Eq. (43), we move the maxima originally located at $d = d_{\max}$ and $d = 0$ to new crest locations c_p and c_n, respectively, and set the force to zero at the original maxima. The crests c_p and c_n are

$$c_p(a) = d_{max} - \varrho \cdot (d_{max} - k_p(a)) , \tag{53}$$

$$c_n(a) = \varrho \cdot k_n(a) , \tag{54}$$

where $\varrho \in (0, 1)$ denotes the ratio, measured relative to the original active force range, of the distance from the original maximum point to the new maximum point (crest). In our experiments, we set $\varrho = 0.3$. Then, we can define the non-monotonic forces $F'_P(d_{ap})$ and $F'_N(d_{an})$ as

$$F'_P(d_{ap}) = \left[\sin^2\left(\frac{\pi(d_{ap} - k_p(a))}{2(c_p(a) - k_p(a))} \right) \mathbf{1}_{[k_p(a), c_p(a)]}(d_{ap}) + \cos^2\left(\frac{\pi(d_{ap} - c_p(a))}{2(d_{max} - c_p(a))} \right) \mathbf{1}_{(c_p(a), d_{max}]}(d_{ap}) \right] \hat{r}_{ap}, \tag{55}$$

$$F'_N(d_{an}) = \left[\sin^2\left(\frac{\pi d_{an}}{2c_n(a)} \right) \mathbf{1}_{[0, c_n(a))}(d_{an}) + \cos^2\left(\frac{\pi(d_{an} - c_n(a))}{2(k_n(a) - c_n(a))} \right) \mathbf{1}_{[c_n(a), k_n(a)]}(d_{an}) \right] \hat{r}_{an}. \tag{56}$$

The corresponding potential energies, $V'_P(d_{ap})$ and $V'_N(d_{an})$, are expressed as

$$V'_P(d_{ap}) = V'_{P1}(d_{ap}) \mathbf{1}_{[k_p(a), c_p(a)]}(d_{ap}) + V'_{P2}(d_{ap}) \mathbf{1}_{(c_p(a), d_{max}]}(d_{ap}) , \tag{57}$$

$$V'_N(d_{an}) = V'_{N1}(d_{an}) \mathbf{1}_{[0, c_n(a))}(d_{an}) + V'_{N2}(d_{an}) \mathbf{1}_{[c_n(a), k_n(a)]}(d_{an}) . \tag{58}$$

Table 6
CIFAR-100 top-1 accuracy (%) for PEML variants obtained by selectively replacing the original monotone potentials V_P and V_N with their non-monotone counterparts V'_P and V'_N.

Positive potential	Negative potential	Top-1 acc. (%)
V_P	V_N	77.61
V'_P	V_N	74.61
V_P	V'_N	75.74
V'_P	V'_N	74.80

Here,

$$V'_{P1}(d_{ap}) = \frac{d_{ap} - k_p(a)}{2} - \frac{c_p(a) - k_p(a)}{2\pi} \sin\left(\frac{\pi\left(d_{ap} - k_p(a)\right)}{c_p(a) - k_p(a)}\right), \tag{59}$$

$$V'_{P2}(d_{ap}) = \frac{d_{ap} - k_p(a)}{2} + \frac{d_{\max} - c_p(a)}{2\pi} \sin\left(\frac{\pi\left(d_{ap} - c_p(a)\right)}{d_{\max} - c_p(a)}\right), \tag{60}$$

$$V'_{N1}(d_{an}) = \frac{k_n(a) - d_{an}}{2} + \frac{c_n(a)}{2\pi} \sin\left(\frac{\pi d_{an}}{c_n(a)}\right), \tag{61}$$

$$V'_{N2}(d_{an}) = \frac{k_n(a) - d_{an}}{2} - \frac{k_n(a) - c_n(a)}{2\pi} \sin\left(\frac{\pi\left(d_{an} - c_n(a)\right)}{k_n(a) - c_n(a)}\right). \tag{62}$$

Following the loss formulation procedure in Section 3.4, we then construct losses for the three combinations obtained by replacing the original pairwise potentials $V_P(d_{ap})$ and $V_N(d_{an})$ in Eqs. (46) and (47) with their non-monotonic counterparts $V'_P(d_{ap})$ and $V'_N(d_{an})$, namely $(V'_P(d_{ap}), V_N(d_{an}))$, $(V_P(d_{ap}), V'_N(d_{an}))$, and $(V'_P(d_{ap}), V'_N(d_{an}))$. For the margin constraint and the cardinality-weighted force balancing, we retain the original PEML formulation. Using these newly formulated losses, we train models under the same protocol as in Section 4.1 and measure the resulting image classification accuracies.

Table 6 reports CIFAR-100 top-1 accuracy for four variants of PEML obtained by selectively replacing the original monotone pairwise potentials V_P and V_N with their non-monotone counterparts V'_P and V'_N. All three variants that break monotonicity suffer a degradation in accuracy. This behavior is consistent with the analysis in Section 3.2.1. The systematic drop in accuracy across all non-monotone configurations thus provides empirical evidence that monotonicity of the force is not merely an incidental modeling choice but a key mechanism for effective hard mining in metric learning. The degradation is also asymmetric between the positive and negative branches. Making only the positive potential non-monotone, (V'_P, V_N), yields the largest performance drop (-3.00 pp), whereas making only the negative potential non-monotone, (V_P, V'_N), leads to a smaller decrease (-1.87 pp). These results are also consistent with the observation that SupCon, whose attractive force is monotonic even though its repulsive force is not, nevertheless remains highly competitive in practice. This suggests that the monotonicity of the attractive force F_P is particularly critical: once F'_P becomes non-monotone, medium-distance positives around the crest $c_p(a)$ dominate the gradient, whereas truly hard positives near the margin receive little or no force, which in turn prevents the formation of tight class clusters around the most challenging intra-class relations. When both branches are made non-monotone, (V'_P, V'_N), the performance slightly exceeds that of the variant in which only the positive branch is modified. This behavior is consistent with a degree of coupling between the attractive and repulsive forces. Once the positive branch loses its intrinsic hard-mining behavior, partially mis-weighting the negatives in a similar manner may help rebalance the overall force field, although it still does not recover the performance of the original monotone specification.

Table 7
Monotone force-magnitude kernels evaluated in the ablation and their zero-baseline primitives on normalized pair hardness $u \in [0, 1]$. All four kernels satisfy the common endpoint normalization $g(0) = 0$ and $g(1) = 1$.

Kernel	$g(u)$	$G(u)$
Trigonometric	$\sin^2\left(\frac{\pi u}{2}\right)$	$\frac{u}{2} - \frac{\sin(\pi u)}{2\pi}$
Linear	u	$\frac{u^2}{2}$
Quadratic	u^2	$\frac{u^3}{3}$
Normalized exponential	$\frac{e^u - 1}{e - 1}$	$\frac{e^u - 1 - u}{e - 1}$

Table 8
CIFAR-100 comparison of four admissible monotone PEML force kernels, with SupCon included as a reference baseline. Across the PEML variants, only g and its corresponding primitive G differ. Accuracy and the clustering metrics are reported in percent, and a lower Q is better.

Method	Accuracy	NMI	AMI	ARI	Ratio $Q \downarrow$
SupCon	76.49	72.21	68.38	41.68	0.7174
Linear	73.68	72.48	68.74	38.31	0.2398
Quadratic	77.45	76.85	73.66	50.68	**0.1960**
Normalized exponential	76.02	74.08	70.54	39.17	0.2183
Trigonometric (PEML)	**77.61**	**77.69**	**74.59**	**56.46**	0.2188

4.6. Ablation study of alternative monotone force kernels

We assess the sensitivity of the learned representation to force-profile shape while retaining the common admissibility conditions discussed in Section 3.4. We compare the squared trigonometric kernel used by PEML with linear, quadratic, and normalized exponential alternatives under common monotonicity, boundedness, and endpoint-normalization requirements.

Let $u \in [0, 1]$ denote normalized pair hardness, with $u = 0$ corresponding to the inactive boundary and $u = 1$ to the hardest endpoint. We write $g(u) \in [0, 1]$ for a continuously differentiable, non-decreasing force-magnitude kernel satisfying $g(0) = 0$ and $g(1) = 1$, and define its zero-baseline primitive as $G(u) = \int_0^u g(t)\,dt$. After rescaling to the positive and negative active intervals, g specifies the pairwise force magnitude and G induces the corresponding pairwise potential. The four kernel–primitive pairs are summarized in Table 7; their unified force–potential construction and distance-domain derivation are provided in Section C.

Although the four kernels share the same force-magnitude range and endpoint normalization, they are not normalized to have the same integrated magnitude $G(1)$. The comparison therefore evaluates the complete monotone force profile; the observed differences cannot be attributed to endpoint slopes or integrated magnitude in isolation. To isolate sensitivity to the exact force profile, only the kernel g and its corresponding primitive G are changed among the four PEML variants. The adaptive thresholds, cardinality-weighted aggregation, annulus-scaled projection, model architecture, and optimization settings remain unchanged, and no kernel-specific shape parameter is tuned. All variants are trained and evaluated using the same CIFAR-100 protocols as in Sections 4.1 and 4.3. In addition to frozen-encoder linear-evaluation accuracy and the clustering metrics NMI, AMI, and ARI, we report the centroid-relative encoder-space ratio $Q = D_{\text{intra}}/D_{\text{inter}}$, where D_{intra} and D_{inter} denote within-class dispersion and inter-centroid separation, respectively. The ratio is computed from the raw pre-projection encoder outputs, and a smaller value indicates lower within-class dispersion relative to inter-centroid separation. SupCon is included as a reference baseline.

As shown in Table 8, the trigonometric kernel achieves the highest observed accuracy, NMI, AMI, and ARI. The quadratic kernel is the clos-

Table 9
Top-1 accuracy (%) on CIFAR-100 for different Euclidean metric losses with either ℓ_2 normalization or annulus-scaled projection applied at the projection-head output.

Projection-head output mapping	PEML	NCA	Contrastive	Triplet	Margin
ℓ_2 normalization	74.12	69.39	66.07	68.91	69.59
Annulus-scaled projection	77.61	71.30	68.01	76.07	72.45

est alternative, attaining 77.45% accuracy compared with 77.61% for the trigonometric kernel, while remaining lower on all three clustering metrics. Its largest difference from the trigonometric kernel is 5.78 pp in ARI. The linear and normalized exponential kernels produce lower values than the trigonometric kernel on all four performance metrics. Across the four admissible kernels, the observed ranges are 3.93 pp in accuracy, 5.21 pp in NMI, 5.85 pp in AMI, and 18.15 pp in ARI. Thus, within this controlled single-run comparison, the learned representation remains sensitive to the exact monotone force profile despite the shared admissibility conditions.

The encoder-space ratio Q provides a complementary geometric view, but its ordering does not coincide with the performance-metric ordering. All four force-kernel variants obtain considerably lower Q values than SupCon, whereas the quadratic kernel attains the lowest Q despite the trigonometric kernel achieving the highest accuracy and clustering scores. Moreover, the trigonometric and normalized exponential kernels have nearly identical ratios, 0.2188 and 0.2183, respectively, while their ARI values differ by 17.29 pp. Accordingly, Q should be interpreted as a global diagnostic of centroid-relative class geometry rather than as a direct surrogate for linear separability or clustering-assignment quality. The SupCon row provides an additional reference to the comparisons reported in Sections 4.1 and 4.3. The quadratic kernel records higher observed values than SupCon on all four performance metrics. The linear and normalized exponential kernels each record higher NMI and AMI but lower accuracy and ARI; the normalized exponential kernel remains within 0.47 pp of SupCon in accuracy. Overall, the three non-trigonometric kernels exceed SupCon in eight of the twelve metric-wise comparisons across accuracy, NMI, AMI, and ARI, although the pattern is not uniform across all kernels and metrics.

Overall, the ablation shows that multiple monotone kernels are compatible with the PEML construction but yield different empirical outcomes. Among the candidates evaluated here, the trigonometric kernel attains the highest observed accuracy and all three clustering scores, providing empirical support for its use as the default PEML profile. This result does not establish its optimality over the broader class of admissible kernels; determining whether other profiles offer stronger theoretical or empirical properties remains a subject for future study.

4.7. Ablation study on annulus-scaled projection for Euclidean metric losses

We empirically assess whether applying the annulus-scaled projection introduced in Section 3.3 to PEML is appropriate. We then examine its general suitability for Euclidean-distance-based metric losses by conducting experiments with Neighborhood Components Analysis (NCA)-style loss (Goldberger et al., 2004), contrastive loss (Hadsell et al., 2006), batch-hard triplet loss (Hermans et al., 2017), and margin loss (Wu et al., 2017). The mathematical formulations of these losses are described in Section G. For all losses, we adopt a common protocol: a ResNet-50 encoder is followed by a projection head with either ℓ_2 normalization or annulus-scaled projection applied at the final layer, and the resulting model is trained on CIFAR-100. Hyperparameters are set according to the recommended configurations in the original references for each loss.

Table 9 shows that replacing ℓ_2 normalization with the annulus-scaled projection consistently improves CIFAR-100 top-1 accuracy for all Euclidean metric losses considered. The gains range from +1.91 pp for NCA-style loss to +7.16 pp for the batch-hard triplet loss, with an average improvement of approximately +3.47 pp over ℓ_2 normalization. In particular, although the NCA loss shares the same softmax-type functional form as SupCon, it operates directly on Euclidean distances and also benefits from the annulus-scaled projection. This observation supports our theoretical argument that, for Euclidean metric losses, the effective force acting on embeddings should have not only a tangential component but also a radial component. The annulus-scaled projection preserves this radial component while retaining the numerical stability advantages of ℓ_2 normalization, whereas pure ℓ_2 normalization suppresses radial updates. Although PEML was originally derived together with the annulus-scaled projection, the fact that the same projection yields consistent gains for three classical Euclidean losses (NCA, contrastive, triplet) and the margin loss under a common training protocol indicates that the proposed projection is a generally useful methodological choice for Euclidean-distance-based metric learning, rather than a mechanism specific to the proposed loss.

4.8. Ablation study on adaptive margin parameters ρ_p and ρ_n

We assess the sensitivity of representation quality to the choice of ρ_p and ρ_n. Using the same training protocol as in Section 4.1, we train ResNet-50 models on CIFAR-10 and CIFAR-100 while varying only ρ_p and ρ_n over representative values between 0.1 and 0.9. For each evaluated (ρ_p, ρ_n) pair, we perform five training and frozen-encoder linear-evaluation runs using different random seeds while retaining the same predefined dataset partition and all other experimental settings. Each reported grid value is the arithmetic mean of the resulting five top-1 accuracies.

Fig. 5 reports the five-seed mean image classification performance under varying margin constraints induced by ρ_p and ρ_n in the range from 0.1 to 0.9. The settings $(\rho_p, \rho_n) = (0.1, 0.2)$ and $(\rho_p, \rho_n) = (0.5, 0.4)$ attain the highest five-seed mean accuracies on CIFAR-10 and CIFAR-100, respectively, and can be viewed as attaining the tightest mutual-information (MI) approximation. Because $d_p^{\max}$ and $d_n^{\min}$ are dynamic and anchor-dependent, the same (ρ_p, ρ_n) can induce different effective interaction ranges across datasets. CIFAR-10 remains in a safe regime across the grid, whereas CIFAR-100 can enter a regime in which repulsion becomes overly sparse.

To summarize the sensitivity along each parameter axis, we first use the five-seed mean at each grid point and then compute the marginal mean for each value of one parameter by averaging over the evaluated values of the other parameter. On CIFAR-10, the marginal mean associated with ρ_n ranges from a maximum of 95.75% to a minimum of 94.84%, a difference of 0.91 pp, whereas the corresponding range associated with ρ_p is only 0.23 pp. On CIFAR-100, the marginal mean associated with ρ_n decreases from 76.14% to 10.70%, a difference of 65.4 pp, whereas the corresponding range associated with ρ_p is only 8.67 pp. These observations indicate that ρ_n is the primary driver of performance on both datasets. Interpreting this trend from a force-balance perspective, consider a fixed current embedding configuration. From Eq. (41), increasing ρ_n contracts $k_n(a)$ toward $d_p^{\max}(a)$ whenever $d_p^{\max}(a) < d_{\max}$. This weakens the pairwise negative branch in two coupled ways. In Eq. (43), a negative pair becomes inactive once $d_{an} > k_n(a)$, while the squared-cosine force magnitude of a fixed negative pair with $d_{an} > 0$ also decreases as $k_n(a)$ contracts, as long as the pair remains active. By contrast, ρ_n does not directly enter the positive-force rule in Eq. (42). Although the cardinality-weighted aggregation in Eq. (34) prevents a represented negative class from receiving a larger aggregate weight solely because it contributes more sampled examples, its normalizers are not recomputed according to the number or magnitude of the interactions that remain active. It therefore does not compensate for negative interactions that are attenuated or made inactive by the contraction of $k_n(a)$. When many negative pairs lie near or beyond the contracted range, the aggregate repulsive contribution can become insufficient relative to attraction, shifting the instantaneous force balance toward an attraction-dominated regime and thereby hindering inter-class separation. This

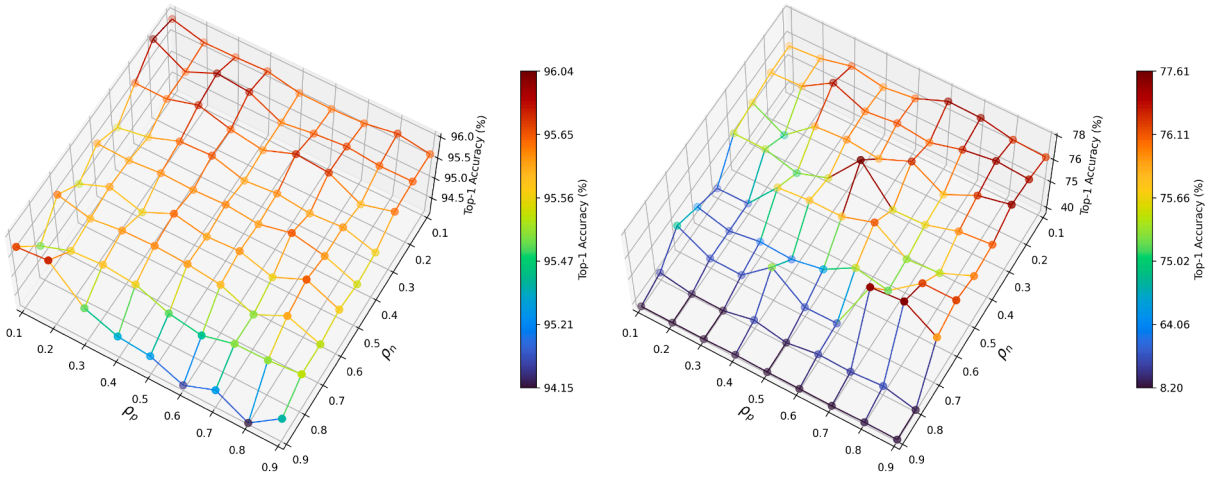


Fig. 5. Top-1 frozen-encoder linear-evaluation accuracy as a function of the adaptive margin parameters (ρ_p, ρ_n) for CIFAR-10 (left) and CIFAR-100 (right). Each marker reports the arithmetic mean over five runs with different random seeds at the indicated (ρ_p, ρ_n) pair, and both its height and color encode this mean. The same predefined dataset partition is used for all runs. For CIFAR-100, where the performance variation is substantially larger, a rescaled accuracy axis is used for clearer visualization.

provides a force-balance explanation for the sharp empirical degradation observed at excessively large ρ_n.

For routine use without a dataset-specific two-dimensional search, the evaluated results support $(\rho_p, \rho_n) = (0.5, 0.4)$ as a practical empirical default. This pair attains the best observed CIFAR-100 result in Fig. 5 and lies within the high-accuracy regime observed on CIFAR-10. The same pair was also used, without a dataset-specific grid search, for the ImageNet-100 classification result in Table 1 and was retained unchanged across the four class-disjoint retrieval benchmarks described in Section 4.2. Moreover, substituting $\rho_n = 0.4$ into Eq. (41) gives $k_n(a) = 0.6d_{\max} + 0.4d_p^{\max}(a) \geq 0.6d_{\max}$ and $k_n(a) - d_p^{\max}(a) = 0.6\big(d_{\max} - d_p^{\max}(a)\big)$. Thus, this setting retains 60% of the anchor-specific interval between $d_p^{\max}(a)$ and $d_{\max}$, providing a transparent geometric guardrail against the extreme contraction obtained as $\rho_n \to 1$. Together, this geometric bound and the cross-dataset empirical evidence provide a principled basis for adopting $(\rho_p, \rho_n) = (0.5, 0.4)$ as a practical default. Therefore, we recommend this setting when dataset-specific tuning is unavailable, while dataset-specific validation can be used to assess whether values of ρ_n closer to 1 are appropriate in a particular setting. The dependence of the effective interaction ranges and pair-distance distributions on the dataset further motivates the development of more refined dataset-adaptive selection criteria for ρ_p and ρ_n in future work.

5. Conclusion

In this work, we introduced a force-first framework that specifies desired embedding-space interactions before constructing a scalar deep metric-learning objective. We identified and analytically characterized three key force-design principles—intrinsic hard mining, margin control through anchor-adaptive interaction ranges, and cardinality-weighted force balancing—and expressed them as quantitative prescriptions for interaction direction, magnitude, active range, and aggregation. Through a Lagrangian-inspired force–potential correspondence, the prescribed conservative forces are integrated into pairwise potentials whose cardinality-weighted aggregation defines the potential energy metric loss (PEML). Under a fixed interaction configuration, the negative gradient of the resulting potential recovers the target interaction field. We further introduced the annulus-scaled projection (ASP), which preserves radial variation for Euclidean-distance-based interactions while bounding the embedding norms.

Empirically, PEML achieved the highest observed frozen-encoder linear-evaluation accuracy among the compared objectives on CIFAR-10, CIFAR-100, and ImageNet-100. Clustering and centroid-relative analyzes further associated this improved separability with more coherent class-level organization than SupCon. On CIFAR100-LT, PEML attained the highest accuracy at every evaluated imbalance ratio in the comparison with SupCon and the evaluated representation objective of ProCo; at an imbalance ratio of 200, it achieved 55.54% and exceeded SupCon by 22.11 percentage points. Across four class-disjoint retrieval benchmarks, PEML achieved the highest Recall@K at every evaluated depth on CUB and Cars under the shared protocol. MS led at every evaluated depth on SOP, while MS and PEML attained the highest values at different retrieval depths on In-Shop. ASP also improved all five Euclidean metric-learning objectives evaluated under the common CIFAR-100 projection comparison, supporting its utility beyond PEML. The observed sensitivity to force-profile shape motivates further study of additional admissible force profiles and principled criteria for selecting among them. Extending the matched two-stage classification evaluation to the full ImageNet-1K dataset and other million-image-scale benchmarks remains an important direction for future work. Overall, this work establishes force-first objective construction as a principled and empirically effective route for translating desired metric-learning interactions into scalar losses, providing a constructive foundation for future deep metric-loss design.

CRediT authorship contribution statement

Byung Do Cheon: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization; **Jeongmin Yu:** Writing – review & editing, Software, Investigation; **Jungeun Kim:** Writing – review & editing, Investigation; **Ha Young Kim:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

CIFAR-10 and CIFAR-100 were obtained from their official public source, and ImageNet was obtained from the official ImageNet resource subject to its access terms. CUB-200-2011, Cars196, Stanford Online Products, and In-Shop Clothes Retrieval were obtained from their respective public benchmark releases and evaluated using the standard class-disjoint partitions described in Section 4.2. The ImageNet-100 setting used in this study follows exactly the 100-class ImageNet synset list used by Barbano et al. (2023). The code and reproducibility materials supporting this study, including dataset construction utilities and the ImageNet-100 synset list, are available at: https://github.com/Superlunary-c/PEML-reproducibility.

Appendix A. Proofs and descent analysis

A.1. Proof of Proposition 1

We give the proof directly for vector-valued embeddings. All gradients are treated as column vectors. Let S denote the input domain, let $w \in \mathbb{R}^m$ be the model parameter vector, and let $f(\cdot; w) : S \to \mathbb{R}^{d_{\text{emb}}}$ be the embedding map. For two samples $s, q \in S$, define

$$a(w) = f(s; w), \qquad t(w) = f(q; w), \qquad \mathbf{z}_{s,q}(w) = \begin{pmatrix} a(w) \\ t(w) \end{pmatrix} \in \mathbb{R}^{2d_{\text{emb}}}. \tag{A.1}$$

Let $\ell : \mathbb{R}^{d_{\text{emb}}} \times \mathbb{R}^{d_{\text{emb}}} \to \mathbb{R}$ be a differentiable scalar pair objective in embedding space, and define its parameter-space composition by

$$\widetilde{\ell}_{s,q}(w) = \ell\big(a(w), t(w)\big) = \ell\big(\mathbf{z}_{s,q}(w)\big). \tag{A.2}$$

This distinction is important: ℓ treats the two embeddings as variables, whereas $\widetilde{\ell}_{s,q}$ accounts for their shared dependence on w.

Suppose that $f(s;\cdot)$ and $f(q;\cdot)$ are Fréchet differentiable at the current parameter point w. Define

$$\begin{aligned} J_s(w) &= D_w f(s; w) \in \mathbb{R}^{d_{\text{emb}} \times m}, \\ J_q(w) &= D_w f(q; w) \in \mathbb{R}^{d_{\text{emb}} \times m}, \\ J_{s,q}(w) &= D_w \mathbf{z}_{s,q}(w) = \begin{pmatrix} J_s(w) \\ J_q(w) \end{pmatrix} \in \mathbb{R}^{2d_{\text{emb}} \times m}. \end{aligned} \tag{A.3}$$

At the current embeddings, let

$$\mathbf{g}_{s,q} = \nabla_{\mathbf{z}_{s,q}} \ell = \begin{pmatrix} \mathbf{g}_a \\ \mathbf{g}_t \end{pmatrix}, \quad \begin{aligned} \mathbf{g}_a &= \nabla_a \ell(a, t), \\ \mathbf{g}_t &= \nabla_t \ell(a, t). \end{aligned} \tag{A.4}$$

The multivariate chain rule gives the exact parameter gradient

$$\nabla_w \widetilde{\ell}_{s,q}(w) = J_{s,q}(w)^\top \mathbf{g}_{s,q} = J_s(w)^\top \mathbf{g}_a + J_q(w)^\top \mathbf{g}_t. \tag{A.5}$$

No Taylor approximation is involved in Eq. (A.5).

Consider one direct, unpreconditioned parameter-gradient step

$$w^+ = w - \eta \nabla_w \widetilde{\ell}_{s,q}(w) = w - \eta J_{s,q}(w)^\top \mathbf{g}_{s,q}, \qquad \Delta w = w^+ - w, \qquad \eta > 0. \tag{A.6}$$

Fréchet differentiability of the vector-valued map $\mathbf{z}_{s,q}$ yields

$$\begin{aligned} &\mathbf{z}_{s,q}(w + \Delta w) = \mathbf{z}_{s,q}(w) + J_{s,q}(w)\Delta w + \mathbf{r}(\Delta w), \\ &\lim_{\|\Delta w\|_2 \to 0} \frac{\|\mathbf{r}(\Delta w)\|_2}{\|\Delta w\|_2} = 0. \end{aligned} \tag{A.7}$$

The vector $\mathbf{r}(\Delta w)$ contains all higher-order terms. More explicitly, suppose that $J_{s,q}$ is locally Lipschitz on a differentiable neighborhood containing the line segment from w to $w + \Delta w$, with

$$\big\| J_{s,q}(u) - J_{s,q}(v) \big\|_{\text{op}} \le L_J \|u - v\|_2. \tag{A.8}$$

The integral remainder then satisfies

$$\mathbf{r}(\Delta w) = \int_0^1 \big[J_{s,q}(w + \tau \Delta w) - J_{s,q}(w) \big] \Delta w \, d\tau,$$

$$\|\mathbf{r}(\Delta w)\|_2 \le \frac{L_J}{2} \|\Delta w\|_2^2. \tag{A.9}$$

If $\mathbf{z}_{s,q}$ is twice continuously differentiable along the update segment, the same remainder can equivalently be expressed coordinate-wise as

$$\begin{aligned} [\mathbf{r}(\Delta w)]_k &= \frac{1}{2} \Delta w^\top \nabla_w^2 [\mathbf{z}_{s,q}]_k (w + \theta_k \Delta w) \Delta w, \\ &\text{for some } \theta_k \in (0, 1), \quad k = 1, \dots, 2d_{\text{emb}}. \end{aligned} \tag{A.10}$$

Thus, the higher-order object is the collection of coordinate-Hessian contractions in Eq. (A.10); these terms are retained in the vector remainder rather than being identified with the first-order Jacobian chain rule.

Substitution of Eq. (A.6) into Eq. (A.7) gives the induced joint embedding update

$$\begin{aligned} \mathbf{z}_{s,q}^+ &= \mathbf{z}_{s,q}(w^+) = \mathbf{z}_{s,q} - \eta J_{s,q}(w) J_{s,q}(w)^\top \mathbf{g}_{s,q} + \mathbf{r}_\eta, \\ \mathbf{r}_\eta &= \mathbf{r}\big(-\eta J_{s,q}(w)^\top \mathbf{g}_{s,q}\big). \end{aligned} \tag{A.11}$$

For a fixed current parameter point, $\mathbf{r}_\eta = o(\eta)$ as $\eta \to 0$. Under the Lipschitz condition in Eq. (A.8), the explicit bound is

$$\big\| \mathbf{r}_\eta \big\|_2 \le \frac{L_J \eta^2}{2} \big\| J_{s,q}(w)^\top \mathbf{g}_{s,q} \big\|_2^2. \tag{A.12}$$

The Jacobian–Gram matrix in Eq. (A.11) is symmetric positive semidefinite because, for every $v \in \mathbb{R}^{2d_{\text{emb}}}$,

$$v^\top J_{s,q} J_{s,q}^\top v = \big\| J_{s,q}^\top v \big\|_2^2 \ge 0. \tag{A.13}$$

Consequently,

$$\mathbf{g}_{s,q}^\top \Big(\mathbf{z}_{s,q}^+ - \mathbf{z}_{s,q} \Big) = -\eta \big\| J_{s,q}(w)^\top \mathbf{g}_{s,q} \big\|_2^2 + o(\eta). \tag{A.14}$$

Equivalently, differentiability of the composite objective gives

$$\widetilde{\ell}_{s,q}(w^+) - \widetilde{\ell}_{s,q}(w) = -\eta \big\| J_{s,q}(w)^\top \mathbf{g}_{s,q} \big\|_2^2 + o(\eta). \tag{A.15}$$

Hence, if $J_{s,q}(w)^\top \mathbf{g}_{s,q} \neq 0$, the pair objective decreases for every sufficiently small positive step size. If the embeddings were optimized as free variables, the corresponding joint DMLS step would instead be $\mathbf{z}_{s,q}^+ = \mathbf{z}_{s,q} - \eta \mathbf{g}_{s,q}$. Therefore, Eq. (A.11) establishes a Jacobian–Gram-transformed first-order correspondence, rather than an unconditional identity, between the parameter-induced embedding update and the direct DMLS update.

We next specialize the joint result to the Euclidean pair objective used in this work. Let

$$\ell(a, t) = \mathcal{L}(d_{at}), \qquad d_{at} = \|a - t\|_2, \tag{A.16}$$

and suppose that $a \neq t$ and that $\mathcal{L}$ is differentiable at d_{at}. Direct differentiation gives

$$\mathbf{g}_a = \mathcal{L}'(d_{at}) \frac{a - t}{d_{at}}, \qquad \mathbf{g}_t = -\mathbf{g}_a. \tag{A.17}$$

The opposing-gradient identity in Eq. (A.17) follows from the dependence on the difference $a - t$; metric symmetry alone would not establish this identity. In this case, Eq. (A.5) becomes

$$\nabla_w \widetilde{\ell}_{s,q}(w) = \big(J_s(w) - J_q(w) \big)^\top \mathbf{g}_a. \tag{A.18}$$

Partitioning $\mathbf{r}_\eta = (\mathbf{r}_{a,\eta}^\top, \mathbf{r}_{t,\eta}^\top)^\top$, and writing $a^+ = a(w^+)$ and $t^+ = t(w^+)$, the two blocks of Eq. (A.11) are

$$\begin{aligned} a^+ &= a - \eta J_s(w) \big(J_s(w) - J_q(w) \big)^\top \mathbf{g}_a + \mathbf{r}_{a,\eta}, \\ t^+ &= t - \eta J_q(w) \big(J_s(w) - J_q(w) \big)^\top \mathbf{g}_a + \mathbf{r}_{t,\eta}. \end{aligned} \tag{A.19}$$

In particular, the first-order operator acting on the anchor gradient is

$$M_{s,q}(w) = J_s(w) \big(J_s(w) - J_q(w) \big)^\top \in \mathbb{R}^{d_{\text{emb}} \times d_{\text{emb}}}. \tag{A.20}$$

Unlike the joint Jacobian–Gram matrix, $M_{s,q}(w)$ need not be symmetric or positive semidefinite and has no general direction-preservation guarantee. For $\mathbf{g}_a \neq 0$, the first-order anchor displacement has the same direction as the direct DMLS displacement $-\mathbf{g}_a$ precisely when

$$M_{s,q}(w) \mathbf{g}_a = \lambda \mathbf{g}_a \qquad \text{for some } \lambda > 0. \tag{A.21}$$

Without this condition, the shared parameterization can transform or suppress the anchor-only direction. This blockwise observation does not alter the joint first-order descent relation in Eq. (A.14).

When $d_{\text{emb}} = 1$, Eq. (A.19) reduces to

$$a^+ = a - \eta\, \zeta_{s,q}(w) \frac{\partial \ell(a,t)}{\partial a} + r_{a,\eta}, \tag{A.22}$$

where

$$\zeta_{s,q}(w) = J_s(w)\big(J_s(w) - J_q(w)\big)^\top = \sum_{i=1}^{m} \frac{\partial f(s;w)}{\partial w_i}\left(\frac{\partial f(s;w)}{\partial w_i} - \frac{\partial f(q;w)}{\partial w_i}\right). \tag{A.23}$$

Thus, the previous scalar expression is recovered as a special case, but its first-order anchor direction agrees with the direct DMLS direction only when $\zeta_{s,q}(w) > 0$, rather than merely when $\zeta_{s,q}(w) \neq 0$. The notation also makes explicit that this factor depends on the samples, parameters, and network Jacobians, and is not in general determined solely by the embedding values (a, t).

For a complete mini-batch objective, the same argument applies by stacking all embeddings entering the objective into a vector $\mathbf{Z}(w)$ and defining $J_{\mathbf{Z}}(w) = D_w \mathbf{Z}(w)$. At any differentiable point, the induced update is

$$\mathbf{Z}^+ = \mathbf{Z} - \eta J_{\mathbf{Z}}(w) J_{\mathbf{Z}}(w)^\top \nabla_{\mathbf{Z}} \mathcal{L}_{\text{batch}}(\mathbf{Z}) + \mathbf{r}_{\text{batch},\eta}, \qquad \mathbf{r}_{\text{batch},\eta} = o(\eta). \tag{A.24}$$

This result is local to the current parameter point and, in stochastic training, conditional on the current mini-batch. It concerns a direct, unpreconditioned parameter-gradient step. Momentum and adaptive optimization introduce additional optimizer states or preconditioning, and their effects on the induced trajectory are discussed separately in Section I.5. □

A.2. Proof of Proposition 2

In the DMLS, the change in a feature f is

$$\Delta f = -\eta \frac{\partial \mathcal{L}}{\partial f}\,. \tag{A.25}$$

We compare the change in the generalized coordinate x induced by the force derived from the potential with Eq. (A.25). A metric-learning batch update can be viewed as the displacement of x over a time interval Δt. To model the discrete update, we assume that x returns to its stationary state after each displacement. Over a sufficiently short interval, the force and, consequently, the acceleration a can be treated as constant. Newton's second law then gives

$$\begin{aligned} m\mathrm{a} &= F \\ &= -\frac{\partial V}{\partial x}\,, \end{aligned} \tag{A.26}$$

where m is the mass of the charge. Under constant acceleration, the corresponding displacement is

$$\Delta x = v_0 \Delta \mathrm{t} + \frac{1}{2}\mathrm{a}(\Delta \mathrm{t})^2\,, \tag{A.27}$$

where Δx denotes the displacement of x and v_0 denotes the initial velocity. Because x is returned to a stationary state before each displacement, $v_0 = 0$, and hence

$$\begin{aligned} \Delta x &= \frac{(\Delta \mathrm{t})^2}{2}\mathrm{a} \\ &= -\frac{(\Delta \mathrm{t})^2}{2m}\frac{\partial V}{\partial x}\,. \end{aligned} \tag{A.28}$$

Comparing Eq. (A.25) and Eq. (A.28) shows that, with $\eta = (\Delta \mathrm{t})^2/(2m)$ and $\mathcal{L} = V$, the two updates share the same negative-gradient form under the stated idealized short-time, uniform-force approximation. This correspondence does not imply that the full stochastic mini-batch trajectory is a physically exact continuous Lagrangian evolution. A direct discrete descent analysis that does not rely on this physical-displacement analogy is provided in Section A.3. □

A.3. Descent under directed asymmetric pair potentials

Let $X = (x_1, \ldots, x_N) \in \mathbb{R}^{Nd}$ denote the collection of generalized coordinates. For each ordered pair (i, j) with $i \neq j$, let $V_{ij}(X)$ denote a directed pair-potential contribution. We do not assume pairwise symmetry; that is, $V_{ij}(X)$ and $V_{ji}(X)$ may be different. The scalar total potential is defined as

$$V_{\text{total}}(X) = \sum_{i \neq j} V_{ij}(X)\,. \tag{A.29}$$

Because the summation is taken over ordered pairs, V_{ij} and V_{ji} are treated as distinct scalar contributions. Therefore, no factor $1/2$ is introduced, unlike in symmetric pair-potential formulations. The net effective force acting on the whole coordinate vector X is defined by the negative gradient of this scalar potential:

$$\mathcal{F}(X) = -\nabla_X V_{\text{total}}(X)\,. \tag{A.30}$$

Equivalently, the force acting on coordinate x_i is

$$\begin{aligned} \mathcal{F}_i(X) &= -\nabla_{x_i} V_{\text{total}}(X) \\ &= -\sum_{p \neq q} \nabla_{x_i} V_{pq}(X)\,. \end{aligned} \tag{A.31}$$

Thus, the descent direction is determined by the gradient of the scalar total potential, not by Newton's third-law action–reaction symmetry between each ordered pair. In particular, neither $V_{ij} = V_{ji}$ nor a pairwise action–reaction relation between directed force contributions is required for descent.

Consider the discrete generalized-coordinate update

$$X^+ = X + \alpha \mathcal{F}(X) = X - \alpha \nabla_X V_{\text{total}}(X)\,, \tag{A.32}$$

where $\alpha > 0$ is the effective step size. In the mechanical discretization used in Proposition 2, $\alpha = (\Delta t)^2/(2m)$. Suppose that V_{total} is differentiable and has an L-Lipschitz continuous gradient in a neighborhood of X. Then, by the standard descent lemma for L-smooth functions,

$$\begin{aligned} V_{\text{total}}(X^+) \leq V_{\text{total}}(X) &+ \nabla_X V_{\text{total}}(X)^\top (X^+ - X) \\ &+ \frac{L}{2}\|X^+ - X\|^2\,. \end{aligned} \tag{A.33}$$

Substituting Eq. (A.32) into Eq. (A.33) gives

$$\begin{aligned} V_{\text{total}}(X^+) &\leq V_{\text{total}}(X) - \alpha \|\nabla_X V_{\text{total}}(X)\|^2 + \frac{L\alpha^2}{2}\|\nabla_X V_{\text{total}}(X)\|^2 \\ &= V_{\text{total}}(X) - \alpha\left(1 - \frac{\alpha L}{2}\right)\|\nabla_X V_{\text{total}}(X)\|^2\,. \end{aligned} \tag{A.34}$$

Therefore, if $0 < \alpha < 2/L$, then

$$V_{\text{total}}(X^+) < V_{\text{total}}(X) \quad \text{whenever} \quad \nabla_X V_{\text{total}}(X) \neq 0\,. \tag{A.35}$$

This establishes one-step monotonic decrease of the scalar total potential under the discrete update. Importantly, the proof does not use pairwise force symmetry. It only requires that the directed pair-potential contributions define a scalar total potential V_{total} and that the update direction is given by $-\nabla_X V_{\text{total}}$.

This result should be understood as a local descent property for a fixed scalar potential, such as the potential associated with a given mini-batch and a fixed active interaction configuration. It does not imply that the stochastic mini-batch training trajectory monotonically decreases the full empirical objective at every iteration.

Appendix B. Examination and derivation of the monotonicity properties of SupCon in Section 3.2.1

The softmax-based losses considered here, including SupCon and InfoNCE, can be expressed as

$$\mathcal{L}_s = \sum_{a \in \mathcal{A}} \frac{-1}{|P(a)|} \sum_{p \in P(a)} \log \frac{\exp(a \cdot p/\tau)}{\sum_{t \in A(a)} \exp(a \cdot t/\tau)}\,, \tag{B.1}$$

where $\tau > 0$ is a temperature and $A(a) = P(a) \cup N(a)$. Note that in SupCon, $P(a)$ denotes the set of batch samples that share the same class label as a (consistent with our notation), whereas in InfoNCE, $P(a)$ comprises the other views of a; with two views, $|P(a)| = 1$. Let $\ell(a)$ denote the per-anchor contribution, so that $\mathcal{L}_s = \sum_{a\in\mathcal{A}} \ell(a)$. The gradient of $\ell(a)$ is

$$\frac{\partial \ell(a)}{\partial a} = \frac{1}{\tau}\left(\sum_{t\in A(a)} \pi_a(t)t - \frac{1}{|P(a)|} \sum_{p\in P(a)} p \right) \tag{B.2}$$

$$= \frac{1}{\tau}\left(\Psi_{pos} + \Psi_{neg}\right), \tag{B.3}$$

where

$$\pi_a(t) = \frac{\exp(a\cdot t/\tau)}{\sum_{u\in A(a)} \exp(a\cdot u/\tau)}, \tag{B.4}$$

and

$$\Psi_{pos} = \sum_{p\in P(a)} \left(\pi_a(p) - \frac{1}{|P(a)|} \right) p, \tag{B.5}$$

$$\Psi_{neg} = \sum_{n\in N(a)} \pi_a(n) n. \tag{B.6}$$

Then, to investigate the intrinsic hard mining property, we apply ℓ_2 normalization, $(a = \alpha/\|\alpha\|)$:

$$\frac{\partial \ell}{\partial \alpha} = \frac{I - aa^\top}{\|\alpha\|} \frac{\partial \ell(a)}{\partial a} \tag{B.7}$$

$$= \frac{1}{\tau\|\alpha\|}\left(\Pi_{pos} + \Pi_{neg}\right), \tag{B.8}$$

where

$$\Pi_{pos} = \sum_{p\in P(a)} (p - (a\cdot p)a)\left(\pi_a(p) - \frac{1}{|P(a)|} \right), \tag{B.9}$$

$$\Pi_{neg} = \sum_{n\in N(a)} (n - (a\cdot n)a)\pi_a(n). \tag{B.10}$$

For ℓ_2-normalized a and t, define $\theta_{at} = \arccos(a^\top t) \in [0, \pi]$. We restrict the following monotonicity analysis to the nonnegative-similarity regime $\theta_{at} \in [0, \pi/2]$. Since

$$\left\| t - (a^\top t)a \right\|_2 = \sin\theta_{at}, \tag{B.11}$$

the magnitudes of the individual pairwise terms appearing in Π_{pos} and Π_{neg} are

$$\mu_p(a, p; \mathcal{A}) = \left| \pi_a(p) - \frac{1}{|P(a)|} \right| \sin\theta_{ap}, \tag{B.12}$$

$$\mu_n(a, n; \mathcal{A}) = \pi_a(n) \sin\theta_{an}. \tag{B.13}$$

These quantities are deterministic for a fixed anchor and current mini-batch. The corresponding terms in $\partial\ell/\partial\alpha$ additionally share the common positive factor $1/(\tau\|\alpha\|_2)$.

To examine their angular dependence in the nonnegative-similarity regime, select a candidate $t \in A(a)$, hold all remaining anchor–candidate logits fixed, and write

$$S_{a,-t} = \sum_{u\in A(a)\setminus\{t\}} \exp(a^\top u/\tau), \tag{B.14}$$

$$\pi_a(\theta) = \frac{\exp(\cos\theta/\tau)}{\exp(\cos\theta/\tau) + S_{a,-t}}, \qquad \theta \in [0, \pi/2]. \tag{B.15}$$

Here, $S_{a,-t} > 0$ in the contrastive setting considered. Differentiating the softmax weight gives

$$\frac{d\pi_a(\theta)}{d\theta} = -\frac{1}{\tau}\pi_a(\theta)\left(1 - \pi_a(\theta)\right)\sin\theta. \tag{B.16}$$

For a negative pair, $\mu_n(\theta) = \pi_a(\theta)\sin\theta$, and hence

$$\frac{d\mu_n}{d\theta} = \pi_a(\theta)\left(\cos\theta - \frac{1 - \pi_a(\theta)}{\tau} \sin^2\theta \right). \tag{B.17}$$

Let

$$h_n(\theta) = \cos\theta - \frac{1 - \pi_a(\theta)}{\tau} \sin^2\theta. \tag{B.18}$$

For $\theta \in (0, \pi/2)$,

$$\frac{dh_n}{d\theta} = -\sin\theta - \frac{\pi_a(\theta)\left(1 - \pi_a(\theta)\right)}{\tau^2} \sin^3\theta \tag{B.19}$$

$$- \frac{2\left(1 - \pi_a(\theta)\right)}{\tau} \sin\theta\cos\theta < 0. \tag{B.20}$$

Moreover, $h_n(0) = 1$ and $h_n(\pi/2) = -(1 - \pi_a(\pi/2))/\tau < 0$. Therefore, h_n has a unique zero on $(0, \pi/2)$, and μ_n increases from zero, attains a unique interior maximum, and then decreases toward $\theta = \pi/2$. In particular, its magnitude approaches zero as $\theta \to 0$.

For a positive pair, suppose that $\pi_a(\theta) \le 1/|P(a)|$ throughout $\theta \in [0, \pi/2]$. Then

$$\mu_p(\theta) = \left(\frac{1}{|P(a)|} - \pi_a(\theta) \right) \sin\theta, \tag{B.21}$$

and

$$\frac{d\mu_p}{d\theta} = \frac{\pi_a(\theta)\left(1 - \pi_a(\theta)\right)}{\tau} \sin^2\theta + \left(\frac{1}{|P(a)|} - \pi_a(\theta) \right) \cos\theta \ge 0. \tag{B.22}$$

Thus, under the stated condition, the positive-pair contribution is non-decreasing with angular separation on $[0, \pi/2]$.

Appendix C. Alternative monotone force kernels

The main-text ablation compares the trigonometric force kernel used by PEML with linear, quadratic, and normalized exponential alternatives. This appendix formalizes their common admissibility conditions and derives a unified force–potential construction while retaining the adaptive interaction ranges and cardinality-weighted aggregation of the original PEML formulation. Across this family, only the scalar kernel g and its corresponding primitive G change. The candidate kernel–primitive pairs are summarized in Table 7, and their empirical comparison is reported in Section 4.6.

C.1. Admissibility conditions

Let g denote a scalar force-magnitude profile defined on a normalized active interval. We consider continuously differentiable functions satisfying

$$\begin{aligned} &g \in C^1([0,1]), \qquad g : [0,1] \to [0,1], \\ &g(0) = 0, \qquad g(1) = 1, \qquad g'(u) \ge 0 \quad \text{for all } u \in [0,1]. \end{aligned} \tag{C.1}$$

The monotonicity condition preserves the intrinsic hard-mining behavior described in Section 3.2.1: the scalar force magnitude cannot decrease as normalized pair hardness increases. The conditions $g : [0,1] \to [0,1]$, $g(0) = 0$, and $g(1) = 1$ place the instantaneous per-pair force magnitude on the common normalized range $0 \le g(u) \le 1$ across kernels and anchors. In particular, the force vanishes continuously at the inactive boundary and reaches unit magnitude at the hardest endpoint. This endpoint normalization makes the attainable per-pair ranges comparable under the cardinality weighting in Eqs. (33) and (34); it neither equalizes the integrated force magnitude across kernels nor requires equality of the resulting positive and negative force vectors, which also depend on pair geometry.

For any admissible g, define its zero-baseline primitive as

$$G(u) = \int_0^u g(t)\,dt, \qquad G(0) = 0. \tag{C.2}$$

Because g is continuous on the compact interval $[0,1]$, G exists and is finite. A closed-form primitive is therefore not an additional admissibility condition. The candidates considered here are selected to admit closed-form primitives so that their induced pairwise potentials can be implemented and compared explicitly. Defining g on a dimensionless

interval also allows the same profile to be applied to anchor-dependent interaction ranges without introducing a kernel-specific range parameter.

C.2. Unified margin-aware force and potential formulation

We retain the adaptive thresholds $k_p(a)$ and $k_n(a)$ defined in Eqs. (40) and (41). In the following derivative calculations, these thresholds are computed from the current embedding configuration and then held fixed, consistently with the detached-threshold implementation analyzed in Section I. We work in the non-degenerate regime $k_p(a) < d_{\max}$ and $k_n(a) > 0$, corresponding to the strict-interior settings $0 < \rho_p, \rho_n < 1$ used in our experiments. For distances within their respective active intervals, define the normalized hardness variables

$$u_{ap} = \frac{d_{ap} - k_p(a)}{d_{\max} - k_p(a)}, \qquad d_{ap} \in [k_p(a), d_{\max}], \tag{C.3}$$

$$u_{an} = 1 - \frac{d_{an}}{k_n(a)}, \qquad d_{an} \in [0, k_n(a)]. \tag{C.4}$$

Both variables lie in $[0, 1]$, with $u = 0$ corresponding to the inactive boundary and $u = 1$ to the hardest endpoint. Because g is non-decreasing, the positive force magnitude increases with d_{ap}, whereas the negative force magnitude decreases with d_{an}.

For $d_{ap}, d_{an} > 0$, let $\hat{r}_{ap} = (p - a)/d_{ap}$ and $\hat{r}_{an} = (a - n)/d_{an}$, following the attractive and repulsive direction conventions in Eqs. (42) and (43). For an admissible kernel g, the positive and negative forces are

$$F_P^{(g)}(d_{ap}) = g(u_{ap})\mathbf{1}_{[k_p(a), d_{\max}]}(d_{ap})\hat{r}_{ap}, \tag{C.5}$$

$$F_N^{(g)}(d_{an}) = g(u_{an})\mathbf{1}_{[0, k_n(a)]}(d_{an})\hat{r}_{an}. \tag{C.6}$$

These analytic vector expressions apply on the non-collision domain; the inclusion convention at the inactive boundary is immaterial because $g(0) = 0$. As in the original formulation, the pairwise-distance denominators in the unit directions are regularized by a small numerical constant in the implementation.

Applying the force–potential construction used in Section 3.4, together with Eq. (C.2), gives the following continuous zero-baseline potentials on the feasible distance domain:

$$V_P^{(g)}(d_{ap}) = \begin{cases} 0, & 0 \le d_{ap} \le k_p(a), \\ (d_{\max} - k_p(a))\, G\left(\dfrac{d_{ap} - k_p(a)}{d_{\max} - k_p(a)}\right), & k_p(a) < d_{ap} \le d_{\max}, \end{cases} \tag{C.7}$$

$$V_N^{(g)}(d_{an}) = \begin{cases} k_n(a)\, G\left(1 - \dfrac{d_{an}}{k_n(a)}\right), & 0 \le d_{an} < k_n(a), \\ 0, & k_n(a) \le d_{an} \le d_{\max}. \end{cases} \tag{C.8}$$

Holding the thresholds fixed, the interval-length factors cancel the factors introduced by the chain rule. Within the interiors of the active intervals, these potentials satisfy

$$\frac{\partial V_P^{(g)}(d_{ap})}{\partial d_{ap}} = g(u_{ap}), \tag{C.9}$$

$$\frac{\partial V_N^{(g)}(d_{an})}{\partial d_{an}} = -g(u_{an}). \tag{C.10}$$

Since $G(0) = g(0) = 0$, each scalar potential is continuously differentiable at its active–inactive junction. With the thresholds held fixed, $\nabla_a d_{ap} = -\hat{r}_{ap}$ and $\nabla_a d_{an} = \hat{r}_{an}$. Consequently, on the interiors of the active intervals, $-\nabla_a V_P^{(g)} = g(u_{ap})\hat{r}_{ap}$ and $-\nabla_a V_N^{(g)} = g(u_{an})\hat{r}_{an}$, which are precisely the attractive and repulsive forces in Eqs. (C.5) and (C.6). If $k_p(a)$ and $k_n(a)$ were differentiated through, additional chain-rule and cross-pair terms would arise; the identities above therefore describe the fixed- or detached-threshold force field used by PEML. Substituting $V_P^{(g)}$ and $V_N^{(g)}$ for V_P and V_N in Eq. (48) yields the corresponding kernel-specific PEML objective without changing its cardinality-weighted aggregation.

C.3. Candidate kernels and endpoint properties

The four kernel–primitive pairs used in the experiment are summarized in Table 7. Each candidate satisfies Eq. (C.1) and has a finite closed-form primitive. In particular, substituting the trigonometric pair into Eqs. (C.5) and (C.6) recovers the original forces in Eqs. (42) and (43); substituting its primitive into Eqs. (C.7) and (C.8) similarly recovers the original potentials in Eqs. (46) and (47).

Although all four profiles are admissible, their endpoint slopes differ:

$$\left(g'_{\text{tri}}(0), g'_{\text{tri}}(1)\right) = (0, 0), \tag{C.11}$$

$$\left(g'_{\text{lin}}(0), g'_{\text{lin}}(1)\right) = (1, 1), \tag{C.12}$$

$$\left(g'_{\text{quad}}(0), g'_{\text{quad}}(1)\right) = (0, 2), \tag{C.13}$$

$$\left(g'_{\text{exp}}(0), g'_{\text{exp}}(1)\right) = \left(\frac{1}{e-1}, \frac{e}{e-1}\right). \tag{C.14}$$

The trigonometric profile is the only candidate considered here whose scalar force magnitude has zero first-order variation with respect to normalized hardness at both endpoints. Because the hardness transformations in Eqs. (C.3) and (C.4) are affine and have strictly positive interval-length denominators, the corresponding distance-domain slopes are $g'(u_{ap})/(d_{\max} - k_p(a))$ and $-g'(u_{an})/k_n(a)$, respectively. Hence, the trigonometric profile also has zero one-sided force-magnitude slope at both the inactive and hardest endpoints in the distance domain. This is a local endpoint property; it does not by itself imply a smaller global Lipschitz constant, lower gradient variance, improved convergence, or greater numerical stability.

The linear profile provides the simplest reference, while the quadratic and normalized exponential profiles provide fixed polynomial and exponential alternatives, respectively. Although all candidates share the same endpoint normalization, their normalized areas differ: $G_{\text{tri}}(1) = G_{\text{lin}}(1) = 1/2$, $G_{\text{quad}}(1) = 1/3$, and $G_{\text{exp}}(1) = (e-2)/(e-1)$. Thus, endpoint normalization does not equalize their integrated force magnitude.

Appendix D. Class-imbalanced data construction

This section specifies the two class-imbalanced data constructions used in Section 4.4. The synthetic construction directly optimizes sample coordinates and is intended to isolate the embedding-space effect of cardinality-weighted force balancing. The CIFAR100-LT construction instead evaluates the same mechanism in an end-to-end encoder-training setting. Because these experiments operate on different sample spaces and use different evaluation protocols, their constructions are described separately.

D.1. Synthetic embedding-evolution construction

We use $C = 10$ classes in $d = 3$ dimensions. The class means are placed on a sphere of radius $R = 5$ using deterministic midpoint Fibonacci directions on $\mathbb{S}^2$ (Marques et al., 2013). For $j = 1, \dots, C$, the directions are

$$\begin{aligned} z_j &= 1 - \frac{2(j - \frac{1}{2})}{C}, \\ \varphi_j &= (j-1)\pi(3 - \sqrt{5}), \\ \boldsymbol{u}_j &= \left(\sqrt{1 - z_j^2}\cos\varphi_j,\ \sqrt{1 - z_j^2}\sin\varphi_j,\ z_j\right), \end{aligned} \tag{D.1}$$

and $\boldsymbol{\mu}_j = R\boldsymbol{u}_j$. For each seed, a balanced master pool of $n_{\max} = 1{,}000$ samples per class is generated as

$$\boldsymbol{x}_{j,i} \sim \mathcal{N}\left(\boldsymbol{\mu}_j, \delta^2 I_3\right), \qquad \delta = 2. \tag{D.2}$$

Here, I_3 denotes the 3×3 identity matrix. The use of class-conditional Gaussian mixtures follows controlled analyzes of contrastive representation learning (Bansal et al., 2025; Saunshi et al., 2019).

Following standard long-tail constructions (Cao et al., 2019; Cui et al., 2019; Liu et al., 2019; Ye et al., 2023), the number of samples assigned to frequency rank j is

$$n_j(\mathrm{IR}) = \max\left\{n_{\min}, \mathrm{round}_{\mathrm{even}}\left(n_{\max}\,\mathrm{IR}^{-\frac{j-1}{C-1}}\right)\right\}, \qquad j = 1,\dots,C, \tag{D.3}$$

where $n_{\min} = 2$ and $\mathrm{round}_{\mathrm{even}}$ denotes rounding to the nearest integer with half-integer ties resolved to the nearest even integer, matching the behavior of `numpy.round` used in the dataset construction. We evaluate $\mathrm{IR} \in \{1, 10, 50, 100, 200, 500\}$. A seed-specific permutation assigns the frequency ranks to the fixed center directions, preventing a particular direction or local center configuration from being systematically associated with the head or tail. Each imbalanced dataset is formed using class-wise prefixes of the same balanced master pool; consequently, the datasets within a seed are nested as the IR increases.

Dimension-aware overlap calibration. For $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \delta^2 I_3)$, the root-mean-square noise magnitude is $\sqrt{\mathbb{E}\|\boldsymbol{\varepsilon}\|_2^2} = \delta\sqrt{3}$. The selected configuration therefore gives

$$\frac{R}{\delta\sqrt{3}} = \frac{5}{2\sqrt{3}} \approx 1.443. \tag{D.4}$$

The actual midpoint-Fibonacci center geometry has nearest and median pairwise separations of 4.8837 and 7.4365, respectively. For two equal-prior Gaussian components with covariance $\delta^2 I_3$, the corresponding pairwise Bayes error is

$$P_{e,jk} = \Phi\left(-\frac{\|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\|_2}{2\delta}\right), \tag{D.5}$$

where Φ is the standard normal cumulative distribution function. This gives an error of 11.11% for the nearest center pair and 3.15% for the median center pair. These values are pairwise equal-prior overlap diagnostics, not the Bayes error of the full imbalanced multiclass mixture. They show that the selected geometry retains appreciable overlap for the most confusable classes and nonzero overlap for a typical class pair, rather than producing a trivially separable dataset. The experiment is explicitly three-dimensional; for another dimension or center layout, R and δ should be recalibrated using both $R/(\delta\sqrt{d})$ and the resulting standardized center separations. The complete seed-specific construction of the nested datasets is summarized in Algorithm 1.

Optimization and evaluation protocol. Each selected coordinate is treated as a trainable embedding. For every sampled coordinate $\boldsymbol{x}_i$, two independent stochastic views are generated as

$$\boldsymbol{v}_i^{(a)} = \left(1 + \epsilon_i^{(a)}\right)\boldsymbol{x}_i + \boldsymbol{\xi}_i^{(a)}, \qquad a \in \{1, 2\}, \tag{D.6}$$

where $\epsilon_i^{(a)} \sim \mathcal{N}(0, 0.05^2)$ and $\boldsymbol{\xi}_i^{(a)} \sim \mathcal{N}(\mathbf{0}, 0.05^2 I_3)$. The perturbations are independent across views. Within every $(\mathrm{IR}, \mathrm{seed})$ pair, SupCon and PEML share the initial coordinates, retained samples, mini-batch schedule, and all augmentation random numbers.

Algorithm 1 Synthetic embedding-evolution dataset construction.

Require: Number of classes C, radius R, isotropic standard deviation δ, maximum count $n_{\max}$, lower bound $n_{\min}$, IR set $\mathcal{I}$, and seed s.

Generate midpoint Fibonacci directions $\{\boldsymbol{u}_j\}_{j=1}^{C}$ on $\mathbb{S}^2$ and set $\boldsymbol{\mu}_j \leftarrow R\boldsymbol{u}_j$.
for $j = 1$ to C **do**
  for $i = 1$ to $n_{\max}$ **do**
    Sample $\boldsymbol{x}_{j,i} \sim \mathcal{N}(\boldsymbol{\mu}_j, \delta^2 I_3)$.
  end for
end for
Draw a seed-specific permutation π_s assigning frequency ranks to center directions.
for $\mathrm{IR} \in \mathcal{I}$ **do**
  for $r = 1$ to C **do**
    Compute $n_r(\mathrm{IR})$ using Eq. (D.3).
    Add the prefix $\{(\boldsymbol{x}_{\pi_s(r),i}, \pi_s(r)) : 1 \le i \le n_r(\mathrm{IR})\}$ to $\mathcal{D}_{\mathrm{IR}}$.
  end for
end for
return $\{\mathcal{D}_{\mathrm{IR}} : \mathrm{IR} \in \mathcal{I}\}$.

Training uses a batch size of 1024 for 500 epochs. We use SGD with zero momentum and weight decay, no warmup, a cosine schedule updated at every optimizer step, and `drop_last=true`. Training is performed in FP32 without gradient clipping, coordinate clipping, or a norm penalty. SupCon uses temperature 0.1 and applies sample-wise ℓ_2 normalization to both views. PEML uses $(\rho_p, \rho_n) = (0.5, 0.4)$ and applies the annulus-scaled projection with $(R_A, R_B, \tau_r) = (1, 2, 0.5)$ to the joint two-view mini-batch. The method-specific optimization settings are selected using direct gradient- and raw-coordinate-update diagnostics so that the reported comparison uses the closest empirically balanced scale among the evaluated controls.

The primary evaluation uses the unique, unaugmented raw coordinates at the training endpoint. We apply k-means with `n_init=50`, `max_iter=500`, `random_state=1`, and the Lloyd algorithm, and report NMI, AMI, ARI, Q_{final}, and G_Q. All reported results are arithmetic means over seeds $\{0, 1, 2, 3, 4\}$.

D.2. Long-tailed resampling of CIFAR-100

For CIFAR100-LT, we apply the exponential class-count schedule in Eq. (D.3) to the 100 CIFAR-100 training classes, with $n_{\max} = 500$ and $n_{\min} = 2$. We evaluate nominal $\mathrm{IR} \in \{10, 50, 100, 200\}$ while retaining the balanced CIFAR-100 test set. Because the class counts are integer-valued, a nominal imbalance ratio need not equal the realized ratio af-

Nominal IR	Total samples	Min count
10	19629	50
50	12655	10
100	10899	5
200	9548	2

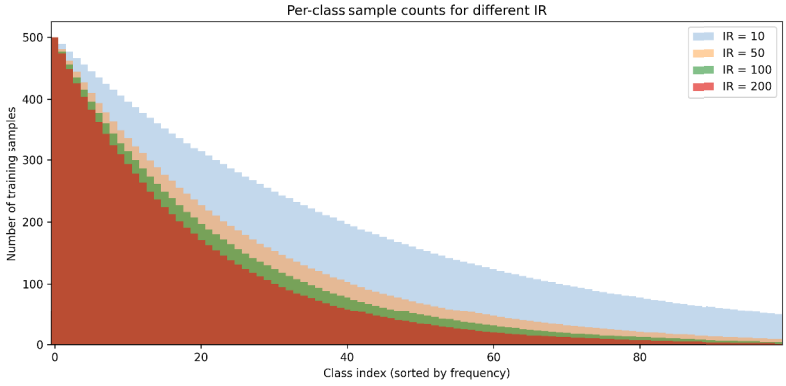


Fig. D.1. Total and per-class training sample counts for the evaluated nominal imbalance ratios.

ter rounding. In particular, nominal IR = 200 yields a head-class count of 500 and a tail-class count of 2, corresponding to a realized head-to-tail ratio of 250. The resulting total and per-class training counts are summarized below.

Fig. D.1 summarizes the class-imbalanced CIFAR-100 variants used in our experiments. The left table reports the total number of training samples and the minimum per-class count for each nominal IR, while the right panel shows the complete per-class count profiles with classes ordered by decreasing frequency. As the nominal IR increases, the head classes retain comparatively large sample counts while the tail classes receive progressively fewer samples. The distinction between nominal and realized imbalance is relevant only where integer rounding changes an endpoint count; all methods at a given setting use the same realized split.

Appendix E. Objective-level comparison with dedicated long-tailed contrastive objectives

This section compares how KCL, DSCL, PaCo, ProCo, and PEML address cardinality in their representation objectives. The comparison is mechanistic rather than empirical for KCL, DSCL, and PaCo, which are not evaluated in Section 4.4.2. Moreover, it isolates the representation objective of each method from additional components in its native training framework. For an anchor i, let $\boldsymbol{z}_i$ be its normalized representation, $\boldsymbol{z}_i^+$ its paired augmented view, P_i the set of all same-class positives (including $\boldsymbol{z}_i^+$), $P_i^\circ = P_i \setminus \{\boldsymbol{z}_i^+\}$ the remaining same-class representations, $N_{i,c}$ the negative representations of class c, and C_i^- the set of negative classes represented in the contrast set. We write $s_{it} = \boldsymbol{z}_i^\top \boldsymbol{z}_t/\tau$.

K-positive contrastive learning. KCL controls positive cardinality by selecting at most k additional same-class representations for each anchor (Kang et al., 2021). Let $S_i^{(k)} \subseteq P_i^\circ$, $|S_i^{(k)}| \le k$, and $\widetilde{P}_i = \{\boldsymbol{z}_i^+\} \cup S_i^{(k)}$. Its positive aggregation can be written as

$$\mathcal{L}_{\mathrm{KCL},i} = -\frac{1}{|\widetilde{P}_i|} \sum_{p \in \widetilde{P}_i} \log \frac{\exp(s_{ip})}{Z_i^{\mathrm{KCL}}}, \tag{E.1}$$

where Z_i^{KCL} is the softmax denominator over the paired key and the labeled memory queue in the native implementation. Restricting $|\widetilde{P}_i|$ prevents the number of selected positives from increasing with head-class frequency. The denominator, however, remains a flat sum over sample representations; it does not average negative contributions within each class and then equally across negative classes.

Decoupled supervised contrastive loss. The DSCL objective distinguishes the paired augmented view from the remaining same-class positives (Xuan & Zhang, 2024). If

$$q_i(t) = \frac{\exp(s_{it})}{Z_i^{\mathrm{DSCL}}}, \tag{E.2}$$

where Z_i^{DSCL} is the sample-wise contrastive denominator, its loss can be expressed as

$$\mathcal{L}_{\mathrm{DSCL},i} = -\alpha \log q_i(i^+) - \frac{1-\alpha}{|P_i^\circ|} \sum_{p \in P_i^\circ} \log q_i(p), \qquad \alpha \in [0,1]. \tag{E.3}$$

Thus, the paired view receives total target weight α, while all other same-class positives jointly receive weight $1-\alpha$, independently of $|P_i^\circ|$. This corrects the relative weighting of the two positive types, but Z_i^{DSCL} retains sample-wise negative aggregation. The full DSCL framework additionally uses a momentum encoder, a memory queue, and patch-based self-distillation; the comparison here concerns only the DSCL representation loss in Eq. (E.3).

Parametric contrastive learning. PaCo combines non-parametric sample contrast with one learnable center $\boldsymbol{c}_j$ for each class j (Cui et al., 2021). Let

$$D_i^{\mathrm{PaCo}} = \sum_{a \in A_i} \exp(s_{ia}) + \sum_{j=1}^{C} \phi_j(i), \tag{E.4}$$

where A_i denotes the sample candidate pool. In the long-tailed center-learning formulation, $\phi_j(i) = q(j)\exp(\boldsymbol{c}_j^\top \boldsymbol{h}_i/\tau)$, where $q(j)$ is the training class frequency and $\boldsymbol{h}_i$ is the encoder representation. The corresponding objective has the form

$$\mathcal{L}_{\mathrm{PaCo},i} = -\frac{1}{1+\alpha|P_i|}\left[\log \frac{\phi_{y_i}(i)}{D_i^{\mathrm{PaCo}}} + \alpha \sum_{p \in P_i} \log \frac{\exp(s_{ip})}{D_i^{\mathrm{PaCo}}}\right]. \tag{E.5}$$

The learnable centers supply one parametric representative per class, and the long-tailed variant incorporates class-frequency information into the center terms. By contrast, the sample-negative component in Eq. (E.4) remains a flat sum and is not explicitly normalized within and across negative classes.

Probabilistic contrastive learning. ProCo replaces finite contrast sets with expectations under online class-conditional feature distributions (Du et al., 2024). It models normalized representations by a mixture of von Mises–Fisher distributions with class prior π_j, mean direction $\boldsymbol{\mu}_j$, and concentration κ_j. Defining

$$r_{ij} = \frac{C_p\left(\left\|\kappa_j \boldsymbol{\mu}_j + \boldsymbol{z}_i/\tau\right\|_2\right)}{C_p(\kappa_j)}, \tag{E.6}$$

Table E.1
Objective-level comparison of cardinality mechanisms in dedicated long-tailed contrastive objectives and PEML. Additional components refer to native training formulations; the comparison isolates their representation objectives.

Method	Main cardinality mechanism	Additional training structure	Negative aggregation relative to PEML
KCL	Caps selected same-class positives at k	Momentum encoder and labeled queue	Flat sample-wise queue denominator
DSCL-loss	Fixes aggregate weights of the paired view and other positives	Momentum encoder and queue in the native framework	Flat sample-wise denominator
PaCo	Adds one learnable center per class and weights sample/center contrast	Learnable centers, momentum encoder, and queue	Class-complete center terms; sample negatives remain unnormalized by class
ProCo-Rep	Replaces finite pairs by online class-distribution expectations	Online $(\boldsymbol{\mu}_j, \kappa_j)$ estimators	One term per class, weighted by the class prior π_j
PEML	Averages positives and applies within-class/among-class averaging to negatives	No center, queue, prior, or distribution estimator	Equal outer weight for each represented negative class

where p is the representation dimension and, following the convention used in the ProCo formulation,

$$C_p(\kappa) = \frac{(2\pi)^{p/2} I_{p/2-1}(\kappa)}{\kappa^{p/2-1}}, \tag{E.7}$$

with I_ν denoting the modified Bessel function of the first kind. Thus, C_p is the normalization integral appearing in the denominator of the von Mises–Fisher density. Under this convention, the adopted representation objective is

$$\mathcal{L}_{\text{ProCo},i} = -\log\left(\pi_{y_i} r_{i,y_i}\right) + \log\left(\sum_{j=1}^{C} \pi_j r_{ij}\right). \tag{E.8}$$

The distribution parameters are estimated from class-wise first moments accumulated online rather than learned as proxy parameters. This construction addresses finite-pair scarcity by replacing empirical pair sums with a closed-form distributional expectation. Its class terms are nevertheless weighted by π_j rather than averaged uniformly over the negative classes represented for an anchor. The matched experiment in Section 4.4.2 uses Eq. (E.8) alone and therefore denotes it as ProCo-Rep. The full ProCo framework instead jointly optimizes

$$\mathcal{L}_{\text{full-ProCo}} = \mathcal{L}_{\text{LA}} + \alpha \mathcal{L}_{\text{ProCo}}, \tag{E.9}$$

where $\mathcal{L}_{\text{LA}}$ is a logit-adjusted classification loss.

PEML. PEML retains finite sample-wise pair interactions and applies cardinality normalization directly to their aggregate forces. For each anchor, its total attractive and repulsive contributions are

$$\begin{aligned} F_P^{\text{tot}}(i) &= \frac{1}{|P_i|} \sum_{p \in P_i} F_P(d_{ip}), \\ F_N^{\text{tot}}(i) &= \frac{1}{|C_i^-|} \sum_{c \in C_i^-} \left(\frac{1}{|N_{i,c}|} \sum_{n \in N_{i,c}} F_N(d_{in}) \right). \end{aligned} \tag{E.10}$$

This restates the aggregation in Eqs. (33) and (34). The positive term is averaged over samples, whereas the negative term is averaged first within each negative class and then across the represented negative classes. Therefore, adding further samples from an already represented negative class changes its within-class estimate but does not directly multiply that class's total outer weight. This mechanism requires neither a fixed positive-set cap, learnable class centers, dataset-level class priors, a memory queue, nor online class-distribution estimators. Its scope is interaction-level: it balances the contrast set conditional on an anchor, but does not alter the frequency with which anchors from different dataset classes are sampled. Table E.1 summarizes the resulting differences in cardinality mechanisms, additional training structures, and negative aggregation.

The methods therefore address class-frequency skew at different levels. KCL modifies positive-set construction, DSCL controls the relative mass assigned to two positive types, PaCo introduces parametric class representatives, and ProCo replaces finite pair sampling with prior-weighted distributional expectations. The novel distinction of PEML among these objectives is the explicit two-level normalization of finite sample-wise repulsion: every represented negative class receives equal outer weight after its member contributions are averaged. This comparison does not imply empirical superiority over KCL, DSCL, or PaCo, which are not evaluated here; it identifies the mechanism tested by the controlled CIFAR100-LT comparison in Section 4.4.2.

Appendix F. Embedding geometry and retrieval under severe class imbalance

This section complements the balanced-data representation analysis in Section 4.3 and the CIFAR100-LT linear-evaluation results in Section 4.4.2 by examining the learned representation at the most severe evaluated imbalance, $IR = 200$. The analysis considers qualitative embedding structure, centroid-relative class geometry, and instance-retrieval behavior. Together, these views distinguish class-level organization from local neighborhood quality, which need not respond identically to severe class imbalance.

F.1. Experimental protocol and analysis space

PEML and SupCon use the same CIFAR100-LT split and the same loss-independent training conditions. These include the ResNet-50 backbone, a two-layer MLP projection head, data augmentation, a global batch of 1024 original images with two augmented views per image (2,048 embeddings entering the loss), the same number of effective optimizer updates, the same number of image exposures, SGD, and cosine learning-rate scheduling with warmup. SupCon uses the same temperature, $\tau = 0.1$, as in the balanced baseline experiments. The data split and frozen-encoder linear-evaluation protocol are also identical. As an implementation consistency check, training SupCon within the PEML codebase reproduced the result obtained with the original SupCon implementation.

All subsequent analyzes use the raw frozen-encoder representation from the $IR = 200$ frozen-encoder linear evaluation in Section 4.4.2. The balanced CIFAR-100 test set contains 100 images from each of the 100 classes. We define Tail-10 as the ten classes with the smallest numbers of training samples in the realized $IR = 200$ split, according to the class-frequency construction in Section D.2. The same Tail-10 class set is used throughout the visualization, geometry, and retrieval analyzes.

F.2. Qualitative embedding structure

For visualization, the frozen-encoder representations are first reduced to 50 dimensions by PCA and then projected to two dimensions by t-SNE. For each method, PCA and t-SNE are fitted independently using identical settings (perplexity 30, 2000 iterations, and random seed 1). All 10,000 test images are included; Tail-10 classes are distinguished by color and the remaining 90 classes are shown in gray.

As shown in Fig. F.1, the PEML tail representations occupy multiple class-dependent regions rather than concentrating at a single location. Several tail classes retain visibly concentrated regions, while other samples overlap with neighboring classes, reflecting the difficulty of the $IR = 200$ setting. SupCon likewise exhibits both class-specific regions and substantial overlap. Because t-SNE preserves local relations only approximately, we use it as qualitative context and evaluate the original high-dimensional representation quantitatively below.

F.3. Centroid-relative class geometry

Let $\mathbf{h}_i \in \mathbb{R}^d$ denote the frozen-encoder representation of test image i, let $\mathcal{I}_c = \{i : y_i = c\}$, and let $n_c = |\mathcal{I}_c|$. The centroid of class c is

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} \mathbf{h}_i. \tag{F.1}$$

For a selected class set C, we define the macro-averaged intra-class distance, mean inter-centroid distance, and their ratio as

$$D_{\text{intra}}(C) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} \|\mathbf{h}_i - \boldsymbol{\mu}_c\|_2, \tag{F.2}$$

$$D_{\text{inter}}(C) = \frac{2}{|C|(|C|-1)} \sum_{\substack{c,c' \in C \\ c<c'}} \|\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}\|_2, \tag{F.3}$$

$$Q(C) = \frac{D_{\text{intra}}(C)}{D_{\text{inter}}(C)}. \tag{F.4}$$

The ratio in Eq. (F.4) is invariant to a uniform rescaling of the representation. A smaller value indicates that within-class dispersion is lower relative to the separation among class centroids. We report both component distances with the ratio so that compactness and separation are not conflated.

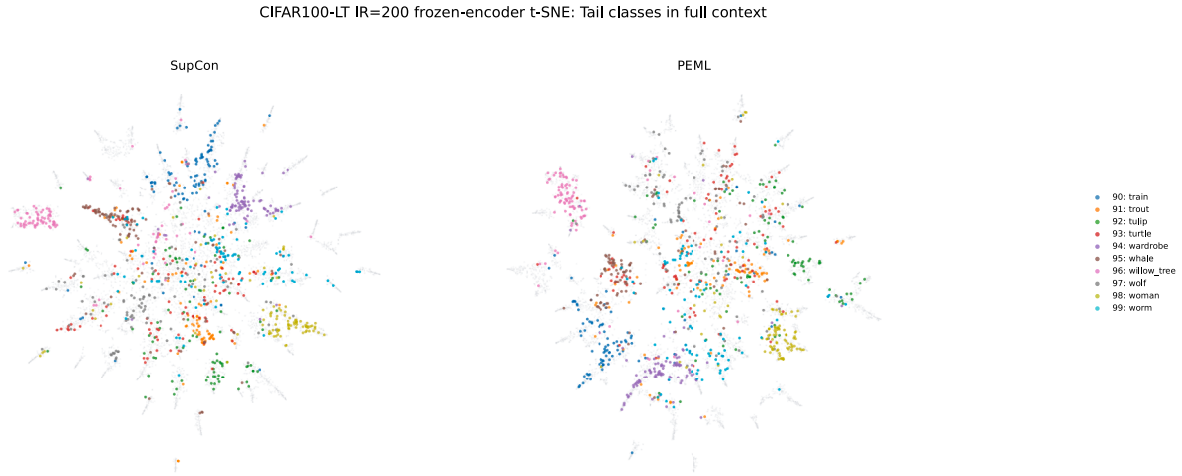


Fig. F.1. t-SNE visualization of the frozen-encoder representations learned by SupCon (left) and PEML (right) on CIFAR100-LT at $IR = 200$. The ten lowest-frequency training classes are shown in color, while all other classes are shown in gray to retain the full embedding context. Each panel is fitted independently under identical settings; therefore, absolute coordinates are not directly comparable across panels.

Table F.1
Geometry of the frozen-encoder representation on the balanced CIFAR-100 test set after training on CIFAR100-LT at $IR = 200$. Overall uses all 100 classes, whereas Tail-10 uses the same ten lowest-frequency training classes as Fig. F.1.

Method	Group	D_{intra}	D_{inter}	Q
SupCon	Overall	1.7140	2.0474	0.8372
SupCon	Tail-10	1.3824	1.0634	1.3000
PEML	Overall	0.2175	0.2945	**0.7387**
PEML	Tail-10	0.2021	0.1905	**1.0607**

Table F.2
Euclidean retrieval performance (%) in the frozen-encoder representation after CIFAR100-LT training at $IR = 200$. Results are class-macro averages, and the query itself is excluded from the gallery.

Method	Query	Gallery	R@1	R@2	R@4	R@8	mAP@R
SupCon	Overall	Full	38.29	48.11	57.17	66.98	12.05
SupCon	Tail-10	Tail-10	70.30	80.80	89.10	94.60	22.81
PEML	Overall	Full	33.97	43.24	53.46	63.06	12.25
PEML	Tail-10	Tail-10	63.30	75.80	83.50	90.00	24.76

The absolute distance scales in Table F.1 are specific to each learned representation and should therefore be interpreted within each method rather than compared directly across methods. The scale-independent ratio is lower for PEML both overall (0.7387 versus 0.8372) and within Tail-10 (1.0607 versus 1.3000). Thus, relative to its own inter-centroid scale, PEML maintains lower within-class dispersion. For both methods, the Tail-10 ratio is higher than the overall ratio, indicating that severe cardinality skew degrades the relative organization of the rarest classes. Nevertheless, the PEML tail ratio remains lower than the SupCon tail ratio.

F.4. Retrieval behavior and relation to downstream separability

We next evaluate Euclidean test-to-test retrieval in the same frozen-encoder space. For query q, let $g_{q,k}$ be the item at rank k after excluding the query itself, and define $r_q(k) = \mathbf{1}[y_{g_{q,k}} = y_q]$. Let $\mathcal{Q}_c$ denote the queries from class c, and let R_q be the number of same-class items available in the gallery. We compute class-macro Recall@K and mAP@R as

$$\mathrm{R@K}(C) = \frac{1}{|C|}\sum_{c\in C}\frac{1}{|\mathcal{Q}_c|}\sum_{q\in\mathcal{Q}_c}\mathbf{1}\left[\sum_{k=1}^{K} r_q(k) > 0\right], \tag{F.5}$$

$$P_q(k) = \frac{1}{k}\sum_{j=1}^{k} r_q(j), \tag{F.6}$$

$$\mathrm{AP@R}_q = \frac{1}{R_q}\sum_{k=1}^{R_q} P_q(k) r_q(k), \tag{F.7}$$

$$\mathrm{mAP@R}(C) = \frac{1}{|C|}\sum_{c\in C}\frac{1}{|\mathcal{Q}_c|}\sum_{q\in\mathcal{Q}_c}\mathrm{AP@R}_q\,. \tag{F.8}$$

We use two complementary protocols. Overall/Full uses all 10,000 test images as both queries and gallery. Tail-10/Tail-10 restricts both sets to the 1000 test images from the same Tail-10 classes and directly probes discrimination within the tail group. Since the test set contains 100 images per class and the query itself is excluded, $R_q = 99$ in both protocols. Because the gallery composition and task difficulty differ, numerical values should be compared between methods within a protocol, not across protocols.

As reported in Table F.2, SupCon achieves higher Recall@K in both protocols, whereas PEML achieves a slightly higher mAP@R overall and a larger mAP@R advantage within Tail-10. These findings are not contradictory because the two metrics emphasize different neighborhood properties. Recall@K records whether at least one positive appears among the first K neighbors, while mAP@R evaluates the ordering of all same-class gallery items over the first R ranks. SupCon therefore provides more frequent early positive hits, whereas PEML provides a more favorable ranking of the broader positive set.

The relationship to the balanced-data results in Section 4.3 is informative. In the balanced setting, PEML improves global clustering scores while matching or improving local k-NN quality, so class-level organization and local neighborhood quality improve together. At $IR = 200$, these properties separate: PEML exhibits the lower centroid-relative geometry ratio in Table F.1 and the higher mAP@R in Table F.2, whereas SupCon retains higher Recall@K. A frozen linear classifier depends on the class-level organization of the representation rather than on obtaining the best value for every instance-retrieval criterion. The observed geometry and retrieval results are therefore consistent with the PEML–SupCon difference in downstream linear separability under severe cardinality skew reported in Table 5.

Finally, the tail-class structures in Fig. F.1, the nonzero intra-class dispersion and inter-centroid structure in Table F.1, and the substantial Tail-10/Tail-10 retrieval performance in Table F.2 do not exhibit the signature of a single-point tail-class collapse. At the same time, the overlap visible in the t-SNE plot and PEML's lower Recall@K show that severe imbalance still degrades local instance neighborhoods. The ad-

ditional analysis therefore supports a specific conclusion rather than uniform superiority: under $IR = 200$, PEML better preserves centroid-relative class organization and broader positive-set ranking, which accompanies its substantially improved linear separability, while SupCon remains stronger for immediate positive-neighbor retrieval.

Appendix G. Euclidean-distance-based metric losses

We describe the Euclidean-distance-based metric losses used in Section 4.7 using unified notation. Let a mini-batch contain B images, each with two augmented views, yielding $N = 2B$ embeddings $\{\boldsymbol{x}_i\}_{i=1}^{N} \subset \mathbb{R}^d$ with corresponding class labels $y_i \in \{1, \dots, C\}$. We denote the Euclidean distance between two embeddings by

$$d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2, \qquad d_{ij}^2 = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2,$$

and define the sets of positive and negative indices for anchor i as

$$P(i) = \{j \neq i : y_j = y_i\}, \qquad N(i) = \{k : y_k \neq y_i\}.$$

We also use the hinge operator $[x]_+ = \max(0, x)$.

NCA-style loss. Following Neighborhood Components Analysis (Goldberger et al., 2004), we define a Gaussian kernel over squared Euclidean distances with temperature $\tau > 0$,

$$p_{ij} = \begin{cases} 0, & j = i, \\ \dfrac{\exp(-d_{ij}^2/\tau)}{\sum_{k \neq i} \exp(-d_{ik}^2/\tau)}, & j \neq i, \end{cases} \tag{G.1}$$

and minimize the negative log-likelihood of positive neighbors in the batch:

$$\mathcal{L}_{\mathrm{NCA}} = -\frac{1}{\sum_{i=1}^{N} |P(i)|} \sum_{i=1}^{N} \sum_{j \in P(i)} \log p_{ij}. \tag{G.2}$$

Contrastive loss. The contrastive loss (Hadsell et al., 2006) penalizes positive pairs with large distances and negative pairs that lie within a margin $m > 0$. For an unordered pair (i, j) we set

$$y_{ij} = \begin{cases} 1, & y_i = y_j,\ i \neq j, \\ 0, & y_i \neq y_j, \end{cases}$$

and define

$$\mathcal{L}_{\mathrm{con}} = \frac{1}{2M} \sum_{i<j} \Big[y_{ij}\, d_{ij}^2 + (1 - y_{ij})\, [m - d_{ij}]_+^2 \Big], \tag{G.3}$$

where M is the number of considered pairs.

Batch-hard triplet loss. We also use a batch-hard triplet loss (Hermans et al., 2017; Schroff et al., 2015) with Euclidean distance. For each anchor i, we select the hardest positive and hardest negative within the batch:

$$d_i^+ = \max_{j \in P(i)} d_{ij}, \qquad d_i^- = \min_{k \in N(i)} d_{ik}, \tag{G.4}$$

and minimize a margin violation:

$$\mathcal{L}_{\mathrm{triplet}} = \frac{1}{|A|} \sum_{i \in A} \big[d_i^+ - d_i^- + m \big]_+, \tag{G.5}$$

where $A = \{i : P(i) \neq \emptyset,\ N(i) \neq \emptyset\}$ is the set of anchors with at least one positive and one negative in the batch.

Margin loss. The margin loss (Wu et al., 2017) extends the contrastive formulation by introducing a learnable distance boundary. We define a signed pair label

$$m_{ij} = \begin{cases} +1, & y_i = y_j,\ i \neq j, \\ -1, & y_i \neq y_j, \end{cases}$$

and associate each class c with a learnable boundary parameter $\beta_c \in \mathbb{R}$. With a fixed margin $\alpha > 0$, the loss for a set S of considered pairs is

$$\mathcal{L}_{\mathrm{margin}} = \frac{1}{|S|} \sum_{(i,j) \in S} \big[\alpha + m_{ij} \left(d_{ij} - \beta_{y_i} \right) \big]_+. \tag{G.6}$$

Positive pairs ($m_{ij} = +1$) are encouraged to have d_{ij} below $\beta_{y_i} - \alpha$, while negative pairs ($m_{ij} = -1$) are pushed above $\beta_{y_i} + \alpha$. Following Wu et al. (2017), we do not use all negative pairs in the batch, but apply distance-weighted negative sampling. Let d_{emb} denote the embedding dimension. For each anchor i we keep all positive pairs

$$S^+(i) = \{(i, j) : j \in P(i)\}, \tag{G.7}$$

and consider all candidate negatives (i, k) with $k \in N(i)$. Among these candidates we retain only those whose distance lies in a prescribed interval $d_{\min} \leq d_{ik} \leq d_{\max}$, and assign an unnormalized sampling weight

$$w(d_{ik}) \propto d_{ik}^{2-d_{\mathrm{emb}}} \left(1 - \tfrac{1}{4} d_{ik}^2 \right)^{\frac{3-d_{\mathrm{emb}}}{2}}, \tag{G.8}$$

which is inversely proportional to the density of pairwise distances on the unit hypersphere. A fixed number of negative pairs $S^-(i) \subset \{(i, k) : k \in N(i)\}$ is then sampled according to the normalized weights $w(d_{ik}) / \sum_{k' \in N(i)} w(d_{ik'})$. The final pair set used in Eq. (G.6) is the union

$$S = \bigcup_{i=1}^{N} \big(S^+(i) \cup S^-(i) \big), \tag{G.9}$$

and the normalization $|S|$ in Eq. (G.6) reflects the number of sampled pairs rather than all possible pairs $i < j$ in the batch.

Appendix H. Computational complexity and empirical overhead

H.1. Theoretical complexity

We analyze the computational complexity of PEML, SupCon, and the NCA-style loss using their dense mini-batch implementations. Let B denote the number of original images in the global mini-batch, let $V = 2$ denote the number of augmented views per image, and let N denote the number of embeddings entering the loss. Each embedding has dimension d_{emb}. Accordingly,

$$N = VB = 2B. \tag{H.1}$$

All three losses therefore operate on N embeddings and construct a dense $N \times N$ pairwise matrix. For SupCon, all N embeddings are used as both anchors and contrastive candidates. Forming the $N \times N$ similarity matrix from the matrix product of an $N \times d_{\mathrm{emb}}$ embedding matrix and its transpose requires $\mathcal{O}(N^2 d_{\mathrm{emb}})$ operations. The subsequent masking, exponentiation, log-normalization, positive averaging, and reduction require $\mathcal{O}(N^2)$ operations. Therefore, the time complexity of SupCon is

$$T_{\mathrm{SupCon}} = \mathcal{O}(N^2 d_{\mathrm{emb}} + N^2) = \mathcal{O}(N^2 d_{\mathrm{emb}}). \tag{H.2}$$

The NCA-style objective in Eqs. (G.1) and (G.2) similarly constructs a dense $N \times N$ matrix of squared Euclidean distances. Computing these distances using embedding norms and a matrix multiplication requires $\mathcal{O}(N^2 d_{\mathrm{emb}})$ operations. The subsequent exponentiation, row-wise normalization, logarithm, positive-pair masking, and reduction collectively require $\mathcal{O}(N^2)$ time. Consequently,

$$T_{\mathrm{NCA}} = \mathcal{O}(N^2 d_{\mathrm{emb}} + N^2) = \mathcal{O}(N^2 d_{\mathrm{emb}}). \tag{H.3}$$

PEML first constructs a dense $N \times N$ Euclidean-distance matrix, which requires $\mathcal{O}(N^2 d_{\text{emb}})$ operations. Each subsequent operation, including mask construction, nearest-negative and farthest-positive selection, adaptive-threshold computation, potential evaluation, and cardinality-weighted reduction, requires at most $\mathcal{O}(N^2)$ time. Hence,

$$T_{\text{PEML}} = \mathcal{O}(N^2 d_{\text{emb}} + N^2) = \mathcal{O}(N^2 d_{\text{emb}}). \tag{H.4}$$

In the theoretical analysis, F + B denotes the forward and backward computations of the loss. For the pairwise operations considered here, the backward pass has the same asymptotic order as the forward pass. Thus, for $\mathcal{X} \in \{\text{PEML}, \text{SupCon}, \text{NCA}\}$,

$$T_{\mathcal{X}}^{\text{F+B}} = \mathcal{O}(N^2 d_{\text{emb}}) = \mathcal{O}\big((2B)^2 d_{\text{emb}}\big) = \mathcal{O}(B^2 d_{\text{emb}}). \tag{H.5}$$

When the embedding dimension d_{emb} and the number of views V are fixed, the F + B time complexity of all three losses is therefore quadratic in the global batch size:

$$T_{\mathcal{X}}^{\text{F+B}} = \mathcal{O}(B^2). \tag{H.6}$$

All three implementations also materialize one or more dense $N \times N$ distance, similarity, mask, probability, or potential matrices. Their loss-specific auxiliary memory complexity is therefore

$$M_{\mathcal{X}}^{\text{aux}} = \mathcal{O}(N^2) = \mathcal{O}(B^2). \tag{H.7}$$

This expression excludes the common embedding tensors, model parameters, optimizer states, and backbone activations. Although the asymptotic memory order is identical, the constant factor can differ because each loss retains a different number of pairwise intermediate tensors for backpropagation.

The class-wise reduction in PEML does not introduce an additional dependence on the dataset-wide number of classes. For each anchor, the class set $C(a)$ contains only the distinct negative classes represented among the N embeddings in the current global mini-batch, so $|C(a)| \le N - 1$, and every $N_c(a)$ is a subset of the same mini-batch. Moreover, PEML maintains no loss-specific trainable parameters or persistent buffers indexed over the dataset label space. Consequently, for fixed B, V, and d_{emb}, its Stage 1 loss-specific per-step F + B time and auxiliary-memory complexities are determined by $N = VB$, rather than by the total number of classes or images in the dataset. This statement concerns per-step loss evaluation: increasing the training-set size generally increases the number of mini-batches per epoch and hence the total runtime under a fixed-epoch training schedule.

H.2. Empirical evaluation of computational cost

We complement the asymptotic analysis with two wall-clock measurements and one per-GPU memory measurement. The epoch time, denoted by t_{epoch}, is the synchronized elapsed time from immediately before entering the training mini-batch loop to immediately after completing the final training batch. It includes data loading, data augmentation, device transfer, encoder and projection-head forward propagation, distributed feature gathering, loss evaluation, backward propagation, DDP gradient synchronization, and parameter optimization. Validation, checkpoint writing, and other post-epoch operations are excluded. Epoch 0 is used as a warmup epoch, and the reported value is the average over epochs 1–10. The measured F + B interval, denoted by $t_{\text{F+B}}$, begins immediately before evaluating the loss and ends immediately after the backward call. The encoder forward pass and the forward feature-gathering operation occur before the starting timestamp. However, because the input embeddings remain connected to the encoder, the backward pass traverses the entire connected computation graph and includes the associated distributed gradient communication. We therefore refer to this quantity as the measured F + B interval, rather than as an isolated loss-kernel microbenchmark. The reported value is averaged over the training batches in the ten measured epochs. We use SupCon as the reference method. For a measured quantity $q \in \{t_{\text{epoch}}, t_{\text{F+B}}, m_{\text{VRAM}}\}$, the relative overhead of loss $\mathcal{X}$ is defined as

$$\Delta_q(\mathcal{X}) = 100\left(\frac{q_{\mathcal{X}}}{q_{\text{SupCon}}} - 1\right). \tag{H.8}$$

A positive value denotes an increase relative to SupCon, whereas a negative value denotes a reduction.

The measurements use the model, data-augmentation, and optimization settings of the supervised CIFAR-100 representation-quality experiment described in Section 4.1. We use a ResNet-50 encoder followed by a two-layer fully connected projection head. The augmentation pipeline, optimizer, learning-rate schedule, embedding dimension, mixed-precision configuration, and all other loss-independent settings remain fixed across PEML, SupCon, and NCA. All experiments use PyTorch's `DistributedDataParallel` implementation on eight NVIDIA GeForce RTX 2080 Ti GPUs. The global mini-batch contains $B = 1024$ original images, each represented by $V = 2$ independently augmented views. Thus, each optimization step processes $N = VB = 2048$ embeddings. The embeddings and labels are gathered across the eight DDP ranks before loss evaluation; consequently, each method receives the same global set of 2048 embeddings and constructs a 2048×2048 pairwise matrix. The same random seed and data-ordering configuration are used for all three methods. Identical timing instrumentation is applied in every run. The observed steady-state VRAM usage is identical across the eight GPUs within each run and is reported in MiB per GPU.

H.3. Results and discussion

As shown in Table H.1, all three objectives have closely comparable practical computational requirements. Relative to SupCon, PEML increases the average epoch time by 0.1355 s, corresponding to an overhead of 0.89%. Its measured F + B interval is 2.3 ms longer per batch, corresponding to an overhead of 1.31%. Thus, the additional adaptive-threshold computations and trigonometric-potential evaluations introduce only a small constant-factor overhead. The epoch-time overhead is smaller than the measured F + B overhead because a complete epoch also includes substantial computation shared by all methods, including data processing, encoder forward propagation, distributed feature gathering, and parameter optimization. These common operations reduce the contribution of the loss-specific difference to the total training time.

PEML uses 4546 MiB, or approximately 4.44 GiB, per GPU, compared with 4516 MiB, or approximately 4.41 GiB, for SupCon. The resulting difference is 30 MiB per GPU, corresponding to an increase of 0.66%. NCA uses 4500 MiB per GPU and exhibits nearly the same epoch and measured F + B times as SupCon. Relative to NCA, PEML is 0.85% slower per epoch and 0.62% slower over the measured F + B interval, while using 1.02% more VRAM.

These wall-clock and memory measurements are specific to the stated hardware, software environment, batch configuration, and implementation. Accordingly, they provide a controlled relative comparison under identical conditions rather than hardware-independent constants.

The theoretical analysis shows that PEML, SupCon, and NCA share the same asymptotic order, while the empirical measurements quantify their constant-factor costs under the evaluated configuration. All three losses require a dense all-pairs distance or similarity computation and therefore exhibit $\mathcal{O}(B^2 d_{\text{emb}})$ loss-level F + B time complexity and $\mathcal{O}(B^2)$ loss-specific auxiliary memory complexity under the two-view setting. PEML additionally computes adaptive interaction thresholds and closed-form trigonometric potentials, but these operations do not alter its asymptotic order. At a global batch size of 1024, PEML incurs only 0.89% additional epoch time, 1.31% additional measured F + B time, and 0.66% additional per-GPU VRAM relative to SupCon. Therefore, under the evaluated configuration, the representation-quality gains of PEML are not accompanied by a substantial computational or memory overhead.

Table H.1
Computational cost on CIFAR-100 using a global batch of 1024 original images and two views per image, resulting in 2048 embeddings, with DDP on eight NVIDIA GeForce RTX 2080 Ti GPUs. Overheads are computed relative to SupCon. Epoch 0 is excluded as a warmup, and epoch time is averaged over epochs 1–10. The measured F + B interval is averaged over the training batches in the measured epochs. VRAM denotes the observed total training footprint per GPU, rather than only the theoretical loss-specific auxiliary memory.

Loss	Epoch time (s)	Epoch overhead (%)	Measured F + B (ms/batch)	F + B overhead (%)	VRAM (MiB/GPU)	VRAM overhead (%)
SupCon	15.2434	–	176.0	–	4516	–
NCA	15.2489	+0.04	177.2	+0.68	4500	-0.35
PEML	15.3789	+0.89	178.3	+1.31	4546	+0.66

Appendix I. Convergence and stability of the PEML embedding dynamics

We analyze the deterministic embedding dynamics induced by PEML in the post-projection embedding space. Let $\mathcal{A}$ denote the current mini-batch and let $N = |\mathcal{A}|$ be the number of embeddings entering the loss. For each sample, let $z_i \in \mathbb{R}^{d_{\text{emb}}}$ denote the projected embedding obtained after the annulus-scaled projection in Eq. (38). We collect the projected embeddings as

$$\mathbf{z} = (z_1, \dots, z_N) \in \mathbb{R}^{N d_{\text{emb}}}. \tag{I.1}$$

Since Eq. (38) maps each embedding into the annulus, each projected embedding satisfies

$$R_A < \|z_i\| < R_B, \qquad i = 1, \dots, N. \tag{I.2}$$

Therefore, by the triangle inequality, every pairwise distance satisfies

$$0 \le d_{ij} = \|z_i - z_j\| \le d_{\max} = 2R_B, \qquad i, j = 1, \dots, N, \tag{I.3}$$

which is consistent with the bounded distance range in Eq. (39).

We consider the ASP-based admissible parameter range $\rho_p \in [0, 1]$ and $\rho_n \in [0, 1)$, with all experimental settings lying in its interior. We also assume a valid contrastive batch, meaning that every anchor has at least one positive and at least one negative class in the batch. This ensures that the cardinality normalizers in Eq. (48) are finite positive constants. For the theoretical analysis, define the non-collision annular configuration domain

$$\mathcal{D} = \Big\{ \mathbf{z} \in \mathbb{R}^{N d_{\text{emb}}} : R_A < \|z_i\| < R_B \quad \text{for every } i, \|z_i - z_j\| > 0 \quad \text{for every } i \ne j \Big\}. \tag{I.4}$$

The direct vector fields below are considered only on $\mathcal{D}$. Importantly, neither the continuous direct-gradient flow nor its unconstrained discrete-time recurrence automatically preserves this domain. The infinite-horizon conclusions below are therefore conditional on the explicit domain-invariance assumptions stated in the corresponding subsections.

The adaptive thresholds in Eqs. (40) and (41) are computed from the current embedding configuration and treated as constants during differentiation. Accordingly, for a fixed threshold collection

$$K = \{k_p(a), k_n(a)\}_{a \in \mathcal{A}}, \tag{I.5}$$

we define the corresponding frozen-threshold PEML objective as

$$\Phi_K(\mathbf{z}) = \mathcal{L}_{\text{PEML}}(\mathbf{z}; K), \tag{I.6}$$

where the pairwise potentials are defined in Eqs. (46) and (47), and the total PEML objective is defined in Eq. (48). We first analyze idealized direct fixed-threshold dynamics generated by Φ_K under explicit domain-invariance and regularity assumptions, and then establish a conditional one-step descent result for the corresponding modeled detached-threshold update.

I.1. Boundedness and finite-force stability

The annulus-scaled projection directly bounds the embedding domain. Since $\rho_p \in [0, 1]$, $\rho_n \in [0, 1)$, and $0 \le d_n^{\min}(a), d_p^{\max}(a) < d_{\max}$, the adaptive thresholds satisfy

$$0 \le k_p(a) = \rho_p d_n^{\min}(a) \le d_n^{\min}(a) < d_{\max}, \tag{I.7}$$

and

$$0 \le d_p^{\max}(a) < k_n(a) = d_{\max} - \rho_n \Big(d_{\max} - d_p^{\max}(a) \Big) \le d_{\max}. \tag{I.8}$$

Thus, the denominators in the positive and negative force kernels are strictly positive:

$$d_{\max} - k_p(a) > 0, \qquad k_n(a) > 0. \tag{I.9}$$

From Eqs. (42) and (43), the positive and negative pairwise force magnitudes are

$$\|F_P(d_{ap})\| = \sin^2 \left(\frac{\pi (d_{ap} - k_p(a))}{2(d_{\max} - k_p(a))} \right) \mathbf{1}_{[k_p(a), d_{\max}]}(d_{ap}), \tag{I.10}$$

and

$$\|F_N(d_{an})\| = \cos^2 \left(\frac{\pi d_{an}}{2k_n(a)} \right) \mathbf{1}_{[0, k_n(a)]}(d_{an}). \tag{I.11}$$

Since $0 \le \sin^2(\cdot) \le 1$ and $0 \le \cos^2(\cdot) \le 1$, we obtain

$$0 \le \|F_P(d_{ap})\| \le 1, \qquad 0 \le \|F_N(d_{an})\| \le 1. \tag{I.12}$$

Hence, each pairwise force is finite.

The corresponding pairwise potentials are also bounded. For the positive potential in Eq. (46), define $u = d_{ap} - k_p(a)$ and $m = d_{\max} - k_p(a)$. On the active interval, $u \in [0, m]$, and

$$V_P(d_{ap}) = \frac{u}{2} - \frac{m}{2\pi} \sin \left(\frac{\pi u}{m} \right). \tag{I.13}$$

Differentiating with respect to u gives

$$\frac{dV_P}{du} = \frac{1}{2} \left(1 - \cos \left(\frac{\pi u}{m} \right) \right) = \sin^2 \left(\frac{\pi u}{2m} \right) \ge 0. \tag{I.14}$$

Because $V_P = 0$ at $u = 0$, V_P is nonnegative and monotonically increasing on its active interval. Its maximum is attained at $u = m$, yielding

$$0 \le V_P(d_{ap}) \le \frac{m}{2} = \frac{d_{\max} - k_p(a)}{2} \le \frac{d_{\max}}{2}. \tag{I.15}$$

Similarly, for the negative potential in Eq. (47), define $v = d_{an}$ and $\ell = k_n(a)$. On the active interval, $v \in [0, \ell]$, and

$$V_N(d_{an}) = \frac{\ell - v}{2} - \frac{\ell}{2\pi} \sin \left(\frac{\pi v}{\ell} \right). \tag{I.16}$$

Differentiating with respect to v gives

$$\frac{dV_N}{dv} = -\frac{1}{2} \left(1 + \cos \left(\frac{\pi v}{\ell} \right) \right) = -\cos^2 \left(\frac{\pi v}{2\ell} \right) \le 0. \tag{I.17}$$

Since $V_N = \ell/2$ at $v = 0$ and $V_N = 0$ at $v = \ell$, the negative potential is nonnegative and bounded as

$$0 \le V_N(d_{an}) \le \frac{\ell}{2} = \frac{k_n(a)}{2} \le \frac{d_{\max}}{2}. \tag{I.18}$$

Combining Eqs. (I.15) and (I.18) with the finite-batch form of PEML in Eq. (48), the frozen-threshold objective is lower bounded and finite:

$$0 \le \Phi_K(\mathbf{z}) < \infty, \qquad \mathbf{z} \in \mathcal{D}. \tag{I.19}$$

Moreover, because the batch is finite, all cardinality weights are finite positive constants, and all pairwise force magnitudes are bounded by Eq. (I.12), there exists a finite constant $G_K > 0$ such that

$$\|\nabla_{\mathbf{z}} \Phi_K(\mathbf{z})\| \le G_K, \qquad \mathbf{z} \in \mathcal{D}. \tag{I.20}$$

Thus, PEML has bounded potentials and uniformly bounded force magnitudes on $\mathcal{D}$.

I.2. Lyapunov stability of the continuous embedding dynamics

We next consider the continuous-time fixed-threshold direct embedding dynamics

$$\dot{\mathbf{z}}(t) = -\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t)). \tag{I.21}$$

For this subsection, assume that the solution exists for all $t \geq 0$ and that there exist a compact set $\Omega_K \subset D$ and an open neighborhood U_K satisfying

$$\Omega_K \subset U_K \quad \text{and} \quad \overline{U_K} \subset D, \tag{I.22}$$

such that $\mathbf{z}(t) \in \Omega_K$ for every $t \geq 0$ and $\nabla_{\mathbf{z}}\Phi_K$ is Lipschitz continuous on U_K. This is an explicit conditional invariance and regularity assumption for the modeled direct flow.

Following the Lyapunov energy argument (Lyapunov, 1992), we use the frozen-threshold potential Φ_K as a Lyapunov-type energy for the modeled embedding flow. The force–potential correspondence is motivated by the Lagrangian view developed in Section 3.1. The monotonic decrease of the potential, however, follows from the gradient-flow form in Eq. (I.21). Along any smooth trajectory of Eq. (I.21), the chain rule gives

$$\frac{d}{dt}\Phi_K(\mathbf{z}(t)) = \nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))^{\top}\dot{\mathbf{z}}(t). \tag{I.23}$$

Substituting Eq. (I.21) into Eq. (I.23) yields

$$\frac{d}{dt}\Phi_K(\mathbf{z}(t)) = -\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))\right\|^2 \leq 0. \tag{I.24}$$

Therefore, Φ_K is non-increasing along the fixed-threshold embedding flow. Since Φ_K is lower bounded by Eq. (I.19), $\Phi_K(\mathbf{z}(t))$ is non-increasing and converges to a finite value. Integrating Eq. (I.24) over $[0, T]$ gives

$$\int_0^T \left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))\right\|^2 dt = \Phi_K(\mathbf{z}(0)) - \Phi_K(\mathbf{z}(T)) \leq \Phi_K(\mathbf{z}(0)). \tag{I.25}$$

Thus, the squared gradient norm is integrable along the trajectory. Moreover, Eq. (I.20) implies that $\dot{\mathbf{z}}(t)$ is bounded, and hence $\mathbf{z}(t)$ is Lipschitz continuous in time. Together with the assumed Lipschitz continuity of $\nabla_{\mathbf{z}}\Phi_K$ on U_K, this makes $t \mapsto \nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))$ uniformly continuous. The standard uniform-continuity argument for a square-integrable function therefore gives

$$\lim_{t\to\infty} \left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))\right\| = 0. \tag{I.26}$$

Consequently, every limit point of the fixed-threshold continuous embedding dynamics belongs to the stationary set

$$S_K = \left\{\mathbf{z} : \nabla_{\mathbf{z}}\Phi_K(\mathbf{z}) = 0\right\}. \tag{I.27}$$

Compactness of Ω_K guarantees the existence of accumulation points, and continuity of $\nabla_{\mathbf{z}}\Phi_K$ implies that every such accumulation point belongs to S_K. In the force interpretation, Eq. (I.27) corresponds to configurations in which the total PEML force vanishes. This energy-decrease result does not by itself establish Lyapunov stability of a particular stationary point, which would require additional local definiteness conditions.

I.3. Discrete-time descent and asymptotic stationarity

We now analyze the discrete-time fixed-threshold direct embedding update

$$\mathbf{z}_{t+1} = \mathbf{z}_t - \eta\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t), \tag{I.28}$$

where $\eta > 0$ is the learning rate. This direct recurrence does not include a reprojection step and therefore does not itself preserve the annular domain. For this subsection, assume that there exist a compact set $\Omega_K \subset D$ and an open set U_K satisfying

$$\Omega_K \subset U_K \quad \text{and} \quad \overline{U_K} \subset D, \tag{I.29}$$

such that $\mathbf{z}_t \in \Omega_K$ for every t, every update segment

$$[\mathbf{z}_t, \mathbf{z}_{t+1}] = \left\{(1-\lambda)\mathbf{z}_t + \lambda\mathbf{z}_{t+1} : \lambda \in [0,1]\right\} \tag{I.30}$$

is contained in U_K, and Φ_K is L-smooth on U_K for some $L > 0$, namely

$$\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{u}) - \nabla_{\mathbf{z}}\Phi_K(\mathbf{v})\right\| \leq L\|\mathbf{u}-\mathbf{v}\| \tag{I.31}$$

for any $\mathbf{u}, \mathbf{v} \in U_K$. By the descent lemma for L-smooth functions,

$$\Phi_K(\mathbf{z}_{t+1}) \leq \Phi_K(\mathbf{z}_t) + \nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)^{\top}(\mathbf{z}_{t+1}-\mathbf{z}_t) + \frac{L}{2}\|\mathbf{z}_{t+1}-\mathbf{z}_t\|^2. \tag{I.32}$$

Substituting Eq. (I.28) into Eq. (I.32) gives

$$\Phi_K(\mathbf{z}_{t+1}) \leq \Phi_K(\mathbf{z}_t) - \eta\left(1 - \frac{L\eta}{2}\right)\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)\right\|^2. \tag{I.33}$$

Therefore, if

$$0 < \eta < \frac{2}{L}, \tag{I.34}$$

then the coefficient in Eq. (I.33) is positive, and $\Phi_K(\mathbf{z}_t)$ is non-increasing.

Let

$$\Phi_{K,*} = \inf_{\mathbf{z}\in\Omega_K} \Phi_K(\mathbf{z}). \tag{I.35}$$

By Eq. (I.19), $\Phi_{K,*} \geq 0$. Summing Eq. (I.33) from $t = 0$ to $T-1$ yields

$$\eta\left(1 - \frac{L\eta}{2}\right)\sum_{t=0}^{T-1}\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)\right\|^2 \leq \Phi_K(\mathbf{z}_0) - \Phi_K(\mathbf{z}_T) \leq \Phi_K(\mathbf{z}_0) - \Phi_{K,*}. \tag{I.36}$$

Letting $T \to \infty$, we obtain

$$\sum_{t=0}^{\infty}\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)\right\|^2 < \infty. \tag{I.37}$$

Hence,

$$\lim_{t\to\infty}\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)\right\| = 0. \tag{I.38}$$

Moreover, Eq. (I.36) implies the finite-time stationarity bound

$$\min_{0\leq t<T}\left\|\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}_t)\right\|^2 \leq \frac{\Phi_K(\mathbf{z}_0) - \Phi_{K,*}}{\eta\left(1 - \frac{L\eta}{2}\right)T}. \tag{I.39}$$

Therefore, under the stated domain-invariance, smoothness, and step-size assumptions, the gradient norm of the fixed-threshold direct embedding dynamics vanishes asymptotically. Because the iterates remain in the compact set Ω_K, they possess accumulation points; continuity of $\nabla_{\mathbf{z}}\Phi_K$ and Eq. (I.38) imply that every such accumulation point is stationary. These conclusions are conditional and do not assert that the unconstrained recurrence in Eq. (I.28) preserves D by itself.

I.4. Detached adaptive thresholds

The preceding analysis considers a fixed threshold collection K. In PEML training, the adaptive thresholds are recomputed at each step from the current embedding configuration through Eqs. (40) and (41) and are then treated as constants during differentiation. Let

$$K_t = \{k_p^t(a), k_n^t(a)\}_{a\in\mathcal{A}} \tag{I.40}$$

denote the threshold collection computed from $\mathbf{z}_t$. The corresponding frozen-threshold surrogate objective is

$$\Phi_{K_t}(\mathbf{z}) = \mathcal{L}_{\mathrm{PEML}}(\mathbf{z}; K_t). \tag{I.41}$$

At the modeled direct-embedding level, the update at step t can then be written as

$$\mathbf{z}_{t+1} = \mathbf{z}_t - \eta\nabla_{\mathbf{z}}\Phi_{K_t}(\mathbf{z}_t). \tag{I.42}$$

Thus, conditional on the current detached thresholds K_t, the PEML gradient realizes the desired force field derived from the frozen-threshold potential.

For a given step t, assume that the complete update segment $[\mathbf{z}_t, \mathbf{z}_{t+1}]$ is contained in an open set U_t satisfying $\overline{U_t} \subset D$, and that Φ_{K_t} is L_t-smooth on U_t for some $L_t > 0$. If $0 < \eta < 2/L_t$, the same argument as in Section I.3 gives

$$\Phi_{K_t}(\mathbf{z}_{t+1}) \le \Phi_{K_t}(\mathbf{z}_t) - \eta\left(1 - \frac{L_t\eta}{2}\right)\left\|\nabla_{\mathbf{z}}\Phi_{K_t}(\mathbf{z}_t)\right\|^2. \tag{I.43}$$

Therefore, conditional on the stated segment-domain, smoothness, and step-size assumptions, each modeled direct-embedding update is a descent step for the current frozen-threshold surrogate objective. Since the threshold collection K_t changes across iterations, Eq. (I.43) should be interpreted as a one-step surrogate descent result rather than as descent of a single time-invariant objective over the entire training trajectory.

Under the explicit domain-invariance and regularity assumptions stated above, the analysis establishes bounded pairwise potentials and force magnitudes, monotonic potential decrease, and asymptotic stationarity for the idealized deterministic fixed-threshold direct embedding dynamics. It also establishes conditional one-step surrogate descent for the modeled detached-threshold update.

I.5. Effect of momentum and adaptive optimization

The force–potential construction and the optimizer-induced trajectory should be distinguished. At iteration t, conditional on the current detached threshold collection K_t, let

$$\boldsymbol{g}_t = \nabla_{\mathbf{z}}\Phi_{K_t}(\mathbf{z}_t), \qquad \boldsymbol{F}_t^{\mathrm{raw}} = -\boldsymbol{g}_t, \tag{I.44}$$

where K_t is held fixed during differentiation, as in Section I.4. For the current frozen-threshold surrogate defined in Eq. (I.41), the second identity follows from the force–potential relation in Eq. (14). This conditional identity is determined by the PEML objective and is therefore independent of the optimizer. Momentum and adaptive optimizers instead determine how the raw gradient and the accumulated optimizer state are converted into a displacement. Related optimizer-dependent effects on embedding updates have also been examined in deep metric learning (Zhang et al., 2020).

For example, a representative heavy-ball momentum update at the modeled embedding level can be written as

$$\boldsymbol{u}_{t+1} = \beta\boldsymbol{u}_t + \boldsymbol{g}_t, \qquad \mathbf{z}_{t+1} = \mathbf{z}_t - \eta\boldsymbol{u}_{t+1}, \qquad 0 \le \beta < 1. \tag{I.45}$$

When $\beta = 0$, Eq. (I.45) reduces to the direct gradient update in Eq. (I.42). For $\beta > 0$, the displacement additionally depends on the accumulated state $\boldsymbol{u}_t$. For a fixed threshold collection K and under an appropriate small-step scaling, a formal continuous-time analog is the damped inertial system (Qian, 1999)

$$M\ddot{\mathbf{z}}(t) + \gamma\dot{\mathbf{z}}(t) + \nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t)) = 0, \qquad M, \gamma > 0. \tag{I.46}$$

The corresponding total mechanical energy satisfies

$$\mathcal{E}_K(t) = \Phi_K(\mathbf{z}(t)) + \frac{M}{2}\|\dot{\mathbf{z}}(t)\|^2,$$
$$\frac{d}{dt}\mathcal{E}_K(t) = -\gamma\|\dot{\mathbf{z}}(t)\|^2 \le 0. \tag{I.47}$$

Momentum therefore extends, rather than removes, the mechanical interpretation by introducing inertia and damping. The decreasing quantity is now the sum of potential and kinetic energy. Because these two components can exchange energy, the potential term alone may increase transiently, and the potential-only decrease in Eq. (I.24) and the one-step result in Eq. (I.43) are not inherited without optimizer-specific conditions. Other momentum conventions, including Nesterov variants, change the precise discretization but retain this dependence on an accumulated optimizer state.

Adam additionally combines first-moment memory with coordinate-wise second-moment scaling (Kingma & Ba, 2015). At the same modeled embedding level, its moment estimates are

$$\boldsymbol{m}_{t+1} = \beta_1\boldsymbol{m}_t + (1-\beta_1)\boldsymbol{g}_t,$$
$$\boldsymbol{s}_{t+1} = \beta_2\boldsymbol{s}_t + (1-\beta_2)\boldsymbol{g}_t^{\odot 2}, \qquad 0 \le \beta_1, \beta_2 < 1, \tag{I.48}$$

with zero-initialized moments and bias-corrected estimates $\hat{\boldsymbol{m}}_{t+1} = \boldsymbol{m}_{t+1}/(1-\beta_1^{t+1})$ and $\hat{\boldsymbol{s}}_{t+1} = \boldsymbol{s}_{t+1}/(1-\beta_2^{t+1})$. Defining the positive-definite diagonal preconditioner

$$D_{t+1} = \mathrm{diag}_i\left(\frac{1}{\sqrt{\hat{s}_{t+1,i}} + \varepsilon}\right), \qquad \varepsilon > 0,$$
$$\mathbf{z}_{t+1} = \mathbf{z}_t - \eta_t D_{t+1}\hat{\boldsymbol{m}}_{t+1}. \tag{I.49}$$

Positive-definite adaptive scaling does not by itself remove a descent interpretation. If the first-moment history is suppressed so that the update direction is $D\boldsymbol{g}$, the corresponding fixed-threshold preconditioned flow satisfies

$$\dot{\mathbf{z}}(t) = -D(t,\mathbf{z})\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t)),$$
$$\frac{d}{dt}\Phi_K(\mathbf{z}(t)) = -\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t))^{\top}D(t,\mathbf{z})\nabla_{\mathbf{z}}\Phi_K(\mathbf{z}(t)) \le 0 \tag{I.50}$$

whenever $D(t,\mathbf{z})$ is positive definite. Thus, coordinate-wise adaptive scaling can be interpreted as a variable-metric gradient flow.

For full Adam, the local L_t-smoothness condition used in Section I.4 instead gives

$$\Phi_{K_t}(\mathbf{z}_{t+1}) \le \Phi_{K_t}(\mathbf{z}_t) - \eta_t\boldsymbol{g}_t^{\top}D_{t+1}\hat{\boldsymbol{m}}_{t+1} + \frac{L_t\eta_t^2}{2}\|D_{t+1}\hat{\boldsymbol{m}}_{t+1}\|^2. \tag{I.51}$$

When $\hat{\boldsymbol{m}}_{t+1} = \boldsymbol{g}_t$, positive definiteness of D_{t+1} yields a descent direction, and a sufficiently small preconditioner-dependent step size gives descent of the current surrogate. In full Adam, however, $\hat{\boldsymbol{m}}_{t+1}$ also contains gradients from previous iterations. The sign of the first-order term in Eq. (I.51) is therefore not determined solely by the current PEML gradient. When K_t changes, the accumulated moments may additionally combine gradients associated with successive frozen-threshold surrogates. Full Adam is consequently represented more naturally by augmented-state dynamics involving $(\mathbf{z}, \boldsymbol{m}, \boldsymbol{s})$ than by the single gradient flow in Eq. (I.21).

These distinctions concern the optimizer-induced trajectory rather than the PEML objective or its conditional raw force field. The equations above describe embedding-level counterparts consistent with the scope of the preceding analysis. In neural-network training, the optimizer states are maintained in parameter space, and the induced embedding trajectory additionally depends on the evolving network Jacobian and stochastic mini-batches. Establishing optimizer-specific convergence guarantees for the resulting augmented stochastic parameter dynamics, including changing detached thresholds, lies beyond the present loss-design scope and is left for future work.

Appendix J. Empirical training dynamics and encoder-space class separation

This section examines how the optimization dynamics and encoder-space class geometry evolve during the CIFAR-100 training runs underlying the classification and representation-quality evaluations reported in Sections 4.1 and 4.3. Complementing the conditional modeled embedding-space analysis in Section I, we track the training loss, parameter-gradient norm, and centroid-relative encoder geometry of PEML and SupCon. We then examine how the final test samples and linear-classification errors are positioned relative to true and competing training-set centroids. The training setting is already described in the main text; only the additional trajectory and endpoint diagnostic measurements are specified below.

J.1. Trajectory measurements

For mini-batch b in epoch e, let $\mathcal{L}_{e,b}$ denote the objective used by the corresponding method and let $\boldsymbol{w} \in \mathbb{R}^m$ collect all trainable model pa-

rameters. We measure the global parameter-gradient norm after backpropagation and immediately before the optimizer update:

$$G_{e,b} = \|\nabla_{\boldsymbol{w}} \mathcal{L}_{e,b}\|_2 = \left[\sum_{q=1}^{m} \left(\frac{\partial \mathcal{L}_{e,b}}{\partial w_q}\right)^2\right]^{1/2}. \tag{J.1}$$

If epoch e contains M_e mini-batches, the plotted curve and shaded band are the within-epoch mean and standard deviation,

$$\overline{G}_e = \frac{1}{M_e}\sum_{b=1}^{M_e} G_{e,b}, \tag{J.2}$$

$$s_{G,e} = \left[\frac{1}{M_e}\sum_{b=1}^{M_e}\left(G_{e,b} - \overline{G}_e\right)^2\right]^{1/2}. \tag{J.3}$$

Each displayed curve corresponds to the representative seed-1 training trajectory of the respective method; the same qualitative non-divergent loss and parameter-gradient behavior was observed across the additional evaluated random seeds. For encoder geometry, we reuse the raw pre-projection representation $\boldsymbol{h}_i$, the class centroid $\boldsymbol{\mu}_c$, and the centroid-relative quantities D_{intra}, D_{inter}, and $Q = D_{\text{intra}}/D_{\text{inter}}$ defined in Section F.3. At each saved checkpoint, these quantities are evaluated over all 50,000 CIFAR-100 training images and all 100 classes without additional ℓ_2 normalization. Lower Q indicates that within-class dispersion is smaller relative to the between-centroid scale. Moreover, because its numerator and denominator scale identically under a uniform rescaling of the encoder representation, Q is invariant to such global scaling.

J.2. Loss and gradient-norm evolution

Fig. J.1(a,b) shows that the two objectives follow distinct but non-divergent loss trajectories. SupCon decreases smoothly from 7.6151 at epoch 1 to 3.1523 at epoch 1000, with its minimum of 3.1416 occurring at epoch 997. PEML exhibits an early adjustment regime in which the loss rises transiently to 0.1388 at epoch 135 before decreasing to 0.0119 at epoch 1000, close to its minimum of 0.0117 at epoch 996. A plausible interpretation follows from the interaction between the annulus-scaled projection and PEML's adaptive, distance-dependent pair potentials. The projection confines the post-projection embeddings to a bounded annulus while retaining radial variation, as described in Eq. (38); changes in their radial and angular configuration can therefore redistribute Euclidean pair distances within this bounded domain. PEML recomputes $k_p(a)$ and $k_n(a)$ from the current nearest-negative and farthest-positive distances, as defined in Eqs. (40) and (41), so this redistribution can alter both the placement of pairs within the active interaction intervals and their potential contributions. For example, when the farthest positive lies in its active interval, an increase in $d_p^{\max}(a)$ can increase its positive potential and, through $k_n(a)$, extend the distance range over which negative interactions remain active. Because the adaptive thresholds and active pair sets are recomputed from the current embedding configuration in each mini-batch, the epoch-averaged loss reflects not only changes in individual pair distances but also changes in which interactions are active and how strongly they contribute. These coupled effects suggest that the early rise corresponds to an adjustment phase in which positive and negative pairs are redistributed across the active interaction intervals, temporarily increasing the aggregate PEML objective before the subsequent decreasing regime emerges. A direct checkpoint-wise decomposition of this process would require tracking active-pair membership and the full post-projection distance distribution.

The gradient trajectories in Fig. J.1(c,d) likewise remain bounded at the epoch-summary level throughout training, without sustained epoch-to-epoch instability. For PEML, $\overline{G}_e$ decreases sharply during the initial epochs, equals 0.0916 at epoch 20, and evolves gradually thereafter, ending at 0.3939. For SupCon, the raw gradient scale increases more steadily and ends at 2.6321. As with the loss values, the absolute gradient magnitudes depend on the scale and structure of the corresponding objective.

Table J.1
Centroid-relative geometry of the raw CIFAR-100 encoder representations at epoch 1000. The component distances are reported to show the decomposition of Q; cross-method comparison focuses on the scale-invariant ratio.

Metric	SupCon	PEML
D_{intra}	1.3257	0.2845
D_{inter}	1.8480	1.3004
Q	0.7174	**0.2188**

The gradient-norm curves are therefore used to characterize the evolution and boundedness of each training trajectory, rather than to rank the two methods by raw gradient magnitude.

J.3. Evolution of encoder-space class separation

Fig. J.1(e) reveals a clearer difference in the learned encoder geometry. PEML begins with a larger ratio than SupCon at epoch 1 (2.6768 versus 1.3982), but falls below SupCon by epoch 50 (0.7087 versus 0.7948). From epoch 50 onward, PEML has the lower Q at all 96 recorded checkpoints. At epoch 1000, PEML reaches $Q = 0.2188$, whereas SupCon reaches $Q = 0.7174$, corresponding to a 69.5% lower ratio for PEML.

As shown in Table J.1, PEML yields smaller values of both D_{intra} and D_{inter} than SupCon. Relative to SupCon, however, the decrease in D_{intra} is substantially larger than that in D_{inter} (78.5% versus 29.6%). The lower Q of PEML therefore reflects a pronounced reduction in within-class dispersion relative to its own between-centroid scale, rather than an increase in absolute inter-centroid distance. Because Q summarizes global centroid-relative geometry, it characterizes class-level organization but does not directly determine performance under local-neighborhood criteria.

The global ratio Q does not reveal whether individual test samples lie closer to their true-class centroid or to a competing centroid. We therefore complement the trajectory analysis with an endpoint diagnostic that relates sample-level centroid geometry to the errors of the frozen linear probes.

J.4. Endpoint centroid-relative failure analysis

This analysis uses the late-training SupCon and PEML encoder checkpoints underlying the balanced CIFAR-100 results in Table 1, taken from the final selected epoch for each method. For both methods, the Stage-2 model uses the same plain linear-classifier formulation, and all encoder parameters and batch-normalization buffers are kept exactly fixed at their Stage-1 checkpoint values. The class centroids are fitted from the raw 2,048-dimensional pre-projection representations of the deterministic 50,000-image training split. The resulting centroids are then used to analyze all 10,000 test images; test samples do not contribute to centroid fitting. No feature normalization or standardization is applied.

For method $m \in \{\text{SupCon}, \text{PEML}\}$, let $\hat{y}_i^{(m)}$ denote the linear-classifier prediction for test sample i. A method-specific failure occurs when $\hat{y}_i^{(m)} \neq y_i$. We use the nearest-wrong-centroid ratio $r_{\text{wrong}}^{(m)}(i)$ defined in Eq. (49). For a failed sample, we additionally define the predicted-class-centroid ratio

$$r_{\text{pred}}^{(m)}(i) = \frac{\left\|\boldsymbol{h}_i^{(m)} - \boldsymbol{\mu}_{\hat{y}_i^{(m)}}^{(m),\text{tr}}\right\|_2}{\left\|\boldsymbol{h}_i^{(m)} - \boldsymbol{\mu}_{y_i}^{(m),\text{tr}}\right\|_2}. \tag{J.4}$$

Thus, $r_{\text{wrong}}^{(m)}(i) < 1$ denotes a nearest-wrong-centroid inversion, while $r_{\text{pred}}^{(m)}(i) < 1$ means that the sample is closer to the centroid of the class predicted by the linear probe than to its true-class centroid. These events

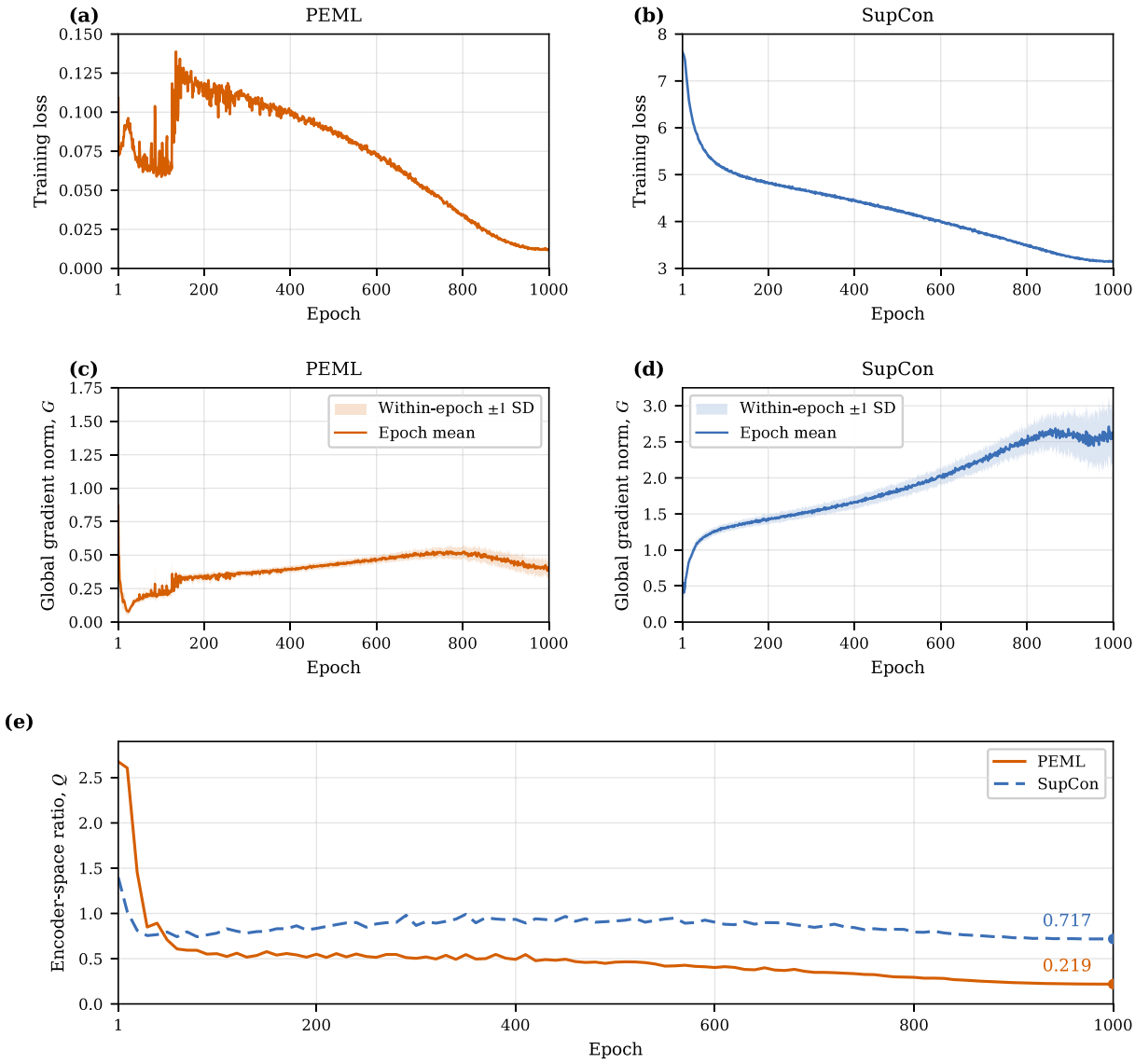


Fig. J.1. Training dynamics and encoder-space class separation on CIFAR-100. (a,b) Epoch-averaged training losses for PEML and SupCon. Separate panels and vertical scales are used because the two objectives have different functional forms, numerical scales, and lower bounds. (c,d) Epoch mean $\overline{G}_e$ of the global parameter-gradient norm, as defined in Eq. (J.2); shading denotes the corresponding within-epoch standard deviation $s_{G,e}$ defined in Eq. (J.3). These panels likewise use separate vertical scales because the raw gradient magnitudes depend on the scale and structure of the corresponding objective. (e) Evolution of the centroid-relative encoder-space ratio $Q = D_{\text{intra}}/D_{\text{inter}}$, computed from the raw pre-projection encoder representations. Lower values indicate stronger within-class compactness relative to the between-centroid scale. Each curve corresponds to the representative seed-1 training trajectory of the respective method.

Table J.2
Paired linear-classification outcomes on the balanced CIFAR-100 test set. Rows report SupCon outcomes and columns report PEML outcomes. Both methods use the same plain linear-probe formulation and exactly frozen encoders. The marginal totals correspond to 76.49% accuracy for SupCon and 77.61% for PEML.

	PEML Correct	PEML Incorrect	Total
SupCon Correct	6977	672	7649
SupCon Incorrect	784	1567	2351
Total	7761	2239	10,000

Table J.3
Centroid-relative diagnostics in the raw encoder space. Percentages in the second and third rows are conditioned on the errors of the corresponding method. "Both methods correct" and "shared failures" use the paired subsets in Table J.2.

Diagnostic	SupCon	PEML
Number of linear-classification errors	2351	2239
Errors with $r_{\text{wrong}} < 1$ (%)	93.9	92.5
Predicted class equals nearest wrong centroid among errors (%)	55.7	69.9
Median r_{wrong}, both methods correct	1.450	3.094
Median r_{wrong}, shared failures	0.848	0.747

need not coincide because a learned affine linear classifier is not a nearest-centroid classifier.

Table J.2 shows that both methods classify 6977 test samples correctly and fail on the same 1567 samples. PEML alone corrects 784 samples missed by SupCon, whereas SupCon alone corrects 672 samples missed by PEML, yielding the net 112-sample advantage reported in the main linear evaluation. The 1456 method-specific outcomes also show that similar aggregate accuracies need not imply identical decision behavior.

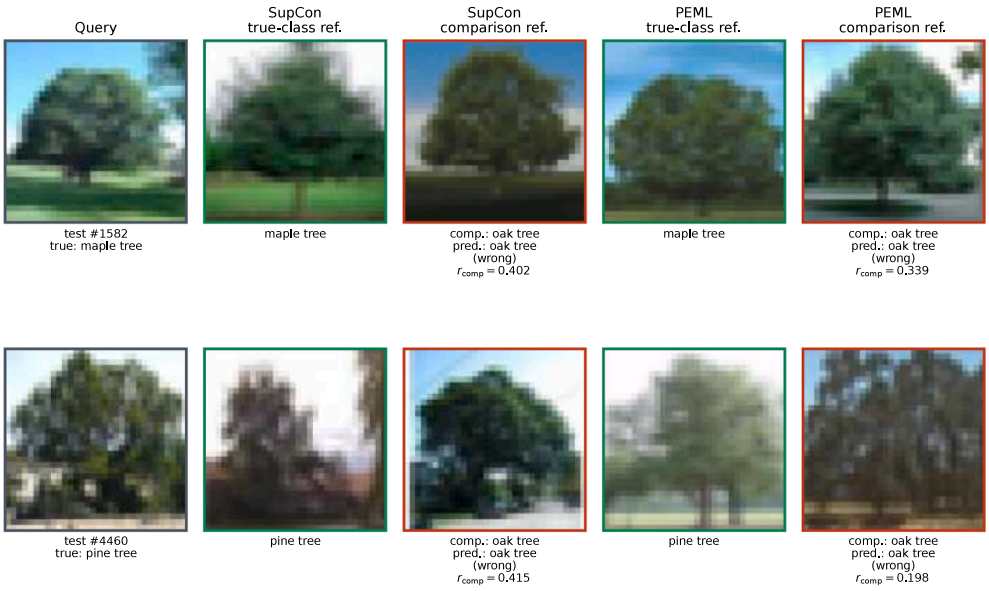


Fig. J.2. Severe centroid-inversion examples among shared failures. Both methods predict oak tree for test image #1582 (maple tree) and test image #4460 (pine tree). Green borders denote nearest training references from the true class, and red borders denote nearest training references from the comparison class. The references are retrieved independently in each method's raw encoder space after case selection and serve only as secondary visual context. The reported r_{comp} is defined in Eq. (J.5); values below one indicate that the comparison centroid is closer than the true-class centroid.

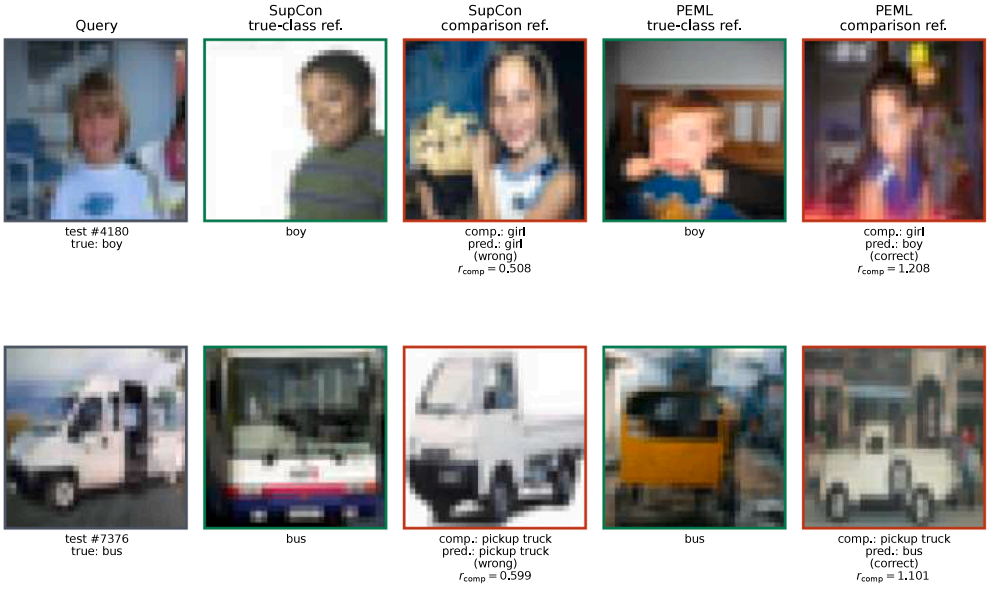


Fig. J.3. Severe centroid-inversion examples for which SupCon fails and PEML is correct. The comparison classes are the wrong SupCon predictions: girl for test image #4180 (boy) and pickup truck for test image #7376 (bus). SupCon has $r_{comp} < 1$ in both cases, whereas PEML places each query closer to its true-class centroid, with $r_{comp} > 1$. Border colors and reference-image roles follow Fig. J.2.

The first failure-conditioned row of Table J.3 shows that more than 92% of the errors of either method occur when the sample is closer to at least one wrong-class centroid than to its true-class centroid. Centroid inversion is therefore the dominant endpoint failure mode for both representations. The class predicted by the linear probe is the nearest wrong centroid for 55.7% of SupCon errors and 69.9% of PEML errors. The remaining cases are expected because the learned affine decision rule depends on class-specific weights and biases rather than only on Euclidean proximity to training centroids.

The ratio distributions reveal a more specific contrast. Among the 6977 samples classified correctly by both methods, the median r_{wrong} is 3.094 for PEML and 1.450 for SupCon, indicating substantially stronger true-to-nearest-wrong centroid separation under PEML for this shared correct subset. Among the 1567 shared failures, however, the median is 0.747 for PEML and 0.848 for SupCon. Thus, PEML improves aggregate class-level organization and the separation of the jointly correct majority, while some of its residual hard cases lie more deeply within a competing centroid region. This conditional behavior does not contradict the higher overall nearest-centroid accuracy in Table 3; it identifies the geometry of the smaller subset that remains difficult after the aggregate improvement.

To visualize severe cases without selecting examples by image appearance, we apply a prespecified centroid-based rule. Within each method-specific failure category, samples are ranked in ascending order of r_{pred} for the failing method. Shared failures are ranked by $\max\{r_{pred}^{(SupCon)}, r_{pred}^{(PEML)}\}$ so that both methods must exhibit a strong in-

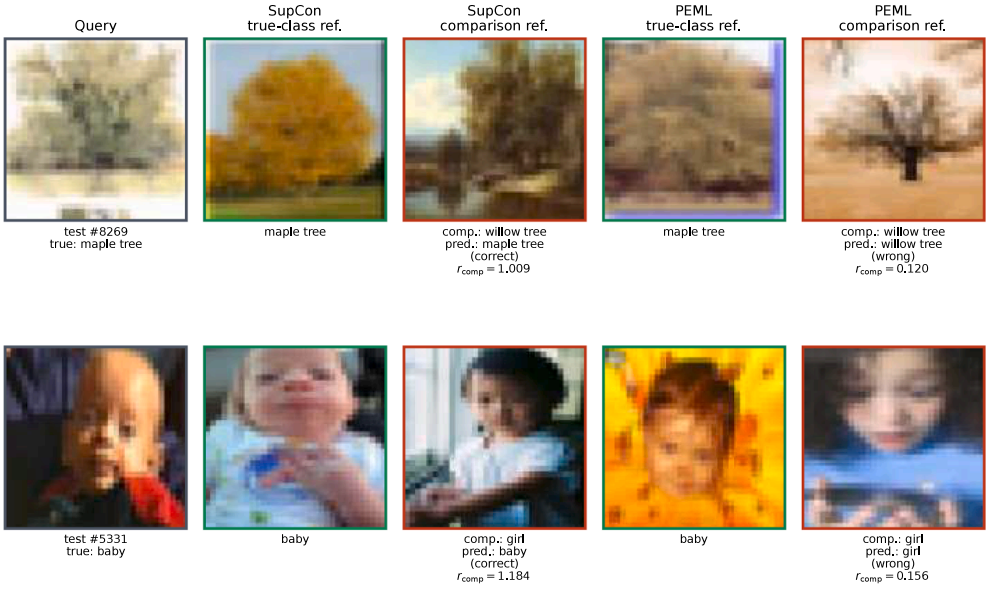


Fig. J.4. Severe centroid-inversion examples for which PEML fails and SupCon is correct. The comparison classes are the wrong PEML predictions: willow tree for test image #8269 (maple tree) and girl for test image #5331 (baby). PEML has $r_{\text{comp}} < 1$ in both cases, whereas SupCon places each query closer to its true-class centroid, with $r_{\text{comp}} > 1$. Border colors and reference-image roles follow Fig. J.2.

version. We select two samples per category while allowing at most one sample from each true class; no fallback case is required. The resulting three-figure collection is therefore category-balanced by construction. Only after this selection do we retrieve the nearest training image from the true class and from a comparison class as secondary visual references.

For each displayed case, let $c^{(m)}_{\text{comp},i}$ be the comparison class for method m. For a shared failure, it is the method's predicted class. For a method-specific failure, the wrong class predicted by the failing method is used as the common comparison class for both representations. The displayed scale-invariant ratio is

$$r^{(m)}_{\text{comp}}(i) = \frac{\left\| \boldsymbol{h}^{(m)}_i - \boldsymbol{\mu}^{(m),\text{tr}}_{c^{(m)}_{\text{comp},i}} \right\|_2}{\left\| \boldsymbol{h}^{(m)}_i - \boldsymbol{\mu}^{(m),\text{tr}}_{y_i} \right\|_2}. \tag{J.5}$$

The six selected examples in Fig. J.2, J.3, J.4 involve visually similar or semantically related CIFAR-100 fine-class pairs, including tree species, child-related categories, and road vehicles. Because the cases are deliberately selected for severe centroid inversion, they illustrate the geometry of pronounced endpoint errors rather than their population frequency. They provide sample-level context for the aggregate failure statistics in Table J.3.

Taken together, the trajectory and endpoint analyzes provide complementary views of the learned representation. Both methods complete the full 1,000-epoch optimization without sustained divergence, while PEML develops a persistently lower global centroid-relative ratio Q from epoch 50 onward. At the late-training checkpoints used for the endpoint analysis, PEML also exhibits higher nearest-centroid accuracy and a larger median nearest-wrong-centroid ratio among samples classified correctly by both methods. The residual errors of both methods are predominantly associated with nearest-wrong-centroid inversion, and the selected severe cases illustrate how such inversions manifest between closely related competing classes. These results provide a temporal and sample-level geometric account consistent with the balanced CIFAR-100 classification and representation-quality improvements reported in Sections 4.1 and 4.3.

Appendix K. Sensitivity analysis of annulus-scaled projection parameters

This section examines the sensitivity of the annulus-scaled projection to its three hyperparameters: the inner radius R_A, the outer radius R_B, and the radius temperature τ_{r} in Eq. (38).

K.1. Interpretation of the projection parameters

For the standardized pre-projection norm $z = (\|x\| - \mu)/s$, the norm of the projected embedding in Eq. (38) can be written as

$$r(z; R_A, R_B, \tau_{\text{r}}) = \|\psi(x)\| = R_A + (R_B - R_A)\sigma\left(\frac{z}{\tau_{\text{r}}}\right). \tag{K.1}$$

The inner radius R_A sets the lower bound on the output norm, whereas the outer radius R_B sets its upper bound. In PEML, R_B also determines the maximum feasible pairwise distance $d_{\max} = 2R_B$, as described in Eq. (39). The annulus width $R_B - R_A$ controls the amount of radial variation available to the projected embeddings. To isolate the effect of τ_{r} from the absolute radius values, we define the normalized radial position within the annulus as

$$q(z; \tau_{\text{r}}) = \frac{r(z; R_A, R_B, \tau_{\text{r}}) - R_A}{R_B - R_A} = \sigma\left(\frac{z}{\tau_{\text{r}}}\right). \tag{K.2}$$

Because z is standardized using the mini-batch mean and standard deviation of the pre-projection norms, τ_{r} controls the transition width of the radial mapping in units of the norm standard deviation. Its limiting behavior is

$$\lim_{\tau_{\text{r}} \to 0^+} q(z; \tau_{\text{r}}) = \begin{cases} 0, & z < 0, \\ \frac{1}{2}, & z = 0, \\ 1, & z > 0, \end{cases} \tag{K.3}$$

$$\lim_{\tau_{\text{r}} \to \infty} q(z; \tau_{\text{r}}) = \frac{1}{2}. \tag{K.4}$$

Therefore, an excessively small τ_{r} produces an approximately two-shell radial distribution concentrated near R_A and R_B, whereas an excessively large τ_{r} compresses the radial distribution toward a single shell at the annulus midpoint $(R_A + R_B)/2$. More quantitatively, the fraction

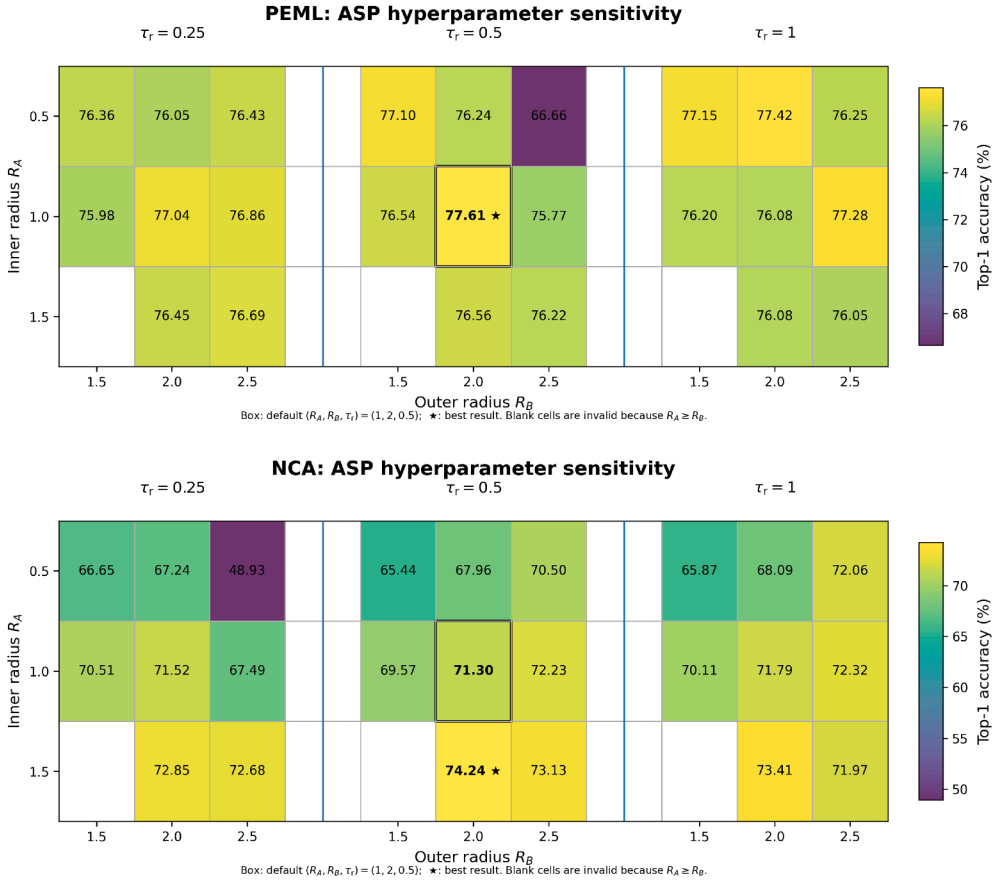


Fig. K.1. Sensitivity of PEML (top) and the NCA-style loss (bottom) to the annulus-scaled projection parameters R_A, R_B, and τ_r on CIFAR-100. Each heatmap cell reports the top-1 frozen-encoder linear-evaluation accuracy (%) obtained from a single training run. The boxed cell denotes the common configuration $(R_A, R_B, \tau_r) = (1, 2, 0.5)$ used in the main-text loss comparison, whereas the star denotes the best-performing configuration observed for each loss. Blank cells correspond to invalid configurations satisfying $R_A \geq R_B$. The color scale is shared across the three values of τ_r within each loss but is independently scaled between PEML and NCA to make the within-loss variation visible.

of the annulus width covered by standardized norms in the interval $z \in [-k, k]$ is

$$W_k(\tau_r) = q(k; \tau_r) - q(-k; \tau_r) = \tanh\left(\frac{k}{2\tau_r}\right). \tag{K.5}$$

For $\tau_r = 0.25$, 0.5, and 1.0, the one-standard-deviation interval covers approximately 96.4%, 76.2%, and 46.2% of the annulus width, respectively. In particular, when $\tau_r = 0.5$, the interval $z \in [-2, 2]$ covers approximately 96.4% of the annulus width. Hence, $\tau_r = 0.5$ represents an intermediate regime that preserves meaningful radial variation around the center of the norm distribution while softly clipping more extreme values. Accordingly, we evaluate representative temperatures on either side of this setting; the limiting regimes are already characterized analytically.

K.2. Experimental protocol

We evaluate the sensitivity of the annulus-scaled projection using PEML and the NCA-style objective. PEML is included because it is the proposed loss and uses R_B to define its feasible distance range. The NCA-style objective is included as a structurally different Euclidean-distance-based objective, allowing us to examine whether the observed parameter sensitivity is specific to PEML or also arises under another Euclidean metric loss.

All experiments use CIFAR-100 and follow the same two-stage ResNet-50 training and frozen-encoder linear-evaluation protocol described in Section 4.1. The model architecture, augmentation procedure, optimizer, learning-rate schedule, batch size, embedding dimension, and all loss-specific hyperparameters other than the annulus-scaled projection parameters are held fixed. In particular, the temperature internal to the NCA-style objective remains fixed; only the radius temperature τ_r in the annulus-scaled projection is varied.

We consider the following parameter grid:

$$R_A \in \{0.5, 1.0, 1.5\}, \qquad R_B \in \{1.5, 2.0, 2.5\},$$
$$\tau_r \in \{0.25, 0.5, 1.0\}. \tag{K.6}$$

Only configurations satisfying $R_A < R_B$ are valid. This yields eight valid radius combinations for each value of τ_r and a total of 24 configurations for each loss. Each heatmap cell in Fig. K.1 reports the CIFAR-100 top-1 linear-evaluation accuracy obtained from a single training run.

For the controlled loss comparison in the main text, all losses use the common annulus-scaled projection configuration $(R_A, R_B, \tau_r) = (1, 2, 0.5)$. Under this common configuration, the NCA-style loss achieves 71.30%, compared with 69.39% under ℓ_2 normalization. The sensitivity analysis additionally identifies the loss-specific best NCA configuration, $(R_A, R_B, \tau_r) = (1.5, 2, 0.5)$, which achieves 74.24%. Accordingly, the boxed cell in both heatmaps denotes the common configuration used in the main-text comparison, whereas the star denotes the best-performing configuration observed for each loss.

K.3. Results and discussion

The PEML results in Fig. K.1 show that the default configuration $(R_A, R_B, \tau_r) = (1, 2, 0.5)$ achieves a top-1 accuracy of 77.61%, which is the highest accuracy among all 24 evaluated configurations. This result directly supports the empirical annulus-scaled projection setting used for PEML in the main experiments.

The observed PEML performance is also generally stable over the tested parameter range. For $\tau_r = 0.25$, the accuracies range from 75.98% to 77.04%, whereas for $\tau_r = 1.0$, they range from 76.05% to 77.42%. Except for the configuration $(R_A, R_B, \tau_r) = (0.5, 2.5, 0.5)$, all evaluated PEML configurations lie within 1.85 percentage points of the best result. Thus, the selected configuration lies within a broad high-performing region rather than constituting an isolated and fragile optimum.

The lower PEML performance at $(R_A, R_B, \tau_r) = (0.5, 2.5, 0.5)$ indicates that simultaneously permitting a small lower norm bound and a wide radial interval can be detrimental. In this configuration, the annulus width is $R_B - R_A = 2$, allowing substantially greater norm and distance variability than the default width of 1. Because PEML defines $d_{\max} = 2R_B$, increasing R_B also expands the feasible distance range used by the adaptive thresholds and force kernels. This result suggests that the lower and upper radial bounds should be selected jointly rather than independently.

For the NCA-style loss, the common configuration $(R_A, R_B, \tau_r) = (1, 2, 0.5)$ achieves 71.30%, improving upon the ℓ_2-normalized result of 69.39% without loss-specific tuning. The highest NCA accuracy in the sensitivity grid is 74.24%, obtained at $(R_A, R_B, \tau_r) = (1.5, 2, 0.5)$. The sensitivity analysis therefore shows both that the common annulus-scaled projection configuration improves NCA and that further gains are possible when the radial bounds are adapted to the distance-scale characteristics of the loss.

Moreover, when $R_B = 2$, increasing R_A from 0.5 to 1.5 consistently improves NCA performance for all three evaluated values of τ_r. This pattern indicates that the inner radius can be particularly important for objectives based on an exponential transformation of squared Euclidean distances, because allowing embeddings with very small norms substantially changes the scale and concentration of the resulting pairwise-distance distribution.

The most pronounced NCA degradation occurs at $(R_A, R_B, \tau_r) = (0.5, 2.5, 0.25)$, where the accuracy decreases to 48.93%. According to Eqs. (K.3) and K.5, the small value $\tau_r = 0.25$ produces a strongly saturated radial mapping, while $(R_A, R_B) = (0.5, 2.5)$ defines the widest annulus considered. Their combination therefore induces an approximately two-shell distribution with a large separation between the inner and outer shells. The observed degradation is consistent with the sensitivity of the NCA exponential kernel to the scale of squared Euclidean distances: a highly dispersed distance distribution can cause the resulting neighborhood probabilities to become excessively concentrated. This result indicates that a small τ_r should not be combined with an unnecessarily wide annulus for NCA-like objectives.

Overall, the results support $\tau_r = 0.5$ as a reasonable intermediate radius temperature. It avoids both the near-two-shell behavior induced by an excessively small temperature and the near-single-shell behavior induced by an excessively large temperature. The results also validate $(R_A, R_B) = (1, 2)$ for PEML, for which the complete default configuration attains the highest observed accuracy. At the same time, the NCA results show that the optimal radial bounds can depend on the formulation and distance-scale sensitivity of the metric loss. The annulus-scaled projection is therefore broadly compatible with Euclidean-distance-based objectives, but its optimal radius configuration should not be interpreted as universal across all losses.

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